

Important Questions for Class 12

Maths

Chapter 7 - Integrals

Very Short Answer Questions

1 Mark

Evaluate the following integrals

1. $\int (\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}) dx$

Ans: We know that $\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}$ is .

This implies $\int (\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}) dx = \int \frac{\pi}{2} dx$

Solving we get,

$$\int \frac{\pi}{2} dx = \frac{\pi}{2} x + C.$$

2. $\int_{-1}^1 e^{|x|} dx$

Ans: Let $I = \int_{-1}^{+1} e^{|x|} dx$

Now, $f(x) = e^{|x|}$

$$f(-x) = e^{|-x|} = e^{|x|} = f(x)$$

So, $f(x)$ is an even function.

$$\int_{-1}^{+1} e^{|x|} dx = 2 \int_0^1 e^{|x|} dx$$

$$= 2 \int_0^1 e^x dx \text{ (as, } |x| = x, x > 0)$$

$$= 2 \left[e^x \right]_0^1$$

$$= 2(e-1)$$

$$= 2e - 2$$

$$3. \int \frac{1}{1 - \sin^2 x} dx$$

Ans:

$$\int \frac{1}{1 - \sin^2 x} dx$$

$$= \int \frac{1}{\cos^2 x} dx$$

$$= \int \sec^2 x dx$$

$$= \tan x + C$$

$$4. \int \left(8^x + x^8 + \frac{8}{x} + \frac{x}{8} \right) dx$$

Ans:

$$\int 8^x dx + \int x^8 dx + \int \frac{8}{x} dx + \int \frac{x}{8} dx$$

$$= \frac{8^x}{\log 8} + \frac{x^9}{9} + 8 \log|x| + \frac{x^2}{16}$$

$$5. \int_{-1}^1 x^{99} \cos^4 x dx$$

Ans:

$$f(x) = x^{99} \cos^4 x$$

$$f(-x) = (-x)^{99} (\cos(-x))^4$$

$$= -x^{99} \cos^4 x = -f(x)$$

We know that property $\int_{-a}^a f(x) dx = 0$, when $f(-x) = -f(x)$

$$\text{So, } \int_{-1}^1 x^{99} \cos^4 x dx = 0 \frac{\pi}{2}$$

$$6. \int \frac{1}{x \log x \log(\log x)} dx$$

Ans:

$$I = \int \frac{1}{x \log x \log(\log x)} dx$$

$$I = \int \frac{1}{\log(\log x)} \left\{ \frac{dx}{x \cdot \log(x)} \right\}$$

$$\text{Let } \log(\log x) = t$$

$$\Rightarrow \frac{1}{\log x} \cdot \frac{1}{x} \cdot dx = dt$$

$$\Rightarrow \frac{dx}{x \cdot \log(x)} = dt$$

Hence,

$$I = \int \frac{dt}{t}$$

$$= \log|t| + C$$

$$= \log|\log(\log x)| + C$$

$$7. \int_0^{\frac{\pi}{2}} \log \left(\frac{4 + 3 \sin x}{4 + 3 \cos x} \right) dx$$

$$\text{Ans: } I = \int_0^{\frac{\pi}{2}} \log \left(\frac{4 + 3 \sin x}{4 + 3 \cos x} \right) \dots (1)$$

$$I = \int_0^{\frac{\pi}{2}} \log \left[\frac{4 + 3\sin\left(\frac{\pi}{2} - x\right)}{4 + 3\cos\left(\frac{\pi}{2} - x\right)} \right] dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \log \left(\frac{4 + 3\cos x}{4 + 3\sin x} \right) dx \dots (2)$$

Adding (1) and (2),

$$2I = \int_0^{\frac{\pi}{2}} \log \left[\left(\frac{4 + 3\sin x}{4 + 3\cos x} \right) \times \left(\frac{4 + 3\cos x}{4 + 3\sin x} \right) \right] dx$$

$$= \int_0^{\frac{\pi}{2}} (\log 1) dx$$

$$= \int_0^{\frac{\pi}{2}} (0) dx$$

$$= 0$$

8. $\int (e^{a \log x} + e^{x \log a}) dx$

Ans: We know that

$$a \log x = \ln x^a$$

$$x \log a = \ln a^x$$

$$\int (e^{a \log x} + e^{x \log a}) dx = \int (e^{\ln x^a} + e^{\ln a^x}) dx$$

Also, we know that $e^{\ln(f(x))} = f(x)$

$$= \int (x^a + a^x) dx$$

$$= \frac{x^{a+1}}{a+1} + \frac{a^x}{\ln a} + c$$

9. $\int \left(\frac{\cos 2x + 2\sin^2 x}{\cos^2 x} \right) dx$

Ans: We know that $\cos 2x = \cos^2 x - \sin^2 x$

$$\begin{aligned}\int \left(\frac{\cos 2x + 2\sin^2 x}{\cos^2 x} \right) dx &= \int \left(\frac{\cos^2 x - \sin^2 x + 2\sin^2 x}{\cos^2 x} \right) dx \\ &= \int \left(\frac{\cos^2 x + \sin^2 x}{\cos^2 x} \right) dx = \int \left(\frac{1}{\cos^2 x} \right) dx\end{aligned}$$

(As $\cos^2 x + \sin^2 x = 1$)

$$= \int \sec^2 x dx = \tan x + C$$

10. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x dx$

Ans: Let $f(x) = \sin^7 x$

$$f(-x) = \sin^7(-x) = -\sin^7 x$$

This implies that $f(x)$ is an odd function.

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x dx = 0$$

$$[\because \int_{-a}^a f(x) dx = 0 \text{ if } f(x) \text{ is an odd function}].$$

11. $\int (x^c + c^x) dx$

Ans: Using the properties,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \int n^x dx = \frac{n^x}{\ln n} + C$$

$$\int (x^c + c^x) dx = \frac{x^{c+1}}{c+1} + \frac{c^x}{\ln c} + C$$

$$12. \frac{d}{dx} \left[\int f(x) dx \right]$$

Ans: Differentiating and integrating a function at a time leaves the function itself.
So,

$$\frac{d}{dx} \left[\int f(x) dx \right] = f(x) + C$$

$$13. \int \frac{1}{\sin^2 x \cos^2 x} dx$$

$$\begin{aligned} \text{Ans: } \int \frac{1}{\sin^2 x \cos^2 x} dx &= \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} dx \quad (\text{Using } \cos^2 x + \sin^2 x = 1) \\ &= \int \frac{\sin^2 x}{\sin^2 x \cos^2 x} dx + \int \frac{\cos^2 x}{\sin^2 x \cos^2 x} dx \\ &= \int \frac{1}{\cos^2 x} dx + \int \frac{1}{\sin^2 x} dx \\ &= \int \sec^2 x dx + \int \operatorname{cosec}^2 x dx \\ &= \tan x - \cot x + C \end{aligned}$$

$$14. \int \frac{1}{\sqrt{x} + \sqrt{x-1}} dx$$

Ans: Rationalizing we have,

$$\begin{aligned} &\int \frac{1}{\sqrt{x} + \sqrt{x-1}} \times \frac{\sqrt{x} - \sqrt{x-1}}{\sqrt{x} - \sqrt{x-1}} dx \\ &= \int \frac{\sqrt{x} - \sqrt{x-1}}{x - x + 1} dx \\ &= \int \sqrt{x} - \sqrt{x-1} dx \\ &= \frac{x^{\frac{3}{2}}}{3/2} - \frac{(x-1)^{\frac{3}{2}}}{3/2} + C \\ &= \frac{2}{3} x^{\frac{3}{2}} - \frac{2}{3} (x-1)^{\frac{3}{2}} + C \end{aligned}$$

15. $\int e^{-\log e^x} dx$

Ans: We know that $\log e^x = x$

Hence, $\int e^{-\log e^x} dx = \int e^{-x} dx$

Now, put $u = -x$

$$\frac{du}{dx} = -1$$

$$\Rightarrow dx = -du$$

$$\begin{aligned}\int e^{-x} dx &= \int -e^u du \\ &= -\int e^u du = -e^u + C \\ &= -e^{-x} + C\end{aligned}$$

16. $\int \frac{e^x}{a^x} dx$

Ans:

$$\begin{aligned}\int \frac{e^x}{a^x} dx &= \int \left(\frac{e}{a} \right)^x dx \\ &= \frac{\left(\frac{e}{a} \right)^x}{\log \left(\frac{e}{a} \right)} + C \\ \left(\int m^x &= \frac{m^x}{\log m} + C \right)\end{aligned}$$

17. $\int 2^x e^x dx$

Ans:

$$\begin{aligned}\int 2^x e^x dx &= \int (2e)^x dx \\ &= \frac{1}{\log(2e)} \cdot (2e)^x + C\end{aligned}$$

$$= \frac{(2e)^x}{\log(2e)} + C$$

$$= \frac{2^x e^x}{\log(2e)} + C$$

18. $\int \frac{x}{\sqrt{x+1}} dx$

Ans: Let $u = x + 1, du = dx$

$$\Rightarrow u - 1 = x$$

$$\int \frac{u-1}{\sqrt{u}} du = \int \frac{u}{\sqrt{u}} du - \int \frac{1}{\sqrt{u}} du$$

$$= \int \frac{u}{u^{\frac{1}{2}}} du - \int \frac{1}{u^{\frac{1}{2}}} du$$

$$= \int u^{\frac{1}{2}} du - \int u^{-\frac{1}{2}} du$$

$$= \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} - \frac{u^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C$$

$$= \frac{u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= \frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} + C$$

Substituting the value of u we have,

$$\int \frac{x}{\sqrt{x+1}} dx = \frac{2}{3} (x+1)^{\frac{3}{2}} - 2(x+1)^{\frac{1}{2}} + C$$

19. $\int \frac{x}{(x+1)^2} dx$

Ans: Let us assume that $u = x + 1, du = dx$

$$\Rightarrow u - 1 = x$$

Now, substituting in the given integral we get,

$$\int \frac{u-1}{u^2} du = \int \left(\frac{1}{u} + \frac{1}{u^2} \right) du$$

Now, by using the power formula of integration we get,

$$= \ln(u) + \frac{u^{-2+1}}{-2+1} + c$$

$$= \ln(u) + \frac{u^{-1}}{-1} + c$$

$$= \ln(u) - \frac{1}{u} + c$$

Now replacing the value of u we get,

$$= \ln(x+1) - \frac{1}{x+1} + c$$

20. $\int \frac{e^{\sqrt{x}}}{x} dx$

Ans: Let us assume $u = \sqrt{x} \Rightarrow u^2 = x$

Now differentiating on both sides,

$$2u du = dx$$

Now, substituting in the given expression we get,

$$= \int \frac{e^u}{u^2} (2u du)$$

$$= 2 \int \frac{e^u}{u} du$$

By using the formula $\int \frac{e^x}{x} dx = Ei(x)$ we get,

$$= 2Ei(u) + c$$

Substituting back in terms of x we have,

$$= 2\text{Ei}(\sqrt{x}) + c$$

21. $\int \cos^2 \alpha dx$

Ans: We know that $\cos^2 \alpha$ is a constant.

So, $\int \cos^2 \alpha dx = x \cos^2 \alpha + C$

22. $\int \frac{1}{x \cos \alpha + 1} dx$

Ans: $I = \int \frac{1}{x \cos \alpha + 1} dx$

Let $x \cos \alpha + 1 = t$

$\cos \alpha dx = dt$

$dx = \frac{dt}{\cos \alpha}$

$I = \int \frac{1}{t} \cdot \frac{dt}{\cos \alpha}$

$\therefore I = \frac{1}{\cos \alpha} \int \frac{dt}{t} = \frac{1}{\cos \alpha} \log|t| + C.$

$I = \frac{\log|x \cos \alpha + 1|}{\cos \alpha} + C$

23. $\int \sec x \cdot \log(\sec x + \tan x) dx$

Ans: Substitute $u = \log(\sec x + \tan x)$

$du = \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} dx$ & $dx = \frac{\sec x + \tan x}{\sec x \tan x + \sec^2 x} du$

$I = \int \sec x \cdot u \cdot \frac{\sec x + \tan x}{\sec x \tan x + \sec^2 x} du$

$$I = \int \sec x \cdot u \cdot \frac{\sec x + \tan x}{\sec x (\tan x + \sec x)} du$$

$$I = \int u du$$

$$I = \frac{1}{2} u^2 + C$$

$$I = \frac{1}{2} \log^2 (\sec x + \tan x) + C$$

$$I = \frac{(\log |\sec x + \tan x|)^2}{2} + C$$

24. $\int \frac{1}{\cos \alpha + x \sin \alpha} dx$

Ans: Substitute $u = \cos \alpha + x \sin \alpha$

$$\begin{aligned} \int \frac{1}{\cos \alpha + x \sin \alpha} dx &= \int \frac{1}{u \sin \alpha} du \\ &= \frac{1}{\sin \alpha} \cdot \int \frac{1}{u} du \left[\int a \cdot f(x) dx = a \cdot \int f(x) dx \right] \end{aligned}$$

Now, use $\int \frac{1}{u} du = \ln(|u|)$

$$= \frac{1}{\sin \alpha} \log |u|$$

$$= \frac{1}{\sin \alpha} \log |\cos \alpha + x \sin \alpha| + C$$

25. $\int \cot x \cdot \log \sin x dx$

Ans: We know the formula of cotangent ratio as,

$$\cot x = \frac{\cos x}{\sin x}$$

By using the above formula for given expression we get,

$$\int \frac{\cos x \log(\sin x)}{\sin x} dx$$

Substitute $u = \sin x$

Now, differentiating on both sides we get,

$$du = \cos x dx$$

Now substituting in the given expression we get,

$$= \int \frac{\log u}{u} du$$

Substitute $v = \log u$ so that by differentiating we have,

$$dv = \frac{1}{u} du$$

Now substituting in the above integral we get,

$$\begin{aligned} &= \int v dv \\ &= \frac{v^2}{2} \\ &= \frac{\log^2(\sin x)}{2} \\ &= \frac{(\log \sin x)^2}{2} + C \end{aligned}$$

$$26. \int \left(x - \frac{1}{2}\right)^3 dx$$

Ans:

$$\begin{aligned} \int \left(x - \frac{1}{2}\right)^3 dx &= \int \left(x^3 - \frac{3x^2}{2} + \frac{3x}{4} - \frac{1}{8}\right) dx \\ &= \int x^3 dx - \int \frac{3x^2}{2} dx + \int \frac{3x}{4} dx - \int \frac{1}{8} dx \\ &= \frac{x^4}{4} - \frac{x^3}{2} + \frac{3x^2}{8} - \frac{1}{8}x + c \end{aligned}$$

$$27. \int \frac{1}{x(2+3\log x)} dx$$

Ans: Substitute $u = 2 + 3\log x$

$$\begin{aligned} \int \frac{1}{x(2+3\log x)} dx &= \int \frac{1}{3u} du \\ &= \frac{1}{3} \int \frac{1}{u} du \\ &= \frac{1}{3} \log u \left[\int \frac{1}{u} du = \log u \right] \\ &= \frac{1}{3} \log |2 + 3\log x| + C \end{aligned}$$

$$28. \int \frac{1 - \sin x}{x + \cos x} dx$$

Ans: Substitute $u = x + \cos x$

$$\begin{aligned} \int \frac{1 - \sin x}{x + \cos x} dx &= \int \frac{1}{u} du \\ &= \log u \\ &= \log |x + \cos x| + C \end{aligned}$$

$$29. \int \frac{1 - \cos x}{\sin x} dx$$

Ans: Separating each term in the numerator with the denominator of the expression $\frac{1 - \cos x}{\sin x}$ we get,

$$\int \frac{1 - \cos x}{\sin x} dx = \int \frac{1}{\sin x} - \frac{\cos x}{\sin x} dx$$

By sum rule,

$$= \int \frac{1}{\sin x} dx - \int \frac{\cos x}{\sin x} dx$$

$$= \int \csc x dx - \int \cot x dx$$

By using the standard results of differentiation we get,

$$= \log \left| \tan \frac{x}{2} \right| - \log |\sin x| + C$$

$$30. \int \frac{x^{e-1} + e^{x-1}}{x^e + e^x} dx$$

Ans: Let us take the given integral as,

$$\begin{aligned} \int \frac{x^{e-1} + e^{x-1}}{x^e + e^x} dx &= \int \frac{x^{e-1} + \frac{e^x}{e}}{x^e + e^x} dx \\ &= \int \frac{ex^{e-1} + e^x}{e(x^e + e^x)} dx \end{aligned}$$

Now, let us take, $x^e + e^x = k$, then differentiating on both sides we have,

$$(ex^{e-1} + e^x) dx = dk$$

So, we have the given expression as,

$$\begin{aligned} &= \frac{1}{e} \int \frac{dk}{k} \\ &= \frac{1}{e} \ln(k) + c \\ &= \frac{1}{e} \ln(x^e + e^x) + c \end{aligned}$$

$$31. \int \frac{(x+1)}{x} (x + \log x) dx$$

Ans: Substitute $u = x + \log x$

Differentiating both sides,

$$\frac{du}{dx} = 1 + \frac{1}{x}$$

$$\left(1 + \frac{1}{x}\right) dx = du$$

$$\int \frac{(x+1)}{x} (x + \log x) dx = \int u dx$$

$$\int u du = \frac{u^2}{2}$$

$$= \frac{(x + \log x)^2}{2} + C$$

$$32. \int \left(\sqrt{ax} - \frac{1}{\sqrt{ax}} \right)^2 dx$$

Ans: Simplifying we have,

$$\int \left(\sqrt{ax} - \frac{1}{\sqrt{ax}} \right)^2 dx = \int ax - 2 + \frac{1}{ax} dx$$

$$= \int ax dx - \int 2 dx + \int \frac{1}{ax} dx$$

$$= \frac{ax^2}{2} - 2x + \frac{1}{a} \log|ax| + C$$

$$33. \int_0^{\pi} |\cos x| dx$$

$$\text{Ans: } \int_0^{\pi} |\cos x| dx = \int_0^{\frac{\pi}{2}} \cos x dx + \int_{\frac{\pi}{2}}^{\pi} -\cos x dx$$

We know that

$$\int_0^{\frac{\pi}{2}} \cos x dx = 1,$$

$$\int_{\frac{\pi}{2}}^{\pi} -\cos x dx = 1$$

$$= 1 + 1$$

$$= 2$$

34. $\int_0^2 [x] dx$ where $[]$ is greatest integer function.

Ans: We know that when $0 < x < 1$ we have,

$$\Rightarrow [x] = 0$$

Similarly when $1 < x < 2$ we have,

$$\Rightarrow [x] = 1$$

Therefore,

$$\therefore \int_0^2 [x] dx = \int_0^1 [x] dx + \int_1^2 [x] dx$$

$$= \int_0^1 0 dx + \int_1^2 1 dx$$

$$= 0 + (2 - 1)$$

$$= 1$$

35. $\int_0^{\sqrt{2}} [x^2] dx$ where $[]$ is greatest integer function.

Ans: When

$$0 < x < 1$$

$$\Rightarrow 0 < x^2 < 1$$

$$\Rightarrow [x^2] = 0$$

Similarly, when

$$1 < x < \sqrt{2}$$

$$1 < x^2 < 2$$

$$[x^2] = 1$$

Therefore, the integral is,

$$\begin{aligned} \therefore \int_0^{\sqrt{2}} [x^2] dx &= \int_0^1 [x^2] dx + \int_1^{\sqrt{2}} [x^2] dx \\ &= \int_0^1 0 dx + \int_1^{\sqrt{2}} 1 dx \\ &= 0 + (\sqrt{2} - 1) \\ &= \sqrt{2} - 1 \end{aligned}$$

$$36. \int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx$$

Ans: Let us assume the given integral as,

$$I = \int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx \dots\dots\dots(1)$$

We know the standard formula of differentiation as,

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

Using this formula for given integral we get,

$$I = \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(x)} dx \dots\dots\dots(2)$$

Now, adding both equations (1) and (2) we get,

$$2I = \int_a^b \frac{f(x) + f(a+b-x)}{f(a+b-x) + f(x)} dx$$

$$2I = \int_a^b dx$$

$$I = \frac{b-a}{2}$$

$$37. \int_{-2}^1 \frac{|x|}{x} dx$$

Ans: Eliminating absolutes we have,

$$\int_{-2}^1 \frac{|x|}{x} dx = \int_{-2}^0 \frac{-x}{x} dx + \int_0^1 \frac{x}{x} dx$$

On solving the above integrals we get,

$$\begin{aligned} \int_{-2}^0 \frac{-x}{x} dx &= -2 + \int_0^1 \frac{x}{x} dx = 1 \\ &= -2 + 1 \\ &= -1 \end{aligned}$$

$$38. \int_{-1}^1 x|x| dx$$

Ans: Eliminating absolutes we have,

$$\int_{-1}^1 x|x| dx = \int_{-1}^0 x(-x) dx + \int_0^1 x.x dx$$

Also,

$$\int_{-1}^0 x(-x) dx = -\frac{1}{3},$$

$$\int_0^1 x.x dx = \frac{1}{3}$$

$$= -\frac{1}{3} + \frac{1}{3}$$

$$= 0$$

$$39. \text{ If } \int_0^a \frac{1}{1+x^2} = \frac{\pi}{4} \text{ then what is value of } a.$$

Ans:

$$\int \frac{1}{1+x^2} = \tan^{-1} x + C$$

$$\int_0^a \frac{1}{1+x^2} = \tan^{-1}(a)$$

$$\tan^{-1}(a) = \frac{\pi}{4}$$

$$\tan \frac{\pi}{4} = a$$

$$\Rightarrow a = 1$$

$$40. \int_a^b f(x) dx + \int_b^a f(x) dx$$

Ans: Using the property $\int_p^q f(x) dx = -\int_q^p f(x) dx$ we have,

$$\int_a^b f(x) dx + \int_b^a f(x) dx = \int_a^b f(x) dx - \int_a^b f(x) dx = 0$$

Long Answer Questions

4 Marks

41. Evaluate:

$$(i) \int \frac{x \csc(\tan^{-1} x^2)}{1+x^4} dx$$

Ans: Let $\tan^{-1} x^2 = t \Leftrightarrow \frac{dt}{dx} = \frac{2x}{1+x^4}$. Substitute the values in the given expression.

$$\int \frac{x \csc(\tan^{-1} x^2)}{1+x^4} dx = \int \csc(t) \times \frac{dt}{2}$$

$$\Rightarrow \int \frac{x \csc(\tan^{-1} x^2)}{1+x^4} dx = \frac{1}{2} [\log |\csc(t) - \cot(t)|] + C$$

Substitute the value of t in the above equation.

$$\int \frac{x \csc(\tan^{-1} x^2)}{1+x^4} dx = \frac{1}{2} \left[\log \left| \csc(\tan^{-1} x^2) - \cot(\tan^{-1} x^2) \right| \right] + C$$

$$\Rightarrow \int \frac{x \csc(\tan^{-1} x^2)}{1+x^4} dx = \frac{1}{2} \left[\log \left| \csc(\tan^{-1} x^2) - \frac{1}{x^2} \right| \right] + C$$

Hence the value of $\int \frac{x \csc(\tan^{-1} x^2)}{1+x^4} dx$ is $\frac{1}{2} \left[\log \left| \csc(\tan^{-1} x^2) - \frac{1}{x^2} \right| \right] + C$.

(ii) $\int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx$.

Ans: Rationalize the fraction by multiplying and dividing with $\sqrt{x+1} - \sqrt{x-1}$.

$$\int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx = \int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} \times \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} - \sqrt{x-1}} dx$$

$$\Rightarrow \int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx = \int \frac{(\sqrt{x+1} - \sqrt{x-1})^2}{(\sqrt{x+1})^2 - (\sqrt{x-1})^2} dx$$

$$\Rightarrow \int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx = \int \frac{2x - 2(\sqrt{x^2 - 1})}{2} dx$$

$$\Rightarrow \int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx = \int x dx - \int \sqrt{x^2 - 1} dx$$

Apply the integration formula

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log |x + \sqrt{x^2 - a^2}| + C \text{ in the above equation.}$$

$$\Rightarrow \int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx = \frac{x^2}{2} - \left[\frac{x}{2} \sqrt{x^2 - 1} - \frac{1}{2} \log |x + \sqrt{x^2 - 1}| \right] + C$$

$$\Rightarrow \int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx = \frac{1}{2} \left[x^2 - x \sqrt{x^2 - 1} + \log |x + \sqrt{x^2 - 1}| \right] + C$$

Hence the value of $\int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx$ is

$$\frac{1}{2} \left[x^2 - x\sqrt{x^2 - 1} + \log|x + \sqrt{x^2 - 1}| \right] + C.$$

$$(iii) \int \frac{1}{\sin(x-a)\sin(x-b)} dx$$

Ans: Write the given expression as

$$\int \frac{1}{\sin(x-a)\sin(x-b)} dx = \frac{1}{\sin(a-b)} \int \frac{\sin[(x-b)-(x-a)]}{\sin(x-a)\sin(x-b)} dx$$

$$\Rightarrow \int \frac{1}{\sin(x-a)\sin(x-b)} dx =$$

$$\frac{1}{\sin(a-b)} \int \frac{\sin(x-b)\cos(x-a) - \sin(x-a)\cos(x-b)}{\sin(x-a)\sin(x-b)} dx$$

$$\Rightarrow \int \frac{1}{\sin(x-a)\sin(x-b)} dx = \frac{1}{\sin(a-b)} \int [\cot(x-a) - \cot(x-b)] dx$$

$$\Rightarrow \int \frac{1}{\sin(x-a)\sin(x-b)} dx = \frac{1}{\sin(a-b)} [\log|\sin(x-a)| - \log|\sin(x-b)|] + C$$

$$\Rightarrow \int \frac{1}{\sin(x-a)\sin(x-b)} dx = \frac{1}{\sin(a-b)} \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + C$$

Hence the value of $\int \frac{1}{\sin(x-a)\sin(x-b)} dx$ is $\frac{1}{\sin(a-b)} \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + C.$

$$(iv) \int \frac{\cos(x+a)}{\cos(x-a)} dx$$

Ans: Modify the above expression as

$$\int \frac{\cos(x+a)}{\cos(x-a)} dx = \int \frac{\cos(x-a+2a)}{\cos(x-a)} dx$$

$$\Rightarrow \int \frac{\cos(x+a)}{\cos(x-a)} dx = \int \frac{\cos(x-a)\cos 2a - \sin(x-a)\sin 2a}{\cos(x-a)} dx$$

$$\Rightarrow \int \frac{\cos(x+a)}{\cos(x-a)} dx = \int \cos 2a dx - \sin 2a \int \tan(x-a) dx$$

$$\Rightarrow \int \frac{\cos(x+a)}{\cos(x-a)} dx = x \cos 2a - \sin 2a \log |\sec(x-a)| + C$$

Hence the value of $\int \frac{\cos(x+a)}{\cos(x-a)} dx$ is $x \cos 2a - \sin 2a \log |\sec(x-a)| + C$.

(v) $\int \cos x \cos 2x \cos 3x dx$

Ans: Modify the given expression as

$$\int \cos x \cos 2x \cos 3x dx = \int \frac{1}{2} (2 \cos x \cdot \cos 3x) \cos 2x dx$$

Use the trigonometric formula $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$ in the above equation.

$$\int \cos x \cos 2x \cos 3x dx = \frac{1}{2} \int (\cos 4x + \cos 2x) \cos 2x dx$$

$$\Rightarrow \int \cos x \cos 2x \cos 3x dx = \frac{1}{2} \int (\cos 4x \cdot \cos 2x + \cos^2 2x) dx$$

$$\Rightarrow \int \cos x \cos 2x \cos 3x dx = \frac{1}{2} \int \frac{1}{2} (2 \cos 4x \cdot \cos 2x + 2 \cos^2 2x) dx$$

Use the trigonometric formula $\cos 2\theta = 2 \cos^2 \theta - 1$ in the above equation.

$$\int \cos x \cos 2x \cos 3x dx = \frac{1}{4} \int (\cos 6x + \cos 2x + \cos 4x + 1) dx$$

$$\Rightarrow \int \cos x \cos 2x \cos 3x dx = \frac{1}{4} \left[x + \frac{\sin 2x}{2} + \frac{\sin 4x}{4} + \frac{\sin 6x}{6} \right] + C$$

$$\Rightarrow \int \cos x \cos 2x \cos 3x dx = \frac{1}{48} [12x + 6 \sin 2x + 3 \sin 4x + 2 \sin 6x] + C$$

Hence the value of $\int \cos x \cos 2x \cos 3x dx$ is

$$\frac{1}{48} [12x + 6\sin 2x + 3\sin 4x + 2\sin 6x] + C.$$

(vi) $\int \cos^5 x dx$

Ans: Rewrite the given expression as

$$\int \cos^5 x dx = \int \cos^4 x \cos x dx$$

$$\Rightarrow \int \cos^5 x dx = \int (\cos^2 x)^2 \cos x dx$$

Use the trigonometric identity $\sin^2 x + \cos^2 x = 1$ in the above equation.

$$\int \cos^5 x dx = \int (1 - \sin^2 x)^2 \cos x dx$$

Assume the substitution $u = \sin x \Leftrightarrow du = \cos x dx$ in the above equation.

$$\int \cos^5 x dx = \int (1 - u^2)^2 du$$

$$\Rightarrow \int \cos^5 x dx = \int (1 - 2u^2 + u^4) du$$

$$\Rightarrow \int \cos^5 x dx = u - \frac{2}{3}u^3 + \frac{1}{5}u^5 + C$$

Resubstitute the value $u = \sin x$ in the above equation.

$$\Rightarrow \int \cos^5 x dx = \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C$$

Hence the value of $\int \cos^5 x dx$ is $\sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C$.

(vii) $\int \sin^2 x \cos^4 x dx$

Ans: Rewrite the given expression as

$$\int \sin^2 x \cos^4 x dx = \int \frac{1}{8} [(4\sin^2 x \cos^2 x)(2\cos^2 x)] dx$$

$$\Rightarrow \int \sin^2 x \cos^4 x dx = \frac{1}{8} \int [(2 \sin x \cos x)^2 (2 \cos^2 x)] dx$$

Use the trigonometric formulas $2 \sin A \cos A = \sin 2A$ and $\cos 2A = 2 \cos^2 A - 1$ in the above equation.

$$\int \sin^2 x \cos^4 x dx = \frac{1}{8} \int (\sin 2x)^2 (1 + \cos 2x) dx$$

$$\Rightarrow \int \sin^2 x \cos^4 x dx = \frac{1}{8} \int \sin^2 2x (1 + \cos 2x) dx$$

Rewrite the above equation as

$$\int \sin^2 x \cos^4 x dx = \frac{1}{16} \int (2 \sin^2 2x) (1 + \cos 2x) dx$$

Use the trigonometric formula $\cos 2x = 1 - 2 \sin^2 x$ in the above equation.

$$\int \sin^2 x \cos^4 x dx = \frac{1}{16} \int (1 - \cos 4x) (1 + \cos 2x) dx$$

$$\Rightarrow \int \sin^2 x \cos^4 x dx = \frac{1}{16} \int [1 - \cos 4x + \cos 2x - \cos 4x \cos 2x] dx$$

Use the trigonometric formula $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$ in the above equation.

$$\int \sin^2 x \cos^4 x dx = \frac{1}{16} \int \left[1 - \cos 4x + \cos 2x - \frac{1}{2} (\cos 6x + \cos 2x) \right] dx$$

$$\Rightarrow \int \sin^2 x \cos^4 x dx = \frac{1}{32} \int (2 + \cos 2x - 2 \cos 4x - \cos 6x) dx$$

$$\Rightarrow \int \sin^2 x \cos^4 x dx = \frac{1}{32} \left[2x + \frac{\sin 2x}{2} - \frac{2 \sin 4x}{4} - \frac{\sin 6x}{6} \right] + C$$

Hence the value of $\int \sin^2 x \cos^4 x dx$ is

$$\frac{1}{32} \left(2x + \frac{1}{2} \sin 2x - \frac{1}{2} \sin 4x - \frac{1}{6} \sin 6x \right) + C.$$

(viii) $\int \cot^3 x \csc^4 x dx$

Ans: Rewrite the given expression by using trigonometric identity

$$\csc^2 x - \cot^2 x = 1.$$

$$\int \cot^3 x \csc^4 x dx = \int -(-\cot^2 x (\cot^2 x + 1)) \csc^2 x dx$$

Assume the substitution $u = \cot x \Leftrightarrow dx = -\frac{1}{\csc^2 x} du$ in the above equation.

$$\int \cot^3 x \csc^4 x dx = -\int u^3 (u^2 + 1) du$$

$$\Rightarrow \int \cot^3 x \csc^4 x dx = -\int (u^5 + u^3) du$$

$$\Rightarrow \int \cot^3 x \csc^4 x dx = -\left[\frac{u^6}{6} + \frac{u^4}{4} \right] + C$$

Resubstitute the value $u = \cot x$ in the above equation.

$$\int \cot^3 x \csc^4 x dx = -\left[\frac{\cot^6 x}{6} + \frac{\cot^4 x}{4} \right] + C$$

Hence the value of $\int \cot^3 x \csc^4 x dx$ is $-\left[\frac{\cot^6 x}{6} + \frac{\cot^4 x}{4} \right] + C$.

(ix) $\int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx$

Ans: Assume the substitution

$u = a^2 \sin^2 x + b^2 \cos^2 x \Leftrightarrow dx = \frac{1}{2a^2 \cos x \sin x - 2b^2 \cos x \sin x} du$ in the above equation.

$$\int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx = \int \frac{1}{2(a^2 - b^2)\sqrt{u}} du$$

$$\Rightarrow \int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx = \frac{1}{2(a^2 - b^2)} \int \frac{1}{\sqrt{u}} du$$

$$\Rightarrow \int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx = \frac{1}{2(a^2 - b^2)} \times 2\sqrt{u} + C$$

Resubstitute the value $u = a^2 \sin^2 x + b^2 \cos^2 x$ in the above equation.

$$\int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx = \frac{1}{a^2 - b^2} \sqrt{a^2 \sin^2 x + b^2 \cos^2 x} + C$$

Hence the value of $\int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx$ is $\frac{1}{a^2 - b^2} \sqrt{a^2 \sin^2 x + b^2 \cos^2 x} + C$

(x) $\int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx$

Ans: Use the trigonometric formula $\cos(A+B) = \cos A \cos B - \sin A \sin B$ in the above expression.

$$\int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx = \int \frac{1}{\sqrt{\cos^3 x (\cos x \cos a - \sin x \sin a)}} dx$$

$$\Rightarrow \int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx = \int \frac{1}{\sqrt{\cos^4 x (\cos a - \tan x \sin a)}} dx$$

$$\Rightarrow \int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx = \int \frac{1}{\cos^2 x \sqrt{\cos a - \tan x \sin a}} dx$$

$$\Rightarrow \int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx = \int \frac{\sec^2 x}{\sqrt{\cos a - \tan x \sin a}} dx$$

Assume the substitution $u = \cos a - \sin a \tan x \Leftrightarrow du = -\sin a \sec^2 x dx$ since the terms $\sin a$, $\cos a$ are constants.

$$\int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx = -\int \frac{1}{\sin a \sqrt{u}} du$$

$$\Rightarrow \int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx = -2 \csc a \sqrt{u} + C$$

Resubstitute the value $u = \cos a - \sin a \tan x$ in the above equation.

$$\int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx = -2 \csc a \sqrt{\cos a - \sin a \tan x} + C$$

Hence the value of $\int \frac{1}{\sqrt{\cos^3 x \cos(x+a)}} dx$ is $-2 \csc a \sqrt{\cos a - \sin a \tan x} + C$.

(xi) $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx$

Ans: Modify the given expression as

$$\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx = \int \frac{(\sin^2 x)^3 + (\cos^2 x)^3}{\sin^2 x \cos^2 x} dx$$

Use the formula $a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$ in the above equation.

$$\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx = \int \frac{(\sin^2 x + \cos^2 x)(\sin^4 x + \cos^4 x - \sin^2 x \cos^2 x)}{\sin^2 x \cos^2 x} dx$$

$$\Rightarrow \int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx$$

$$= \int \frac{(\sin^2 x)^2 + (\cos^2 x)^2 + 2\sin^2 x \cos^2 x - 2\sin^2 x \cos^2 x - \sin^2 x \cos^2 x}{\sin^2 x \cos^2 x} dx$$

$$\Rightarrow \int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx = \int \frac{(\sin^2 x + \cos^2 x)^2 - 3\sin^2 x \cos^2 x}{\sin^2 x \cos^2 x} dx$$

$$\Rightarrow \int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx = \int \left(\frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} - 3 \right) dx$$

$$\Rightarrow \int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx = \int \sec^2 x dx + \int \csc^2 x dx - \int 3 dx + C$$

$$\Rightarrow \int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx = \tan x - \cot x - 3x + C$$

Hence the value of $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx$ is $\tan x - \cot x - 3x + C$.

(xii) $\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx$

Ans: Assume the substitution $\sin x - \cos x = t \Leftrightarrow (\sin x + \cos x)dx = dt$ and the value of t^2 will be

$$t^2 = (\sin x - \cos x)^2$$

$$\Rightarrow t^2 = \sin^2 x + \cos^2 x - 2\sin x \cos x$$

$$\Rightarrow \sin 2x = 1 - t^2$$

Substitute all the values in the given expression.

$$\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \int \frac{1}{\sqrt{1-t^2}} dt$$

$$\Rightarrow \int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \sin^{-1} t + C$$

Substitute the value $t = \sin x - \cos x$ in the above equation.

$$\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \sin^{-1}(\sin x - \cos x) + C$$

Hence the value of $\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx$ is $\sin^{-1}(\sin x - \cos x) + C$.

42. Evaluate

(i) $\int \frac{x}{x^4 + x^2 + 1} dx$

Ans: Assume the substitution $x^2 = t \Leftrightarrow 2x dx = dt$ in the above equation.

$$\int \frac{x}{x^4 + x^2 + 1} dx = \int \frac{1}{2(t^2 + t + 1)} dt$$

$$\Rightarrow \int \frac{x}{x^4 + x^2 + 1} dx = \frac{1}{2} \int \frac{1}{\left(t + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dt$$

$$\Rightarrow \int \frac{x}{x^4 + x^2 + 1} dx = \frac{1}{2} \times \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + C$$

$$\Rightarrow \int \frac{x}{x^4 + x^2 + 1} dx = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) + C$$

Substitute $t = x^2$ in the above equation.

$$\Rightarrow \int \frac{x}{x^4 + x^2 + 1} dx = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x^2 + 1}{\sqrt{3}} \right) + C$$

Hence the value of $\int \frac{x}{x^4 + x^2 + 1} dx$ is $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x^2 + 1}{\sqrt{3}} \right) + C$.

$$(ii) \int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx$$

Ans: Assume the substitution $t = \log x \Leftrightarrow dt = \frac{1}{x} dx$ in the above equation.

$$\int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx = \int \frac{dt}{6t^2 + 7t + 2}$$

$$\Rightarrow \int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx = \int \frac{dt}{6t^2 + 3t + 4t + 2}$$

$$\Rightarrow \int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx = \int \frac{1}{(3t+2)(2t+1)} dt$$

$$\Rightarrow \int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx = \int \frac{2(3t+2) - 3(2t+1)}{(3t+2)(2t+1)} dt$$

$$\Rightarrow \int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx = \int \frac{2dt}{2t+1} - \int \frac{3dt}{3t+2}$$

$$\Rightarrow \int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx = \log(2t+1) - \log(3t+2) + C$$

$$\Rightarrow \int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx = \log \left| \frac{2t+1}{3t+2} \right| + C$$

Substitute the assumed value $t = \log x$ in the above equation.

$$\Rightarrow \int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx = \log \left| \frac{2\log x + 1}{3\log x + 2} \right| + C$$

Hence the value of $\int \frac{1}{x[6(\log x)^2 + 7\log x + 2]} dx$ is $\log \left| \frac{2\log x + 1}{3\log x + 2} \right| + C$.

(iii) $\int \frac{dx}{1+x-x^2}$

Ans: Simplify the given expression as

$$\int \frac{dx}{1+x-x^2} = \int \frac{dx}{-x^2 + 2\left(\frac{1}{2}\right)x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + 1}$$

$$\Rightarrow \int \frac{dx}{1+x-x^2} = \int \frac{1}{-\left(x - \frac{1}{2}\right)^2 + \frac{5}{4}} dx$$

Assume the substitution $u = x - \frac{1}{2} \Leftrightarrow du = dx$ in the above equation.

$$\int \frac{dx}{1+x-x^2} = \int \frac{4}{-4u^2 + 5} du$$

Assume the substitution $u = \frac{\sqrt{5}}{3}v \Leftrightarrow du = dv$ in the above equation.

$$\int \frac{dx}{1+x-x^2} = 4 \int \frac{1}{2\sqrt{5}(-v^2 + 1)} dv$$

$$\Rightarrow \int \frac{dx}{1+x-x^2} = 4 \times \frac{1}{2\sqrt{5}} \int \frac{1}{-v^2+1} dv$$

$$\Rightarrow \int \frac{dx}{1+x-x^2} = \frac{2}{\sqrt{5}} \times \frac{1}{2} (\log|v+1| - \log|v-1|) + C$$

Resubstitute $v = \frac{2}{\sqrt{5}}u$ and $u = x - \frac{1}{2}$ in the above equation.

$$\int \frac{dx}{1+x-x^2} = \frac{1}{\sqrt{5}} \left[\log \left(\frac{2}{\sqrt{5}} \left(x - \frac{1}{2} \right) + 1 \right) - \log \left(\frac{2}{\sqrt{5}} \left(x - \frac{1}{2} \right) - 1 \right) \right] + C$$

$$\Rightarrow \int \frac{dx}{1+x-x^2} = \frac{1}{\sqrt{5}} \log \left| \frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x} \right| + C$$

Hence the value of $\int \frac{dx}{1+x-x^2}$ is $\frac{1}{\sqrt{5}} \log \left| \frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x} \right| + C$.

(iv) $\int \frac{1}{\sqrt{9+8x-x^2}} dx$

Ans: Simplify the above equation as

$$\int \frac{1}{\sqrt{9+8x-x^2}} dx = \int \frac{1}{\sqrt{9+2(4)x-x^2+4^2-4^2}} dx$$

$$\Rightarrow \int \frac{1}{\sqrt{9+8x-x^2}} dx = \int \frac{1}{\sqrt{25-(x-4)^2}} dx$$

$$\Rightarrow \int \frac{1}{\sqrt{9+8x-x^2}} dx = \int \frac{1}{\sqrt{5^2-(x-4)^2}} dx$$

Apply the integration formula $\int \frac{1}{\sqrt{a^2-x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + C$ in the above equation.

$$\int \frac{1}{\sqrt{9+8x-x^2}} dx = \sin^{-1} \left(\frac{x-4}{5} \right) + C$$

Hence the value of $\int \frac{1}{\sqrt{9+8x-x^2}} dx$ is $\sin^{-1}\left(\frac{x-4}{5}\right) + C$.

$$(v) \int \frac{1}{\sqrt{(x-a)(x-b)}} dx$$

Ans: The given integral can be written as

$$\begin{aligned} \int \frac{1}{\sqrt{(x-a)(x-b)}} dx &= \int \frac{dx}{\sqrt{x^2 - bx - ax + ab}} \\ &= \int \frac{dx}{\sqrt{x^2 - x(a+b) + ab}} \\ &= \int \frac{dx}{\sqrt{x^2 - 2x\left(\frac{a+b}{2}\right) + \left(\frac{a+b}{2}\right)^2 - \left(\frac{a+b}{2}\right)^2 + ab}} \\ &= \int \frac{dx}{\sqrt{\left(x - \frac{a+b}{2}\right)^2 - \left(\frac{a^2 + 2ab + b^2}{4} + ab\right)}} \\ &= \int \frac{dx}{\sqrt{\left(x - \frac{a+b}{2}\right)^2 + \frac{-a^2 - b^2 + 2ab}{4}}} \\ &= \int \frac{dx}{\sqrt{\left(x - \frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2}} \\ &= \log \left| \left(x - \frac{a+b}{2}\right) + \sqrt{\left(x - \frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2} \right| + C \end{aligned}$$

After simplifying we get,

$$\int \frac{1}{\sqrt{(x-a)(x-b)}} dx = \log \left| x - \frac{a+b}{2} + \sqrt{x^2 - x(a+b) + ab} \right| + C$$

Thus, we'll have

$$\int \frac{1}{\sqrt{(x-a)(x-b)}} dx = \log \left| x - \frac{a+b}{2} + \sqrt{(x-a)(x+b)} \right| + C, \text{ where } C \text{ is an integration constant.}$$

(vi) $\int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}}$

Ans: Multiply and divide the fraction with $\sin(x-\alpha)$.

$$\begin{aligned} \int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}} dx &= \int \frac{\sin(x-\alpha)}{\sqrt{\sin(x+\alpha)\sin(x-\alpha)}} dx \\ \Rightarrow \int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}} dx &= \int \frac{\sin x \cos \alpha - \cos x \sin \alpha}{\sqrt{\sin^2 x - \sin^2 \alpha}} dx \\ \Rightarrow \int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}} dx &= \cos \alpha \int \frac{\sin x dx}{\sqrt{(1-\cos^2 x) - (1-\cos^2 \alpha)}} - \sin \alpha \int \frac{\cos x}{\sqrt{\sin^2 x - \sin^2 \alpha}} dx \\ \Rightarrow \int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}} dx &= \cos \alpha \int \frac{\sin x dx}{\sqrt{\cos^2 \alpha - \cos^2 x}} - \sin \alpha \int \frac{\cos x}{\sqrt{\sin^2 x - \sin^2 \alpha}} dx \end{aligned}$$

Assume $u = \cos x \Leftrightarrow du = -\sin x dx$ and $v = \sin x \Leftrightarrow dv = \cos x dx$ in the above equation.

$$\begin{aligned} \Rightarrow \int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}} dx &= \cos \alpha \int \frac{-du}{\sqrt{\cos^2 \alpha - u^2}} - \sin \alpha \int \frac{dv}{\sqrt{v^2 - \sin^2 \alpha}} \\ \Rightarrow \int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}} dx &= -\cos \alpha \sin^{-1} \left(\frac{u}{\cos \alpha} \right) - \sin \alpha \log \left| v + \sqrt{v^2 - \sin^2 \alpha} \right| \end{aligned}$$

Substitute the values of u and v in the above equation.

$$\Rightarrow \int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}} dx = -\cos \alpha \sin^{-1}\left(\frac{\cos x}{\cos \alpha}\right) - \sin \alpha \log \left| \sin x + \sqrt{\sin^2 x - \sin^2 \alpha} \right| + C$$

Hence the value of $\int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}} dx$ is

$$-\cos \alpha \sin^{-1}\left(\frac{\cos x}{\cos \alpha}\right) - \sin \alpha \log \left| \sin x + \sqrt{\sin^2 x - \sin^2 \alpha} \right| + C.$$

(vii) $\int \frac{5x-2}{3x^2+2x+1} dx$

Ans: Write $5x-2$ as $\frac{5}{6}(6x+2) - \frac{11}{3}$ and split.

$$\int \frac{5x-2}{3x^2+2x+1} dx = \int \left(\frac{5(6x+2)}{6(3x^2+2x+1)} - \frac{11}{3(3x^2+2x+1)} \right) dx$$

Assume the substitution $u = 3x^2 + 2x + 1 \Leftrightarrow dx = \frac{1}{6x+2} du$ in the above equation.

$$\int \frac{5x-2}{3x^2+2x+1} dx = \frac{5}{6} \int \frac{1}{u} du - \frac{11}{3} \int \frac{1}{\left(\sqrt{3x} + \frac{1}{\sqrt{3}}\right)^2 + \left(\sqrt{\frac{2}{3}}\right)^2} dx$$

$$\Rightarrow \int \frac{5x-2}{3x^2+2x+1} dx = \frac{5}{6} \ln|u| - \frac{11}{3} \times \frac{1}{\left(\sqrt{\frac{2}{3}}\right)} \tan^{-1} \left(\frac{\sqrt{3x} + \frac{1}{\sqrt{3}}}{\sqrt{\frac{2}{3}}} \right) + C$$

$$\Rightarrow \int \frac{5x-2}{3x^2+2x+1} dx = \frac{5}{6} \log|3x^2+2x+1| - \frac{11}{3\sqrt{2}} \tan^{-1} \left(\frac{3x+1}{\sqrt{2}} \right) + C$$

Hence the value of $\int \frac{5x-2}{3x^2+2x+1} dx$ is

$$\frac{5}{6} \log|3x^2+2x+1| - \frac{11}{3\sqrt{2}} \tan^{-1} \left(\frac{3x+1}{\sqrt{2}} \right) + C.$$

(viii) $\int \frac{x^2}{x^2 + 6x + 12} dx$

Ans: Perform long division method and simplify the above fraction as

$$\begin{aligned}\int \frac{x^2}{x^2 + 6x + 12} dx &= \int \left(1 + \frac{-6x - 12}{x^2 + 6x + 12} \right) dx \\ \Rightarrow \int \frac{x^2}{x^2 + 6x + 12} dx &= \int du + \int \frac{-6x - 12}{x^2 + 2(3)x + 3^2 - 3^2 + 12} dx \\ \Rightarrow \int \frac{x^2}{x^2 + 6x + 12} dx &= x + \int \frac{-6x - 12}{(x + 3)^2 + 2} dx\end{aligned}$$

Assume the substitution $u = x + 3 \Leftrightarrow du = dx$ in the above equation.

$$\begin{aligned}\int \frac{x^2}{x^2 + 6x + 12} dx &= x + \int \frac{-6u + 6}{u^2 + 3} du \\ \Rightarrow \int \frac{x^2}{x^2 + 6x + 12} dx &= x - 3 \int \frac{2u}{u^2 + 3} + 6 \int \frac{1}{u^2 + (\sqrt{3})^2} du \\ \Rightarrow \int \frac{x^2}{x^2 + 6x + 12} dx &= x - \log|u^2 + 3| + 6 \times \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{u}{\sqrt{3}} \right) + C \\ \Rightarrow \int \frac{x^2}{x^2 + 6x + 12} dx &= x - 3 \log|x^2 + 6x + 12| + 2\sqrt{3} \tan^{-1} \left(\frac{x + 3}{\sqrt{3}} \right) + C\end{aligned}$$

Hence the value of $\int \frac{x^2}{x^2 + 6x + 12} dx$ is

$$x - 3 \log|x^2 + 6x + 12| + 2\sqrt{3} \tan^{-1} \left(\frac{x + 3}{\sqrt{3}} \right) + C.$$

(ix) $\int \frac{x + 2}{\sqrt{4x - x^2}} dx$

Ans: Simplify the given expression as

$$\int \frac{x+2}{\sqrt{4x-x^2}} dx = \int \frac{x+2}{\sqrt{-x^2+2(2)x+2^2-2^2}} dx$$

$$\Rightarrow \int \frac{x+2}{\sqrt{4x-x^2}} dx = \int \frac{x+2}{\sqrt{-(x-2)^2+4}} dx$$

Assume the substitution $u = x - 2 \Leftrightarrow du = dx$ in the above equation.

$$\int \frac{x+2}{\sqrt{4x-x^2}} dx = \int \frac{u+4}{\sqrt{-u^2+4}} du$$

$$\Rightarrow \int \frac{x+2}{\sqrt{4x-x^2}} dx = \int \frac{u}{\sqrt{-u^2+4}} du + 4 \int \frac{1}{\sqrt{-u^2+2^2}} du$$

$$\Rightarrow \int \frac{x+2}{\sqrt{4x-x^2}} dx = -\sqrt{-u^2+4} + 4 \sin^{-1}\left(\frac{u}{2}\right)$$

Substitute the value $u = x - 2$ in the above equation.

$$\int \frac{x+2}{\sqrt{4x-x^2}} dx = -\sqrt{4x-x^2} + 4 \sin^{-1}\left(\frac{x-2}{2}\right) + C$$

Hence the value of $\int \frac{x+2}{\sqrt{4x-x^2}} dx$ is $-\sqrt{4x-x^2} + 4 \sin^{-1}\left(\frac{x-2}{2}\right) + C$.

(x) $\int x\sqrt{1+x-x^2} dx$

Ans: Write the term x as $\frac{1}{2} - \frac{1}{2}(1-2x)$ in the above expression.

$$\int x\sqrt{1+x-x^2} dx = \int \left(\frac{\sqrt{-x^2+x+1}}{2} - \frac{(1-2x)\sqrt{-x^2+x+1}}{2} \right) dx$$

$$\Rightarrow \int x\sqrt{1+x-x^2} dx = \frac{1}{2} \int (2x-1)\sqrt{-x^2+x+1} dx + \frac{1}{2} \int \sqrt{-x^2+x+1} dx$$

Assume the substitution $u = -x^2 + x + 1 \Leftrightarrow dx = \frac{1}{1-2x} du$ in the above equation.

$$\int x\sqrt{1+x-x^2}dx = \frac{1}{2}\int -\sqrt{u}du + \frac{1}{2}\int \sqrt{-x^2 + 2\left(\frac{1}{2}\right)x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} + 1dx$$

$$\Rightarrow \int x\sqrt{1+x-x^2}dx = -\frac{1}{2} \times \frac{2u^{\frac{3}{2}}}{\frac{3}{2}} + \frac{1}{2}\sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2} dx$$

Apply the integration formula $\int \sqrt{a^2 - x^2}dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$ in the above equation.

$$\begin{aligned} \int x\sqrt{1+x-x^2}dx &= -\frac{1}{3}(-x^2+x+1)^{\frac{3}{2}} + \frac{1}{2} \times \frac{\left(x - \frac{1}{2}\right)}{2} \sqrt{-x^2+x+1} + \frac{1}{2} \\ &\times \frac{5}{2} \sin^{-1}\left(\frac{x - \frac{1}{2}}{\frac{\sqrt{5}}{2}}\right) + C \end{aligned}$$

$$\Rightarrow \int x\sqrt{1+x-x^2}dx = -\frac{1}{3}(1+x-x^2)^{\frac{3}{2}} + \frac{1}{8}(2x-1)\sqrt{1+x-x^2} + \frac{5}{16}\sin^{-1}\left(\frac{2x-1}{\sqrt{5}}\right) + C$$

Hence the value of $\int x\sqrt{1+x-x^2}dx$ is

$$-\frac{1}{3}(1+x-x^2)^{\frac{3}{2}} + \frac{1}{8}(2x-1)\sqrt{1+x-x^2} + \frac{5}{16}\sin^{-1}\left(\frac{2x-1}{\sqrt{5}}\right) + C.$$

(xi) $\int (3x-2)\sqrt{x^2+x+1}dx$

Ans: Write the term $3x-2$ as $\frac{3}{2}(2x+1) - \frac{7}{2}$ in the above expression.

$$\int (3x-2)\sqrt{x^2+x+1}dx = \int \left(\frac{3(2x+1)\sqrt{x^2+x+1}}{2} - \frac{7\sqrt{x^2+x+1}}{2} \right) dx$$

$$\Rightarrow \int (3x-2)\sqrt{x^2+x+1}dx = \frac{3}{2} \int (2x+1)\sqrt{x^2+x+1}dx - \frac{7}{2} \int \sqrt{x^2+x+1}dx$$

Assume the substitution $u = x^2 + x + 1 \Leftrightarrow dx = \frac{1}{2x+1} du$ in the above equation.

$$\int (3x-2)\sqrt{x^2+x+1}dx = \frac{3}{2} \times \int \sqrt{u}du - \frac{7}{2} \int \sqrt{x^2 + 2\left(\frac{1}{2}\right)x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + 1}dx$$

$$\Rightarrow \int (3x-2)\sqrt{x^2+x+1}dx = \frac{3}{2} \times \frac{2}{3} u^{\frac{3}{2}} - \frac{7}{2} \int \sqrt{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx$$

Apply the integration formula

$$\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log \left| x + \sqrt{x^2 + a^2} \right| + C \text{ in the above equation.}$$

$$\int (3x-2)\sqrt{x^2+x+1}dx = (x^2+x+1)^{\frac{3}{2}} - \frac{7}{2} \left[\frac{x + \frac{1}{2}}{2} \sqrt{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} + \frac{3}{2} \log \left| x + \frac{1}{2} + \sqrt{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \right| \right] + C$$

$$\Rightarrow \int (3x-2)\sqrt{x^2+x+1}dx = (x^2+x+1)^{\frac{3}{2}} - \frac{7}{2} \left[\left(x + \frac{1}{2}\right) \sqrt{x^2+x+1} + \frac{3}{8} \log \left| x + \frac{1}{2} + \sqrt{x^2+x+1} \right| \right] + C$$

Hence the value of $\int (3x-2)\sqrt{x^2+x+1}dx$ is

$$(x^2+x+1)^{\frac{3}{2}} - \frac{7}{2} \left[\left(x + \frac{1}{2}\right) \sqrt{x^2+x+1} + \frac{3}{8} \log \left| x + \frac{1}{2} + \sqrt{x^2+x+1} \right| \right] + C.$$

(xii) $\int \sqrt{\sec x + 1} dx$

Ans: We know that,

$$\sec x = \frac{1}{\cos x}$$

So, the given integral can be written as

$$\begin{aligned}\int \sqrt{\sec x + 1} \, dx &= \int \sqrt{\frac{1}{\cos x} + 1} \, dx \\ &= \int \sqrt{\frac{1 + \cos x}{\cos x}} \, dx\end{aligned}$$

Since, we have the half angle formulas as,

$$1 + \cos 2x = 2 \cos^2 \frac{x}{2}$$

$$\cos 2x = 1 - 2 \sin^2 \frac{x}{2}$$

Using these formulas in the integral we get,

$$\begin{aligned}&= \int \sqrt{\frac{2 \cos^2 \frac{x}{2}}{1 - 2 \sin^2 \frac{x}{2}}} \, dx \\ &= \sqrt{2} \int \frac{\cos \frac{x}{2}}{\sqrt{1 - 2 \sin^2 \frac{x}{2}}} \, dx\end{aligned}$$

Taking $\sqrt{2}$ out from the denominator we get,

$$= \frac{\sqrt{2}}{\sqrt{2}} \int \frac{\cos \frac{x}{2}}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 - \left(\sin \frac{x}{2}\right)^2}} \, dx$$

Now, let us assume,

$$\sin \frac{x}{2} = k$$

Now, differentiating on both sides we have,

$$\frac{1}{2} \cos \frac{x}{2} \, dx = dk$$

$$\cos \frac{x}{2} \, dx = 2dk$$

Now the integral becomes,

$$= \int \frac{2dk}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 - (k)^2}}$$

We know the formula,

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + c$$

So, we get the integral value as,

$$= 2 \sin^{-1}\left(\frac{k}{\left(\frac{1}{\sqrt{2}}\right)}\right) + c$$

$$= 2 \sin^{-1}\left(\frac{\sin \frac{x}{2}}{\frac{1}{\sqrt{2}}}\right) + c$$

Hence, $\int \sqrt{\sec x + 1} \, dx = 2 \sin^{-1}\left(\sqrt{2} \sin \frac{x}{2}\right) + c$, where c is an integration constant.

43. Evaluate:

(i) $\int \frac{dx}{x(x^7 + 1)}$

Ans: Assume the substitution $u = x^7 \Leftrightarrow du = 7x^6 dx$ in the above equation.

$$\int \frac{dx}{x(x^7 + 1)} = \int \frac{1}{x(x^7 + 1)} \times \frac{1}{7x^6} du$$

$$\Rightarrow \int \frac{dx}{x(x^7 + 1)} = \frac{1}{7} \int \frac{1}{u(u + 1)} du$$

Perform the partial fraction decomposition in the above equation.

$$\int \frac{dx}{x(x^7+1)} = \frac{1}{7} \int \left(\frac{1}{u} - \frac{1}{u+1} \right) du$$

$$\Rightarrow \int \frac{dx}{x(x^7+1)} = \frac{1}{7} [\log|u| - \log|u+1|] + C$$

$$\Rightarrow \int \frac{dx}{x(x^7+1)} = \frac{1}{7} \log \left| \frac{x^7}{x^7+1} \right| + C$$

Hence the value of $\int \frac{dx}{x(x^7+1)}$ is $\frac{1}{7} \log \left| \frac{x^7}{x^7+1} \right| + C$.

(ii) $\int \frac{\sin x}{(1+\cos x)(2+3\cos x)} dx$

Ans: Assume the substitution $\cos x = t \Leftrightarrow \sin x dx = -dt$ in the above equation.

$$\int \frac{\sin x}{(1+\cos x)(2+3\cos x)} dx = \int \frac{-dt}{(1+t)(2+3t)}$$

On performing the partial fraction decomposition, the above fraction is modified as

$$\int \frac{\sin x}{(1+\cos x)(2+3\cos x)} dx = \int \left(\frac{1}{1+t} - \frac{3}{2+3t} \right) dt$$

$$\Rightarrow \int \frac{\sin x}{(1+\cos x)(2+3\cos x)} dx = \log|1+t| - \log|2+3t| + C$$

$$\Rightarrow \int \frac{\sin x}{(1+\cos x)(2+3\cos x)} dx = \log \left| \frac{1+t}{2+3t} \right| + C$$

$$\Rightarrow \int \frac{\sin x}{(1+\cos x)(2+3\cos x)} dx = \log \left| \frac{1+\cos x}{2+3\cos x} \right| + C$$

Hence the value of $\int \frac{\sin x}{(1+\cos x)(2+3\cos x)} dx$ is $\log \left| \frac{1+\cos x}{2+3\cos x} \right| + C$.

$$(iii) \int \frac{\sin \theta \cos \theta}{\cos^2 \theta - \cos \theta - 2} d\theta$$

Ans: Assume the substitution $u = \cos \theta \Leftrightarrow du = -\sin \theta d\theta$ in the above equation.

$$\int \frac{\sin \theta \cos \theta}{\cos^2 \theta - \cos \theta - 2} d\theta = \int -\frac{u}{u^2 - u - 2} du$$

Perform partial fraction decomposition in the above equation.

$$\begin{aligned} \int \frac{\sin \theta \cos \theta}{\cos^2 \theta - \cos \theta - 2} d\theta &= -\int \left(\frac{1}{3(u+1)} + \frac{2}{3(u-2)} \right) dx \\ \Rightarrow \int \frac{\sin \theta \cos \theta}{\cos^2 \theta - \cos \theta - 2} d\theta &= -\frac{1}{3} \int \frac{1}{u+1} du - \frac{2}{3} \int \frac{1}{u-2} du \\ \Rightarrow \int \frac{\sin \theta \cos \theta}{\cos^2 \theta - \cos \theta - 2} d\theta &= -\frac{1}{3} \log|u+1| - \frac{2}{3} \log|u-2| + C \end{aligned}$$

Substitute the value $u = \cos \theta$ in the above equation.

$$\int \frac{\sin \theta \cos \theta}{\cos^2 \theta - \cos \theta - 2} d\theta = -\frac{2}{3} \log|\cos \theta - 2| - \frac{1}{3} \log|1 + \cos \theta| + C.$$

Hence the value of $\int \frac{\sin \theta \cos \theta}{\cos^2 \theta - \cos \theta - 2} d\theta$ is $-\frac{2}{3} \log|\cos \theta - 2| - \frac{1}{3} \log|1 + \cos \theta| + C$

$$(iv) \int \frac{x-1}{(x+1)(x-2)(x+3)} dx$$

Ans: Let us represent the above expression in the partial fractions as,

$$\frac{x-1}{(x+1)(x-2)(x+3)} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{x+3}$$

Taking the LCM in above equation and representing as the coefficients of x we get,

$$x-1 = x^2(A+B+C) + x(A+4B-C) + (-6A+3B-2C)$$

Comparing the coefficients on both sides of equation we get,

$$A + B + C = 0$$

$$A + 4B - C = 1$$

$$-6A + 3B - 2C = -1$$

On solving the above three equations we get,

$$A = \frac{1}{3}$$

$$B = \frac{1}{15}$$

$$C = \frac{-2}{5}$$

So, the integral becomes,

$$\begin{aligned} \int \frac{x-1}{(x+1)(x-2)(x+3)} dx &= \int \left(\frac{1}{3(x+1)} + \frac{1}{15(x-2)} - \frac{2}{5(x+3)} \right) dx \\ \Rightarrow \int \frac{x-1}{(x+1)(x-2)(x+3)} dx &= \frac{1}{3} \int \frac{1}{x+1} dx + \frac{1}{15} \int \frac{1}{x-2} dx - \frac{2}{5} \int \frac{1}{x+3} dx \\ \Rightarrow \int \frac{x-1}{(x+1)(x-2)(x+3)} dx &= \frac{1}{3} \log|x+1| + \frac{1}{15} \log|x-2| - \frac{2}{5} \log|x+3| + C \end{aligned}$$

Hence the value of the integral $\int \frac{x-1}{(x+1)(x-2)(x+3)} dx$ is

$$\frac{1}{3} \log|x+1| + \frac{1}{15} \log|x-2| - \frac{2}{5} \log|x+3| + C.$$

$$(v) \int \frac{x^2 + x + 2}{(x-2)(x-1)} dx$$

Ans: Simplify the given expression by performing long division.

$$\begin{aligned} \int \frac{x^2 + x + 2}{(x-2)(x-1)} dx &= \int \left(\frac{4x}{(x-2)(x-1)} + 1 \right) dx \\ \Rightarrow \int \frac{x^2 + x + 2}{(x-2)(x-1)} dx &= 4 \int \frac{x}{(x-2)(x-1)} dx + \int 1 dx \end{aligned}$$

Perform partial fraction decomposition in the above equation.

$$\begin{aligned}\int \frac{x^2 + x + 2}{(x-2)(x-1)} dx &= 4 \int \left(\frac{2}{x-2} - \frac{1}{x-1} \right) dx + x \\ \Rightarrow \int \frac{x^2 + x + 2}{(x-2)(x-1)} dx &= 4 \left[2 \int \frac{1}{x-2} dx - \int \frac{1}{x-1} dx \right] + x \\ \Rightarrow \int \frac{x^2 + x + 2}{(x-2)(x-1)} dx &= 8 \log|x-2| - 4 \log|x-1| + x + C \\ \Rightarrow \int \frac{x^2 + x + 2}{(x-2)(x-1)} dx &= x + 4 \log \left| \frac{(x-2)^2}{x-1} \right| + C\end{aligned}$$

Hence the value of $\int \frac{x^2 + x + 2}{(x-2)(x-1)} dx$ is $x + 4 \log \left| \frac{(x-2)^2}{x-1} \right| + C$.

(vi) $\int \frac{(x^2 + 1)(x^2 + 2)}{(x^2 + 3)(x^2 + 4)} dx$

Ans: Assume the substitution $t = x^2$ in the given fraction.

$$\begin{aligned}\frac{(x^2 + 1)(x^2 + 2)}{(x^2 + 3)(x^2 + 4)} &= \frac{(t + 1)(t + 2)}{(t + 3)(t + 4)} \\ \Rightarrow \frac{(x^2 + 1)(x^2 + 2)}{(x^2 + 3)(x^2 + 4)} &= \frac{t^2 + 3t + 2}{t^2 + 7t + 12}\end{aligned}$$

Perform long division on the above fraction, then we will get

$$\frac{(x^2 + 1)(x^2 + 2)}{(x^2 + 3)(x^2 + 4)} = 1 + \frac{-4t - 10}{t^2 + 7t + 12}$$

Perform the partial fraction decomposition to the above fraction, then we will have

$$\frac{(x^2 + 1)(x^2 + 2)}{(x^2 + 3)(x^2 + 4)} = 1 - \frac{2}{x^2 + 3} + \frac{6}{x^2 + 4}$$

Apply integration on both sides of the above equation.

$$\int \frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)} dx = \int \left(1 - \frac{2}{x^2+3} + \frac{6}{x^2+4} \right) dx$$

$$\Rightarrow \int \frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)} dx = \int 1 dx + 2 \int \frac{1}{x^2+(\sqrt{3})^2} dx - 6 \int \frac{1}{(x^2+2^2)} dx$$

$$\Rightarrow \int \frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)} dx = x + \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3}} \right) - 3 \tan^{-1} \left(\frac{x}{2} \right) + C$$

Hence the value of $\int \frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)} dx$ is $x + \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3}} \right) - 3 \tan^{-1} \left(\frac{x}{2} \right) + C$.

(vii) $\int \frac{dx}{(2x+1)(x^2+4)}$

Ans: On performing partial fraction decomposition to the given expression we will have

$$\int \frac{dx}{(2x+1)(x^2+4)} = \int \left(\frac{4}{17(2x+1)} - \frac{2x-1}{17(x^2+4)} \right) dx$$

$$\Rightarrow \int \frac{dx}{(2x+1)(x^2+4)} = \frac{4}{17} \int \frac{1}{2x+1} dx - \frac{1}{17} \int \frac{2x-1}{x^2+4} dx$$

$$\Rightarrow \int \frac{dx}{(2x+1)(x^2+4)} = \frac{4}{17} \times \frac{1}{2} \log|2x+1| - \frac{1}{17} \left[\int \frac{2x}{x^2+4} dx - \int \frac{1}{x^2+2^2} dx \right]$$

$$\Rightarrow \int \frac{dx}{(2x+1)(x^2+4)} = \frac{2}{17} \log|2x+1| - \frac{1}{17} \left[\log|x^2+4| - \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) \right] + C$$

$$\Rightarrow \int \frac{dx}{(2x+1)(x^2+4)} = \frac{2}{17} \log|2x+1| - \frac{1}{17} \log|x^2+4| + \frac{1}{34} \tan^{-1} \left(\frac{x}{2} \right) + C$$

Hence the value of $\int \frac{dx}{(2x+1)(x^2+4)}$ is

$$\frac{2}{17} \log|2x+1| - \frac{1}{17} \log|x^2+4| + \frac{1}{34} \tan^{-1}\left(\frac{x}{2}\right) + C.$$

(viii) $\int \frac{dx}{\sin x(1-2\cos x)}$

Ans: Simplify the given fraction as

$$\int \frac{dx}{\sin x(1-2\cos x)} = -\int \frac{dx}{(2\cos x-1)\sin x} \times \frac{\sin x}{\sin x}$$

Use the trigonometric identity $\sin^2 x + \cos^2 x = 1$ in the above equation.

$$\int \frac{dx}{\sin x(1-2\cos x)} = -\int \frac{\sin x dx}{(2\cos x-1)(\cos^2 x-1)}$$

Assume the substitution $u = \cos x \Leftrightarrow du = -\sin x dx$ in the above equation.

$$\int \frac{dx}{\sin x(1-2\cos x)} = \int \frac{1}{(2u-1)(u^2-1^2)} du$$

$$\Rightarrow \int \frac{dx}{\sin x(1-2\cos x)} = \int \frac{1}{(2u-1)(u+1)(u-1)} du$$

Perform partial fraction decomposition in the above equation.

$$\int \frac{dx}{\sin x(1-2\cos x)} = \int \left(-\frac{4}{3(2u-1)} + \frac{1}{6(u+1)} + \frac{1}{2(u-1)} \right) du$$

$$\Rightarrow \int \frac{dx}{\sin x(1-2\cos x)} = -\frac{4}{3} \times \frac{1}{2} \log|2u-1| + \frac{1}{6} \log|u+1| + \frac{1}{2} \log|u-1| + C$$

$$\Rightarrow \int \frac{dx}{\sin x(1-2\cos x)} = -\frac{1}{2} \log|1-\cos x| - \frac{1}{6} \log|1+\cos x| + \frac{2}{3} \log|1-2\cos x| + C$$

Hence the value of $\int \frac{dx}{\sin x(1-2\cos x)}$ is

$$-\frac{1}{2} \log|1-\cos x| - \frac{1}{6} \log|1+\cos x| + \frac{2}{3} \log|1-2\cos x| + C.$$

$$(ix) \int \frac{\sin x}{\sin 4x} dx$$

Ans: Use the trigonometric formulas $\sin 2A = 2\sin A \cos A$ in the above equation.

$$\begin{aligned} \int \frac{\sin x}{\sin 4x} dx &= \int \frac{\sin x}{2\sin 2x \cos 2x} dx \\ \Rightarrow \int \frac{\sin x}{\sin 4x} dx &= \int \frac{\sin x}{2(2\sin x \cos x) \cos 2x} \times \frac{\cos x}{\cos x} dx \\ \Rightarrow \int \frac{\sin x}{\sin 4x} dx &= \frac{1}{4} \int \frac{\cos x}{\cos^2 x \cos 2x} dx \end{aligned}$$

Use the formula $\sin^2 x + \cos^2 x = 1$ and $\cos 2A = 1 - 2\sin^2 A$ in the above equation.

$$\int \frac{\sin x}{\sin 4x} dx = \frac{1}{4} \int \frac{\cos x dx}{(1 - \sin^2 x)(1 - 2\sin^2 x)}$$

Assume the substitution $\sin x = t \Leftrightarrow \cos x dx = dt$ in the above equation.

$$\int \frac{\sin x}{\sin 4x} dx = \frac{1}{4} \int \frac{1}{(1 - t^2)(1 - 2t^2)} dt$$

Perform partial fraction decomposition.

$$\begin{aligned} \int \frac{\sin x}{\sin 4x} dx &= \frac{1}{4} \int \left[-\frac{1}{1 - t^2} + \frac{2}{1 - 2t^2} \right] dt \\ \Rightarrow \int \frac{\sin x}{\sin 4x} dx &= -\frac{1}{4} \int \frac{1}{1 - t^2} dt + \frac{2}{4} \int \frac{1}{1 - (\sqrt{2}t)^2} dt \\ \Rightarrow \int \frac{\sin x}{\sin 4x} dx &= -\frac{1}{4} \times \frac{1}{2} \log \left| \frac{1+t}{1-t} \right| + \frac{1}{2} \times \frac{1}{2\sqrt{2}} \log \left| \frac{1+\sqrt{2}t}{1-\sqrt{2}t} \right| + C \\ \Rightarrow \int \frac{\sin x}{\sin 4x} dx &= -\frac{1}{8} \log \left| \frac{1+\sin x}{1-\sin x} \right| + \frac{1}{4\sqrt{2}} \log \left| \frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x} \right| + C \end{aligned}$$

Hence the value of $\int \frac{\sin x}{\sin 4x} dx$ is $-\frac{1}{8} \log \left| \frac{1+\sin x}{1-\sin x} \right| + \frac{1}{4\sqrt{2}} \log \left| \frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x} \right| + C$.

$$(x) \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx$$

Ans: Simplify the above expression as

$$\begin{aligned} \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx &= \int \frac{1 - \frac{1}{x^2}}{x^2 + 1 + \frac{1}{x^2}} dx \\ \Rightarrow \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx &= \int \frac{\left(1 - \frac{1}{x^2}\right)}{\left(x + \frac{1}{x}\right)^2 - 1} dx \end{aligned}$$

Assume the substitution $t = x + \frac{1}{x} \Leftrightarrow dt = \left(1 - \frac{1}{x^2}\right) dx$ in the above equation.

$$\begin{aligned} \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx &= \int \frac{dt}{t^2 - 1} \\ \Rightarrow \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx &= \frac{1}{2} \int \frac{2dt}{(t+1)(t-1)} \end{aligned}$$

Perform partial fraction decomposition in the above equation.

$$\begin{aligned} \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx &= \frac{1}{2} \int \frac{dt}{t-1} - \frac{1}{2} \int \frac{dt}{t+1} \\ \Rightarrow \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx &= \frac{1}{2} \log|t-1| - \frac{1}{2} \log|t+1| + C \\ \Rightarrow \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx &= \frac{1}{2} \log \left| \frac{t-1}{t+1} \right| + C \\ \Rightarrow \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx &= \frac{1}{2} \log \left| \frac{x^2 - x + 1}{x^2 + x + 1} \right| + C \end{aligned}$$

Hence the value of $\int \frac{x^2 - 1}{x^4 + x^2 + 1} dx$ is $\frac{1}{2} \log \left| \frac{x^2 - x + 1}{x^2 + x + 1} \right| + C$.

(xi) $\int \sqrt{\tan x} dx$

Ans: Assume the substitution $\tan x = t^2 \Leftrightarrow dx = \frac{2tdt}{\sec^2 x} = \frac{2tdt}{1+t^4}$ in the given expression.

$$\int \sqrt{\tan x} dx = \int \left(t \times \frac{2t}{1+t^4} \right) dt$$

$$\Rightarrow \int \sqrt{\tan x} dx = \int \frac{2t^2}{1+t^4} dt$$

$$\Rightarrow \int \sqrt{\tan x} dx = \int \frac{2}{t^2 + \frac{1}{t^2}} dt$$

$$\Rightarrow \int \sqrt{\tan x} dx = \int \frac{\left(1 + \frac{1}{t^2}\right) + \left(1 - \frac{1}{t^2}\right)}{t^2 + \frac{1}{t^2}} dt$$

$$\Rightarrow \int \sqrt{\tan x} dx = \int \frac{1 + \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} dt + \int \frac{1 - \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} dt$$

Assume the substitutions $t - \frac{1}{t} = u \Leftrightarrow \left(1 + \frac{1}{t^2}\right) dt = du$ and

$t + \frac{1}{t} = v \Leftrightarrow \left(1 - \frac{1}{t^2}\right) dt = dv$ in the above equation.

$$\int \sqrt{\tan x} dx = \int \frac{1}{\left(t - \frac{1}{t}\right)^2 + 2} du + \int \frac{1}{\left(t + \frac{1}{t}\right)^2 - 2} dv$$

$$\Rightarrow \int \sqrt{\tan x} dx = \int \frac{1}{u^2 + (\sqrt{2})^2} du + \int \frac{1}{v^2 - (\sqrt{2})^2} dv$$

$$\Rightarrow \int \sqrt{\tan x} dx = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{u}{\sqrt{2}} \right) + \frac{1}{2\sqrt{2}} \log \left| \frac{v - \sqrt{2}}{v + \sqrt{2}} \right| + C$$

Substitute the values of u, v and t in the above equation.

$$\int \sqrt{\tan x} dx = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t - \frac{1}{t}}{\sqrt{2}} \right) + \frac{1}{2\sqrt{2}} \log \left| \frac{t + \frac{1}{t} - \sqrt{2}}{t + \frac{1}{t} + \sqrt{2}} \right| + C$$

$$\Rightarrow \int \sqrt{\tan x} dx = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t^2 - 1}{\sqrt{2}t} \right) + \frac{1}{2\sqrt{2}} \log \left| \frac{t^2 - \sqrt{2}t + 1}{t^2 + \sqrt{2}t + 1} \right|$$

$$\Rightarrow \int \sqrt{\tan x} dx = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\tan x - 1}{\sqrt{2 \tan x}} \right) + \frac{1}{2\sqrt{2}} \log \left| \frac{\tan x - \sqrt{2 \tan x} + 1}{\tan x + \sqrt{2 \tan x} + 1} \right| + C$$

Hence the value of $\int \sqrt{\tan x} dx$ is

$$\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\tan x - 1}{\sqrt{2 \tan x}} \right) + \frac{1}{2\sqrt{2}} \log \left| \frac{\tan x - \sqrt{2 \tan x} + 1}{\tan x + \sqrt{2 \tan x} + 1} \right| + C.$$

(xii) $\int \frac{x^2 + 9}{x^4 + 81} dx$

Ans: Simplify the given expression by taking x^2 as common from the both numerator and denominator.

$$\int \frac{x^2 + 9}{x^4 + 81} = \int \frac{1 + \frac{9}{x^2}}{x^2 + \frac{81}{x^2}} dx$$

$$\Rightarrow \int \frac{x^2 + 9}{x^4 + 81} = \int \frac{1 + \frac{9}{x^2}}{x^2 + \left(\frac{9}{x}\right)^2 + 2(x)\left(\frac{9}{x}\right) - 2(x)\left(\frac{9}{x}\right)} dx$$

$$\Rightarrow \int \frac{x^2 + 9}{x^4 + 81} = \int \frac{1 + \frac{9}{x^2}}{\left(x - \frac{9}{x}\right)^2 + 18} dx$$

Assume the substitution $x - \frac{9}{x} = t \Leftrightarrow \left(1 + \frac{9}{x^2}\right) dx = dt$ in the above equation.

$$\Rightarrow \int \frac{x^2 + 9}{x^4 + 81} = \int \frac{dt}{t^2 + (3\sqrt{2})^2}$$

$$\Rightarrow \int \frac{x^2 + 9}{x^4 + 81} = \frac{1}{3\sqrt{2}} \tan^{-1} \left(\frac{t}{3\sqrt{2}} \right) + C$$

Substitute the value $t = x - \frac{9}{x}$ in the above equation.

$$\Rightarrow \int \frac{x^2 + 9}{x^4 + 81} = \frac{1}{3\sqrt{2}} \tan^{-1} \left(\frac{x - \frac{9}{x}}{3\sqrt{2}} \right) + C$$

$$\Rightarrow \int \frac{x^2 + 9}{x^4 + 81} = \frac{1}{3\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 9}{3\sqrt{2}} \right) + C$$

44.

(i) $\int x^5 \sin x^3 dx$

Ans: Simplify the above expression as

$$\int x^5 \sin x^3 dx = \int x^3 \sin(x^3) x^2 dx$$

Assume the substitution $x^3 = t \Leftrightarrow 3x^2 dx = dt$ in the above equation.

$$\int x^5 \sin x^3 dx = \int t \sin t \times \frac{dt}{3}$$

To use the integration by parts rule, assume the functions $u = t \Leftrightarrow du = dt$ and $dv = \sin t dt \Leftrightarrow v = -\cos t$. Now use the integration by parts formula in the above equation.

$$\int x^5 \sin x^3 dx = \frac{1}{3} \left[t(-\cos t) - \int (-\cos t) dt \right]$$

$$\Rightarrow \int x^5 \sin x^3 dx = \frac{1}{3} \left[-t \cos t + \int \cos t dt \right]$$

$$\Rightarrow \int x^5 \sin x^3 dx = \frac{1}{3}(-t \cos t + \sin t) + C$$

Substitute the value $t = x^3$ in the above equation.

$$\int x^5 \sin x^3 dx = \frac{1}{3}(-x^3 \cos(x^3) + \sin(x^3)) + C$$

Hence the value of $\int x^5 \sin x^3 dx$ is $\frac{1}{3}(-x^3 \cos(x^3) + \sin(x^3)) + C$.

(ii) $\int \sec^3 x dx$

Ans: Separate the function in the given expression as

$$\int \sec^3 x dx = \int \sec x (\sec^2 x) dx$$

To use the integration by parts rule, assume the functions $u = \sec x \Leftrightarrow du = \sec x \tan x dx$ and $dv = \sec^2 x dx \Leftrightarrow v = \tan x$. Now substitute the values in the integration by parts formula.

$$\int \sec^3 x dx = \sec x \tan x - \int \tan x (\sec x \tan x) dx$$

Use the trigonometric identity $\sec^2 x - \tan^2 x = 1$ in the above equation.

$$\int \sec^3 x dx = \sec x \tan x - \int \sec x (\sec^2 x - 1) dx$$

$$\Rightarrow \int \sec^3 x dx = \sec x \tan x - \int \sec^3 x dx + \int \sec x dx$$

$$\Rightarrow \int \sec^3 x dx + \int \sec^3 x dx = \sec x \tan x + \int \sec x dx$$

$$\Rightarrow \int \sec^3 x dx = \frac{1}{2} [\sec x + \tan x + \ln |\sec x + \tan x|] + C$$

Hence the value of $\int \sec^3 x dx$ is $\frac{1}{2} [\sec x + \tan x + \ln |\sec x + \tan x|] + C$.

(iii) $\int e^{ax} \cos(bx + c) dx$

Ans: To use the integration by parts rule assume the functions $u = e^{ax} \Leftrightarrow du = ae^{ax}$ and $dv = \cos(bx + c) \Leftrightarrow v = \frac{1}{b} \sin(bx + c)$. Now substitute the values in the integration by parts formula.

$$\int e^{ax} \cos(bx + c) dx = e^{ax} \left(\frac{1}{b} \sin(bx + c) \right) - \int ae^{ax} \left(\frac{1}{b} \sin(bx + c) \right) dx$$

$$\Rightarrow \int e^{ax} \cos(bx + c) dx = \frac{e^{ax}}{b} \sin(bx + c) - \frac{a}{b} \int e^{ax} \sin(bx + c) dx$$

Assume the integral value $\int e^{ax} \sin(bx + c) dx$ separately from the above equation. For this value also assume the functions to use integration by parts as $u = e^{ax} \Leftrightarrow du = ae^{ax} dx$ and $dv = \sin(bx + c) \Leftrightarrow v = -\frac{1}{b} \cos(bx + c)$. Substitute the values in the assumed value.

$$\int e^{ax} \sin(bx + c) dx = e^{ax} \left(-\frac{1}{b} \cos(bx + c) \right) - \int ae^{ax} \left(-\frac{1}{b} \cos(bx + c) \right) dx$$

$$\Rightarrow \int e^{ax} \sin(bx + c) dx = -\frac{e^{ax}}{b} \cos(bx + c) + \frac{a}{b} \int e^{ax} \cos(bx + c) dx$$

Substitute the above value in the integral value of $\int e^{ax} \cos(bx + c) dx$.

$$\int e^{ax} \cos(bx + c) dx = \frac{e^{ax}}{b} \sin(bx + c) - \frac{a}{b} \left[-\frac{e^{ax}}{b} \cos(bx + c) + \frac{a}{b} \int e^{ax} \cos(bx + c) dx \right]$$

$$\Rightarrow \int e^{ax} \cos(bx + c) dx = \frac{e^{ax}}{b} \sin(bx + c) + \frac{ae^{ax}}{b^2} \cos(bx + c) - \frac{a^2}{b^2} \int e^{ax} \cos(bx + c) dx$$

$$\Rightarrow \int e^{ax} \cos(bx + c) dx + \frac{a^2}{b^2} \int e^{ax} \cos(bx + c) dx$$

$$= \frac{e^{ax}}{b^2} [b \sin(bx + c) + a \cos(bx + c)] + C$$

$$\Rightarrow \left(\frac{a^2 + b^2}{b^2} \right) \int e^{ax} \cos(bx + c) dx = \frac{e^{ax}}{b^2} [b \sin(bx + c) + a \cos(bx + c)] + C$$

$$\Rightarrow \int e^{ax} \cos(bx + c) dx = \frac{e^{ax}}{a^2 + b^2} [b \sin(bx + c) + a \cos(bx + c)] + C$$

Hence the value of $\int e^{ax} \cos(bx + c) dx$ is

$$\frac{e^{ax}}{a^2 + b^2} [b \sin(bx + c) + a \cos(bx + c)] + C.$$

(iv) $\int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx$

Ans: Assume the substitution $3x = \tan t \Leftrightarrow 3dx = \sec^2 t dt$ in the above equation.

$$\int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx = \int \sin^{-1}\left(\frac{2 \tan t}{1 + \tan^2 t}\right) \times \frac{1}{3} \sec^2 t dt$$

Use the formula $\sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$ in the above equation.

$$\int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx = \frac{1}{3} \int \sin^{-1}(\sin 2t) \sec^2 t dt$$

$$\Rightarrow \int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx = \frac{2}{3} \int t \sec^2 t dt$$

To use the integral by parts rule, assume the functions $u = t \Leftrightarrow du = dt$ and $dv = \sec^2 t dt \Leftrightarrow v = \tan t$ in the above equation.

$$\int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx = \frac{2}{3} \left[t(\tan t) - \int \tan t dt \right]$$

$$\Rightarrow \int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx = \frac{2}{3} t \tan t - \frac{2}{3} \log |\sec t| + C$$

Substitute the value $3x = \tan t \Leftrightarrow t = \tan^{-1}(3x)$ in the above equation.

$$\int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx = \frac{2}{3} (\tan^{-1}(3x)) \times 3x - \frac{1}{3} \log |1 + \tan^2 t| + C$$

$$\Rightarrow \int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx = 2x \tan^{-1} 3x - \frac{1}{3} \log |1 + 9x^2| + C$$

Hence the value of $\int \sin^{-1}\left(\frac{6x}{1+9x^2}\right) dx$ is $2x \tan^{-1} 3x - \frac{1}{3} \log |1 + 9x^2| + C$.

(v) $\int \cos \sqrt{x} dx$

Ans: Assume the substitution $\sqrt{x} = t \Leftrightarrow \frac{1}{2\sqrt{x}} dx = dt$ in the given expression.

$$\int \cos \sqrt{x} dx = \int (\cos t) 2t dt$$

$$\Rightarrow \int \cos \sqrt{x} dx = 2 \int t \cos t dt$$

To use the integration by parts rule, assume the functions $u = t \Leftrightarrow du = dt$ and $dv = \cos t dt \Leftrightarrow v = \sin t$.

$$\Rightarrow \int \cos \sqrt{x} dx = 2 \left[t \sin t - \int \sin t dt \right]$$

$$\Rightarrow \int \cos \sqrt{x} dx = 2 \left[t \sin t - (-\cos t) \right] + C$$

Substitute the value of $t = \sqrt{x}$ in the above equation.

$$\int \cos \sqrt{x} dx = 2 \left[\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x} \right] + C$$

Hence the value of $\int \cos \sqrt{x} dx$ is $2 \left[\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x} \right] + C$.

(vi) $\int x^3 \tan^{-1} x dx$

Ans: To use the integration by parts rule assume the functions $u = \tan^{-1} x \Leftrightarrow du = \frac{1}{1+x^2} dx$ and $dv = x^3 dx \Leftrightarrow v = \frac{x^4}{4}$.

$$\int x^3 \tan^{-1} x dx = \left(\tan^{-1} x \right) \left(\frac{x^4}{4} \right) - \int \frac{x^4}{4} \left(\frac{1}{1+x^2} \right) dx$$

$$\Rightarrow \int x^3 \tan^{-1} x dx = \frac{x^4 \tan^{-1} x}{4} - \frac{1}{4} \int \frac{(x^4 - 1) + 1}{1+x^2} dx$$

$$\Rightarrow \int x^3 \tan^{-1} x dx = \frac{x^4 \tan^{-1} x}{4} - \frac{1}{4} \int \frac{(x^2 - 1)(x^2 + 1) + 1}{1+x^2} dx$$

$$\Rightarrow \int x^3 \tan^{-1} x dx = \frac{x^4 \tan^{-1} x}{4} - \frac{1}{4} \left[\int (x^2 - 1) dx + \int \frac{1}{1+x^2} dx \right]$$

$$\Rightarrow \int x^3 \tan^{-1} x dx = \frac{x^4 \tan^{-1} x}{4} - \frac{1}{4} \left[\frac{x^3}{3} - x + \tan^{-1} x \right] + C$$

$$\Rightarrow \int x^3 \tan^{-1} x dx = \frac{x^4}{4} \tan^{-1} x - \frac{x^3}{12} + \frac{x}{4} - \frac{1}{4} \tan x + C$$

$$\Rightarrow \int x^3 \tan^{-1} x dx = \left(\frac{x^4 - 1}{4} \right) \tan^{-1} x - \frac{x^4}{12} + \frac{x}{4} + C$$

Hence the value of $\int x^3 \tan^{-1} x dx$ is $\left(\frac{x^4 - 1}{4} \right) \tan^{-1} x - \frac{x^4}{12} + \frac{x}{4} + C$.

(vii) $\int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx$

Ans: Use the trigonometric formulas $\sin 2x = 2 \sin x \cos x$ and $\cos 2x = 2 \cos^2 x - 1$ in the given expression.

$$\int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx = \int e^{2x} \left(\frac{1 + 2 \sin x \cos x}{2 \cos^2 x} \right) dx$$

$$\Rightarrow \int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx = \int e^{2x} \left(\frac{1}{2 \cos^2 x} + \frac{2 \sin x \cos x}{2 \cos^2 x} \right) dx$$

$$\Rightarrow \int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx = \frac{1}{2} \int e^{2x} \sec^2 x dx + \int e^{2x} \tan x dx$$

Assume the functions $u = e^{2x} \Leftrightarrow du = 2e^{2x} dx$, $dv = \sec^2 x dx \Leftrightarrow v = \tan x$ for the first integral in the above equation.

$$\int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx = \frac{1}{2} \left[e^{2x} \tan x - \int 2e^{2x} \tan x dx \right] + \int e^{2x} \tan x dx$$

$$\Rightarrow \int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx = \frac{1}{2} e^{2x} \tan x - \int e^{2x} \tan x dx + \int e^{2x} \tan x dx$$

$$\Rightarrow \int e^{2x} \tan x dx = \frac{1}{2} e^{2x} \tan x + C$$

Hence the value of $\int e^{2x} \tan x dx$ is $\frac{1}{2} e^{2x} \tan x + C$.

(viii) $\int e^x \left(\frac{x-1}{2x^2} \right) dx$

Ans: Simplify the given expression as

$$\int e^x \left(\frac{x-1}{2x^2} \right) dx = \frac{1}{2} \int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$$

Assume $t = \frac{1}{x} \Leftrightarrow t' = -\frac{1}{x^2}$ in the above equation.

$$\int e^x \left(\frac{x-1}{2x^2} \right) dx = \frac{1}{2} \int e^x (t + t') dx$$

Use the integration formula $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$ in the above equation.

$$\int e^x \left(\frac{x-1}{2x^2} \right) dx = \frac{1}{2} e^x \left(\frac{1}{x} \right) + C$$

$$\Rightarrow \int e^x \left(\frac{x-1}{2x^2} \right) dx = \frac{e^x}{2x} + C$$

Hence the value of $\int e^x \left(\frac{x-1}{2x^2} \right) dx$ is $\frac{e^x}{2x} + C$.

(ix) $\int \sqrt{2ax - x^2} dx$

Ans: Simplify the given expression as

$$\int \sqrt{2ax - x^2} dx = \int \sqrt{2(a)x - x^2 + a^2 - a^2} dx$$

$$\Rightarrow \int \sqrt{2ax - x^2} dx = \int \sqrt{a^2 - (x-a)^2} dx$$

Use the integration formula $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$ in the above equation.

$$\int \sqrt{2ax - x^2} dx = \frac{x-a}{2} \sqrt{2ax - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x-a}{a} \right) + C$$

Hence the value of $\int \sqrt{2ax - x^2} dx$ is $\frac{x-a}{2} \sqrt{2ax - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x-a}{a} \right) + C$.

$$(x) \int e^x \frac{(x^2 + 1)}{(x+1)^2} dx$$

Ans: Simplify the given expression as

$$\int e^x \frac{(x^2 + 1)}{(x+1)^2} dx = \int e^x \frac{(x^2 + 1 + 2x - 2x)}{(x+1)^2} dx$$

$$\Rightarrow \int e^x \frac{(x^2 + 1)}{(x+1)^2} dx = \int e^x \frac{(x+1)^2 - 2x}{(x+1)^2} dx$$

$$\Rightarrow \int e^x \frac{(x^2 + 1)}{(x+1)^2} dx = \int e^x dx - 2 \int e^x \frac{x}{(x+1)^2} dx$$

$$\Rightarrow \int e^x \frac{(x^2 + 1)}{(x+1)^2} dx = e^x - 2 \int e^x \frac{(x+1-1)}{(x+1)^2} dx$$

$$\Rightarrow \int e^x \frac{(x^2 + 1)}{(x+1)^2} dx = e^x - 2 \int e^x \left(\frac{1}{x+1} + \frac{(-1)}{(x+1)^2} \right) dx$$

If $f(x) = \frac{1}{x+1}$, then $f'(x) = \frac{-1}{(x+1)^2}$. So, use the formula

$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$ in the above equation.

$$\int e^x \frac{(x^2 + 1)}{(x+1)^2} dx = e^x - 2e^x \left(\frac{1}{x+1} \right) + C$$

$$\Rightarrow \int e^x \frac{(x^2 + 1)}{(x+1)^2} dx = e^x \left[1 - \frac{2}{x+1} \right] + C$$

$$\Rightarrow \int e^x \frac{(x^2 + 1)}{(x+1)^2} dx = e^x \left(\frac{x-1}{x+1} \right) + C$$

Hence the value of $\int e^x \frac{(x^2+1)}{(x+1)^2} dx$ is $e^x \left(\frac{x-1}{x+1} \right) + C$.

(xi) $\int e^x \frac{(2 + \sin 2x)}{(1 + \cos 2x)} dx$

Ans: Use the trigonometric formulas $\sin 2x = 2 \sin x \cos x$ and $\cos 2x = 2 \cos^2 x - 1$ in the given expression.

$$\int e^x \frac{(2 + \sin 2x)}{(1 + \cos 2x)} dx = \int e^x \frac{(2 + 2 \sin x \cos x)}{2 \cos^2 x} dx$$

$$\Rightarrow \int e^x \frac{(2 + \sin 2x)}{(1 + \cos 2x)} dx = \int e^x [\sec^2 x + \tan x] dx$$

If $f(x) = \tan x$ then $f'(x) = \sec^2 x$. So, use the formula

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C \text{ in the above equation.}$$

$$\int e^x \frac{(2 + \sin 2x)}{(1 + \cos 2x)} dx = e^x \tan x + C$$

Hence the value of $\int e^x \frac{(2 + \sin 2x)}{(1 + \cos 2x)} dx$ is $e^x \tan x + C$.

(xii) $\int \left\{ \log(\log x) + \frac{1}{(\log x)^2} \right\} dx$

Ans: Consider $I = \int \log(\log x) dx$

Apply integration by parts.

$$I = \log(\log x) \int 1 dx - \int [(\log(\log x))' \times \int 1 dx] dx$$

$$\Rightarrow I = x \log(\log x) - \int \frac{1}{\log x} \times \frac{1}{x} \times x dx$$

$$\Rightarrow I = x \log(\log x) - \int (\log x)^{-1} dx$$

Consider $I_1 = \int (\log x)^{-1} dx$

Apply integration by parts.

$$I = x \log(\log x) - \left[(\log x)^{-1} \int 1 dx - \int \left[\frac{d}{dx} ((\log x)^{-1}) \int 1 dx \right] dx \right]$$

$$I = x \log(\log x) - \left[\frac{x}{\log x} - \int -(\log x)^{-2} \times \frac{1}{x} \times x dx \right]$$

$$I = x \log(\log x) - \frac{x}{\log x} - \int \frac{1}{(\log x)^2} dx$$

$$\Rightarrow \int \left\{ \log(\log x) + \frac{1}{(\log x)^2} \right\} dx = x \log(\log x) - \frac{x}{\log x} + C$$

Hence the value of $\int \left\{ \log(\log x) + \frac{1}{(\log x)^2} \right\} dx$ is $x \log(\log x) - \frac{x}{\log x} + C$.

(xiii) $\int (6x + 5) \sqrt{6 + x - x^2} dx$

Ans: Let $6x + 5 = A \cdot \frac{d}{dx} (6 - x^2 + x) + B$

$$\Rightarrow 6x + 5 = A(-2x + 1) + B$$

$$\Rightarrow 6x + 5 = -2Ax + A + B$$

Equating and solving for coefficients, then we will get $A = -3$ and $B = 8$.

Now the given expression can be written as

$$\int (6x + 5) \sqrt{6 + x - x^2} dx = \int [-3(-2x + 1) + 8] \sqrt{6 - x^2 + x} dx$$

$$\Rightarrow \int (6x + 5) \sqrt{6 + x - x^2} dx = -3 \int (-2x + 1) \sqrt{6 + x - x^2} + 8 \int \sqrt{6 + x - x^2} dx$$

Assume $6 + x - x^2 = t \Leftrightarrow (1 - 2x) dx = dt$ in the above equation.

$$\int (6x+5)\sqrt{6+x-x^2}dx = -3\int \sqrt{t}dt + 8\int \sqrt{6+2\left(\frac{1}{2}\right)x-x^2-\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2}dx$$

$$\Rightarrow \int (6x+5)\sqrt{6+x-x^2}dx = -3 \times \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + 8\int \sqrt{\left(\frac{5}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2}dx$$

Use the integration formula $\int \sqrt{a^2-x^2}dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$ and substitute $t = 6+x-x^2$ in the above equation.

$$\int (6x+5)\sqrt{6+x-x^2}dx = -2(6+x-x^2)^{\frac{3}{2}}$$

$$+ 8\left[\frac{x-\frac{1}{2}}{2}\sqrt{6+x-x^2} + \frac{\left(\frac{5}{2}\right)^2}{2}\sin^{-1}\left(\frac{x-\frac{1}{2}}{2}\right)\right] + C$$

$$\Rightarrow \int (6x+5)\sqrt{6+x-x^2}dx = -2(6+x-x^2)^{\frac{3}{2}}$$

$$+ 8\left[\frac{2x-1}{4}\sqrt{6+x-x^2} + \frac{25}{8}\sin^{-1}\left(\frac{2x-1}{4}\right)\right] + C$$

Hence the value of $\int (6x+5)\sqrt{6+x-x^2}dx$ is

$$-2(6+x-x^2)^{\frac{3}{2}} + 8\left[\frac{2x-1}{4}\sqrt{6+x-x^2} + \frac{25}{8}\sin^{-1}\left(\frac{2x-1}{4}\right)\right] + C.$$

(xiv) $\int (x-2)\sqrt{\frac{x+3}{x-3}}dx$

Ans:

$$\int (x-2)\sqrt{\frac{x+3}{x-3}}dx = \frac{\sqrt{x^2-9}\left[(x+2)\sqrt{9-x^2} - 6\sin^{-1}\left(\frac{\sqrt{3-x}}{\sqrt{6}}\right)\right]}{2\sqrt{9-x^2}} + C, \text{ where } C \text{ is an integration constant.}$$

(xv) $\int (2x - 5)\sqrt{x^2 - 4x + 3} dx$

Ans: Simplify the given expression as

$$\int (2x - 5)\sqrt{x^2 - 4x + 3} dx = \int (2x - 4 - 1)\sqrt{x^2 - 4x + 3} dx$$

$$\Rightarrow \int (2x - 5)\sqrt{x^2 - 4x + 3} dx = \int (2x - 4)\sqrt{x^2 - 4x + 3} dx - \int \sqrt{x^2 - 4x + 3} dx$$

Let $u = x^2 - 4x + 3 \Leftrightarrow du = (2x - 4)dx$, now the above equation is modified as

$$\int (2x - 5)\sqrt{x^2 - 4x + 3} dx = \int \sqrt{u} du - \int \sqrt{x^2 - 2(2)x + 2^2 - 2^2 + 3} dx$$

$$\Rightarrow \int (2x - 5)\sqrt{x^2 - 4x + 3} dx = \frac{u^{\frac{3}{2}}}{\frac{3}{2}} - \int \sqrt{(x - 2)^2 - 1^2} dx$$

Use the integration formula $\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log|x + \sqrt{x^2 - a^2}| + C$ and substitute $u = x^2 - 4x + 3$ in the above equation.

$$\int (2x - 5)\sqrt{x^2 - 4x + 3} dx = \frac{2}{3}(x^2 - 4x + 3)^{\frac{3}{2}} - \left[\frac{x - 2}{2} \sqrt{x^2 - 4x + 3} + \frac{1}{2} \log|x - 2 + \sqrt{x^2 - 4x + 3}| \right] + C$$

$$\Rightarrow \int (2x - 5)\sqrt{x^2 - 4x + 3} dx = \frac{2}{3}(x^2 - 4x + 3)^{\frac{3}{2}} - \left(\frac{x - 2}{2} \right) \sqrt{x^2 - 4x + 3} + \frac{1}{2} \log|x - 2 + \sqrt{x^2 - 4x + 3}| + C$$

Hence the value of $\int (2x - 5)\sqrt{x^2 - 4x + 3} dx$ is

$$\frac{2}{3}(x^2 - 4x + 3)^{\frac{3}{2}} - \left(\frac{x - 2}{2} \right) \sqrt{x^2 - 4x + 3} + \frac{1}{2} \log|x - 2 + \sqrt{x^2 - 4x + 3}| + C.$$

(xiv) $\int \sqrt{x^2 - 4x + 8} dx$

Ans: Simplify the above equation as

$$\int \sqrt{x^2 - 4x + 8} dx = \int \sqrt{x^2 - 2(2)x + 2^2 - 2^2 + 8} dx$$

$$\Rightarrow \int \sqrt{x^2 - 4x + 8} dx = \int \sqrt{(x-2)^2 + (2)^2} dx$$

Use the integration formula $\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log|x + \sqrt{x^2 + a^2}| + C$ in the above equation.

$$\int \sqrt{x^2 - 4x + 8} dx = \frac{x-2}{2} \sqrt{x^2 - 4x + 8} + \frac{2^2}{2} \log|x-2 + \sqrt{x^2 - 4x + 8}| + C$$

Hence the value of $\int \sqrt{x^2 - 4x + 8} dx$ is

$$\left(\frac{x-2}{2} \right) \sqrt{x^2 - 4x + 8} + 2 \log|(x-2) + \sqrt{x^2 - 4x + 8}| + C.$$

45. Evaluate the following definite integrals:

(i) $\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$

Ans: Given integral is $-\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$

Say, $\sin x - \cos x = t$

$$\Rightarrow (\cos x + \sin x) dx = dt$$

For the limits –

- When $x = 0 \Rightarrow t = -1$

- When $x = \frac{\pi}{4} \Rightarrow t = 0$

Now,

$$(\sin x - \cos x)^2 = t^2$$

$$\Rightarrow (\sin^2 x + \cos^2 x - 2 \sin x \cos x) = t^2$$

It is known that, $(\sin^2 x + \cos^2 x) = 1$ and $\sin 2x = 2 \sin x \cos x$

$$\Rightarrow (1 - \sin 2x) = t^2$$

$$\Rightarrow 1 - t^2 = \sin 2x$$

Using the above equations, the integral changes to –

$$\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx = \int_{-1}^0 \frac{dt}{9 + 16(1 - t^2)}$$

$$= \int_{-1}^0 \frac{dt}{9 + 16 - 16t^2}$$

$$= \int_{-1}^0 \frac{dt}{25 - 16t^2}$$

$$= \int_{-1}^0 \frac{dt}{(5)^2 - (4t)^2}$$

Using $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$

$$= \frac{1}{4} \left[\frac{1}{2 \times 5} \log \left| \frac{5+4t}{5-4t} \right| \right]_{-1}^0$$

$$= \frac{1}{4} \left[\frac{1}{10} \log \left| \frac{5}{5} \right| - \frac{1}{10} \log \left| \frac{5-4}{5+4} \right| \right]$$

$$= \frac{1}{4} \times \frac{1}{10} \left[\log |1| - \log \left| \frac{1}{9} \right| \right]$$

$$= \frac{1}{40} [\log |1| - \log |1| + \log |9|]$$

$$= \frac{1}{40} \log |9|$$

Thus, $\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx = \frac{1}{40} \log |9|$

(ii) $\int_0^{\frac{\pi}{2}} \cos 2x \log \sin x dx$

Ans: Given integral – $\int_0^{\frac{\pi}{2}} \cos 2x \log \sin x dx$

Using integration by parts – $\int u v dx = u \int v dx - \int u' \left(\int v dx \right) dx$

Here,

$$u = \log \sin x \Rightarrow u' = \frac{1}{\sin x} \times \cos x$$

$$v = \cos 2x \Rightarrow \int v dx = \frac{1}{2} \sin 2x$$

The integral becomes –

$$\int_0^{\frac{\pi}{2}} \cos 2x \log \sin x dx = \left[\log(\sin x) \times \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{1}{\sin x} \cos x \times \sin 2x \right) dx$$

It is known that – $\sin 2x = 2 \sin x \cos x$

$$= \left[\log(\sin x) \times \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{1}{\sin x} \cos x \times 2 \sin x \cos x \right) dx$$

$$= \left[\log(\sin x) \times \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (\cos^2 x) dx$$

It is known that – $\cos 2x = 2 \cos^2 x - 1$

$$= \left[\log(\sin x) \times \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos 2x) dx$$

$$= \left[\log(\sin x) \times \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$$

Substituting the limits –

$$= \left[\log\left(\sin \frac{\pi}{2}\right) \times \frac{1}{2} \sin \pi - \log(\sin 0) \times \frac{1}{2} \sin 0 \right] - \frac{1}{2} \left[\frac{\pi}{2} + \frac{1}{2} \sin \pi - 0 - \frac{1}{2} \sin 0 \right]$$

$$= [0 - 0] - \frac{1}{2} \left[\frac{\pi}{2} + 0 - 0 - 0 \right]$$

$$\Rightarrow I = -\frac{\pi}{4}$$

$$\text{Thus, } \int_0^{\frac{\pi}{2}} \cos 2x \log \sin x dx = -\frac{\pi}{4}$$

$$\text{(iii) } \int_0^1 x \sqrt{\frac{1-x^2}{1+x^2}} dx$$

$$\text{Ans: Given integral} = \int_0^1 x \sqrt{\frac{1-x^2}{1+x^2}} dx$$

$$\text{Say, } x^2 = t \Rightarrow 2x dx = dt$$

The limits –

- When $x = 0 \Rightarrow t = 0$
- When $x = 1 \Rightarrow t = 1$

The integral becomes –

$$\int_0^1 x \sqrt{\frac{1-x^2}{1+x^2}} dx = \frac{1}{2} \int_0^1 \sqrt{\frac{1-t}{1+t}} dt$$

Multiply the numerator and denominator by $\sqrt{1-t}$

$$= \frac{1}{2} \int_0^1 \frac{\sqrt{(1-t)(1-t)}}{\sqrt{(1+t)(1-t)}} dt$$

$$= \frac{1}{2} \int_0^1 \frac{1-t}{\sqrt{1-t^2}} dt$$

$$= \frac{1}{2} \left[\int_0^1 \frac{1}{\sqrt{1-t^2}} dt - \int_0^1 \frac{t}{\sqrt{1-t^2}} dt \right]$$

Integrating the first term –

$$\int_0^1 \frac{1}{\sqrt{1-t^2}} dt = \left[\sin^{-1} t \right]_0^1$$

$$= \left[\sin^{-1} 1 - \sin^{-1} 0 \right]$$

$$= \frac{\pi}{2} - 0$$

$$\Rightarrow \int_0^1 \frac{1}{\sqrt{1-t^2}} dt = \frac{\pi}{2}$$

Integrating the second term – $\int_0^1 \frac{t}{\sqrt{1-t^2}} dt$

Say, $1-t^2 = u \Rightarrow -2t dt = du$

The limits –

- When $t = 0 \Rightarrow u = 1$
- When $t = 1 \Rightarrow u = 0$

$$\therefore \int_0^1 \frac{t}{\sqrt{1-t^2}} dt = -\frac{1}{2} \int_1^0 \frac{du}{\sqrt{u}}$$

Interchanging the limits –

$$= \frac{1}{2} \int_0^1 \frac{du}{\sqrt{u}}$$

$$= \frac{1}{2} \left[\frac{1}{\left(\frac{1}{2}\right)} \sqrt{u} \right]_0^1$$

$$= \sqrt{1} - 0$$

$$\int_0^1 \frac{t}{\sqrt{1-t^2}} dt = 1$$

Using them –

$$\int_0^1 x \sqrt{\frac{1-x^2}{1+x^2}} dx = \frac{1}{2} \left[\int_0^1 \frac{1}{\sqrt{1-t^2}} dt - \int_0^1 \frac{t}{\sqrt{1-t^2}} dt \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{2} - 1 \right]$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

Thus, $\int_0^1 x \sqrt{\frac{1-x^2}{1+x^2}} dx = \frac{\pi}{4} - \frac{1}{2}$

(iv) $\int_0^{\frac{1}{\sqrt{2}}} \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx$

Ans: Given integral – $\int_0^{\frac{1}{\sqrt{2}}} \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx$

$$\int_0^{\frac{1}{\sqrt{2}}} \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx = \int_0^{\frac{1}{\sqrt{2}}} \frac{\sin^{-1} x}{(1-x^2)\sqrt{1-x^2}} dx$$

Say, $\sin^{-1} x = t \Rightarrow \frac{1}{\sqrt{1-x^2}} dx = dt$

The limits –

- When $x = 0 \Rightarrow t = 0$
- When $x = \frac{1}{\sqrt{2}} \Rightarrow t = \frac{\pi}{4}$

Also, $x = \sin t \Rightarrow x^2 = \sin^2 t$

So, $(1-x^2) = (1-\sin^2 t) \Rightarrow (1-x^2) = \cos^2 t$

The integral becomes –

$$\int_0^{\frac{1}{\sqrt{2}}} \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx = \int_0^{\frac{\pi}{4}} \frac{t}{\cos^2 t} dt$$

$$= \int_0^{\frac{\pi}{4}} t \sec^2 t dt$$

Using integration by parts – $\int u v dx = u \int v dx - \int u' (\int v dx) dx$

Here,

$$u = t \Rightarrow u' = 1$$

$$v = \sec^2 t \Rightarrow \int v dx = \tan t$$

The integral becomes –

$$\int_0^{\frac{\pi}{4}} t \sec^2 t dt = \left[t \tan t \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} (1 \times \tan t) dt$$

$$= \left[t \tan t \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} (\tan t) dt$$

$$= \left[t \tan t \right]_0^{\frac{\pi}{4}} - \left[-\log(\cos t) \right]_0^{\frac{\pi}{4}}$$

Applying the limits,

$$= \left[\frac{\pi}{4} \tan \frac{\pi}{4} - 0 \right] + \left[\log \left(\cos \frac{\pi}{4} \right) - \log(\cos 0) \right]$$

$$= \left[\frac{\pi}{4} \right] + \left[\log \left(\frac{1}{\sqrt{2}} \right) - \log(1) \right]$$

$$= \frac{\pi}{4} + \left[\log 1 - \log \sqrt{2} - \log 1 \right]$$

$$= \frac{\pi}{4} + \left[-\log \sqrt{2} \right]$$

$$= \frac{\pi}{4} - \frac{1}{2} \log 2$$

$$\text{Thus, } \int_0^{\frac{1}{\sqrt{2}}} \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx = \frac{\pi}{4} - \frac{1}{2} \log 2$$

$$(v) \int_0^{\frac{\pi}{2}} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$$

$$\text{Ans: Given integral} = \int_0^{\frac{\pi}{2}} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx = \int_0^{\frac{\pi}{2}} \frac{2 \sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

Divide the numerator and denominator by $\cos^4 x$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{2 \sin x \cos x}{\cos^4 x}}{\frac{\sin^4 x}{\cos^4 x} + \frac{\cos^4 x}{\cos^4 x}} dx$$

Simplify,

$$= \int_0^{\frac{\pi}{2}} \frac{2 \tan x \sec^2 x}{\tan^4 x + 1} dx$$

Say, $\tan x = t \Rightarrow \sec^2 x dx = dt$

The limits –

- When $x = 0 \Rightarrow t = 0$
- When $x = \frac{\pi}{2} \Rightarrow t = \infty$

The integral becomes –

$$= \int_0^{\infty} \frac{2tdt}{t^4 + 1}$$

Say, $t^2 = u \Rightarrow 2tdt = du$

The limits –

- When $t = 0 \Rightarrow u = 0$
- When $t = \infty \Rightarrow u = \infty$

The integral becomes –

$$\begin{aligned} &= \int_0^{\infty} \frac{du}{u^2 + 1} \\ &= \left[\tan^{-1} u \right]_0^{\infty} \\ &= \left[\tan^{-1} \infty - \tan^{-1} 0 \right] \end{aligned}$$

$$= \frac{\pi}{2} - 0$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx = \frac{\pi}{2}$$

$$\text{Thus, } \int_0^{\frac{\pi}{2}} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx = \frac{\pi}{2}$$

$$\text{(vi) } \int_1^2 \frac{5x^2}{x^2 + 4x + 3} dx$$

$$\text{Ans: Given integral} = \int_1^2 \frac{5x^2}{x^2 + 4x + 3} dx$$

$$\int_1^2 \frac{5x^2}{x^2 + 4x + 3} dx = \int_1^2 \left\{ 5 - \frac{20x + 15}{x^2 + 4x + 3} \right\} dx$$

$$\int_1^2 \frac{5x^2}{x^2 + 4x + 3} dx = [5x]_1^2 - \int_1^2 \frac{20x + 15}{x^2 + 4x + 3} dx \rightarrow (1)$$

$$\text{Solving } \int_1^2 \frac{20x + 15}{x^2 + 4x + 3} dx ,$$

$$\text{Let } 20x + 15 = A \frac{d}{dx}(x^2 + 4x + 3) + B$$

Equating the coefficients of x and constant term we get,

$$A = 10, B = -25$$

$$\text{Let } x^2 + 4x + 3 = t$$

Differentiating this we get,

$$(2x + 4)dx = dt$$

The integral becomes –

$$10 \int \frac{dt}{t} - 25 \int \frac{dx}{(x+2)^2 - 1^2}$$

$$\begin{aligned}
 10 \int \frac{dt}{t} - 25 \int \frac{dx}{(x+2)^2 - 1^2} &= 10 \log t - 25 \left[\frac{1}{2} \log \left(\frac{x+2-1}{x+2+1} \right) \right] \\
 \Rightarrow \int_1^2 \frac{20x+15}{x^2+4x+3} dx &= \left[10 \log(x^2+4x+3) - 25 \left[\frac{1}{2} \log \left(\frac{x+1}{x+3} \right) \right] \right]_1^2 \\
 \Rightarrow \int_1^2 \frac{20x+15}{x^2+4x+3} dx &= 10 \log 15 - 10 \log 8 - 25 \left[\frac{1}{2} \log \frac{3}{5} - \frac{1}{2} \log \frac{2}{4} \right] \\
 \Rightarrow \int_1^2 \frac{20x+15}{x^2+4x+3} dx &= 10 \log 5 + 10 \log 3 - 10 \log 4 - 10 \log 2 \\
 &\quad - \frac{25}{2} [\log 3 - \log 5 - \log 2 + \log 4] \\
 \Rightarrow \int_1^2 \frac{20x+15}{x^2+4x+3} dx &= \frac{45}{2} \log 5 - \frac{45}{2} \log 4 - \frac{5}{2} \log 3 + \frac{5}{2} \log 2 \\
 \therefore \int_1^2 \frac{20x+15}{x^2+4x+3} dx &= \frac{45}{2} \log \frac{5}{4} - \frac{5}{2} \log \frac{3}{2}
 \end{aligned}$$

Substituting in (1) we get,

$$\int_1^2 \frac{5x^2}{x^2+4x+3} dx = 5 - \left[\frac{45}{2} \log \frac{5}{4} - \frac{5}{2} \log \frac{3}{2} \right]$$

$$\text{Thus, } \int_1^2 \frac{5x^2}{x^2+4x+3} dx = 5 - \frac{5}{2} \left[9 \log \frac{5}{4} - \log \frac{3}{2} \right]$$

$$\text{(vii) } \int_0^{\frac{\pi}{2}} \frac{x + \sin x}{1 + \cos x} dx$$

$$\text{Ans: Given integral} = \int_0^{\frac{\pi}{2}} \frac{x + \sin x}{1 + \cos x} dx$$

It is known that –

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$1 + \cos x = 2 \cos^2 \frac{x}{2}$$

So,

$$\int_0^{\frac{\pi}{2}} \frac{x + \sin x}{1 + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{x + 2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} dx$$

Simplify,

$$= \int_0^{\frac{\pi}{2}} \frac{x}{2 \cos^2 \frac{x}{2}} dx + \int_0^{\frac{\pi}{2}} \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} x \sec^2 \frac{x}{2} dx + \int_0^{\frac{\pi}{2}} \tan \frac{x}{2} dx$$

Evaluating the first term – $\frac{1}{2} \int_0^{\frac{\pi}{2}} x \sec^2 \frac{x}{2} dx$

Using integration by parts – $\int u v dx = u \int v dx - \int u' \left(\int v dx \right) dx$

Here,

$$u = x \Rightarrow u' = 1$$

$$v = \sec^2 \frac{x}{2} \Rightarrow \int v dx = 2 \tan \frac{x}{2}$$

The integral becomes –

$$\int_0^{\frac{\pi}{2}} x \sec^2 \frac{x}{2} dt = \left[2x \tan \frac{x}{2} \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \left(1 \times 2 \tan \frac{x}{2} \right) dt$$

$$= \left[2x \tan \frac{x}{2} \right]_0^{\frac{\pi}{2}} - 2 \int_0^{\frac{\pi}{2}} \left(\tan \frac{x}{2} \right) dt$$

$$= \left[2x \tan \frac{x}{2} \right]_0^{\frac{\pi}{2}} - 2 \int_0^{\frac{\pi}{2}} \left(\tan \frac{x}{2} \right) dt$$

Applying the limits,

$$= \left[\pi \tan \frac{\pi}{4} - 0 \right] - 2 \int_0^{\frac{\pi}{2}} \left(\tan \frac{x}{2} \right) dt$$

$$= [\pi] - 2 \int_0^{\frac{\pi}{2}} \left(\tan \frac{x}{2} \right) dt$$

Using this,

$$\int_0^{\frac{\pi}{2}} \frac{x + \sin x}{1 + \cos x} dx = \left[\frac{1}{2} \int_0^{\frac{\pi}{2}} x \sec^2 \frac{x}{2} dx \right] + \int_0^{\frac{\pi}{2}} \tan \frac{x}{2} dx$$

$$= \frac{1}{2} \left[[\pi] - 2 \int_0^{\frac{\pi}{2}} \left(\tan \frac{x}{2} \right) dt \right] + \int_0^{\frac{\pi}{2}} \tan \frac{x}{2} dx$$

$$= \frac{\pi}{2} - \int_0^{\frac{\pi}{2}} \tan \frac{x}{2} dx + \int_0^{\frac{\pi}{2}} \tan \frac{x}{2} dx$$

$$= \frac{\pi}{2}$$

Thus, $\int_0^{\frac{\pi}{2}} \frac{x + \sin x}{1 + \cos x} dx = \frac{\pi}{2}$

46. Evaluate:

(i) $\int_1^3 \{ |x-1| + |x-2| + |x-3| \} dx$

Ans: Given integral – $\int_1^3 \{ |x-1| + |x-2| + |x-3| \} dx$

Say, $I = \int_1^3 \{ |x-1| + |x-2| + |x-3| \} dx$

Splitting the integral –

$$\Rightarrow I = \int_1^3 |x-1| dx + \int_1^3 |x-2| dx + \int_1^3 |x-3| dx$$

i.e. $I = I_1 + I_2 + I_3$

with –

$$I_1 = \int_1^3 |x-1| dx$$

$$I_2 = \int_1^3 |x-2| dx$$

$$I_3 = \int_1^3 |x-3| dx$$

Now,

$$I_1 = \int_1^3 |x-1| dx$$

Here, $(x-1) \geq 0$ for $1 \leq x \leq 3$

$$\Rightarrow I_1 = \int_1^3 (x-1) dx$$

$$= \left[\frac{x^2}{2} - x \right]_1^3$$

$$= \left[\frac{3^2}{2} - 3 - \frac{1^2}{2} + 1 \right]$$

$$= \left[\frac{9}{2} - \frac{1}{2} - 2 \right]$$

$$= \left[\frac{8}{2} - 2 \right]$$

$$= 4 - 2$$

$$= 2$$

$$\Rightarrow I_1 = 2$$

$$I_2 = \int_1^3 |x-2| dx$$

Here, $(x-2) \geq 0$ for $2 \leq x \leq 3$ and $(x-2) \leq 0$ for $1 \leq x \leq 2$

$$\Rightarrow I_2 = \int_1^2 -(x-2) dx + \int_2^3 (x-2) dx$$

$$= -\left[\frac{x^2}{2} - 2x\right]_1^2 + \left[\frac{x^2}{2} - 2x\right]_2^3$$

$$= -\left[\frac{2^2}{2} - 4 - \frac{1^2}{2} + 2\right] + \left[\frac{3^2}{2} - 6 - \frac{2^2}{2} + 4\right]$$

$$= -\left[\frac{4}{2} - \frac{1}{2} - 2\right] + \left[\frac{9}{2} - \frac{4}{2} - 2\right]$$

$$= -\left[\frac{3}{2} - 2\right] + \left[\frac{5}{2} - 2\right]$$

$$= -\frac{3}{2} + 2 + \frac{5}{2} - 2$$

$$= \frac{2}{2}$$

$$= 1$$

$$\Rightarrow I_2 = 1$$

$$I_3 = \int_1^3 |x-3| dx$$

Here, $(x-3) \leq 0$ for $1 \leq x \leq 3$

$$\Rightarrow I_3 = \int_1^3 -(x-3) dx$$

$$= -\left[\frac{x^2}{2} - 3x\right]_1^3$$

$$= -\left[\frac{3^2}{2} - 9 - \frac{1^2}{2} + 3\right]$$

$$= -\left[\frac{9}{2} - \frac{1}{2} - 6\right]$$

$$= -\left[\frac{8}{2} - 6\right]$$

$$= -4 + 6$$

$$= 2$$

$$\Rightarrow I_3 = 2$$

$$I = I_1 + I_2 + I_3$$

$$\Rightarrow I = 2 + 1 + 2$$

$$\Rightarrow I = 5$$

$$\text{Thus, } \int_1^3 \{|x-1| + |x-2| + |x-3|\} dx = 5$$

$$(ii) \int_0^{\pi} \frac{x}{1 + \sin x} dx$$

$$\text{Ans: Given integral} = \int_0^{\pi} \frac{x}{1 + \sin x} dx$$

$$\text{Say, } I = \int_0^{\pi} \frac{x}{1 + \sin x} dx \rightarrow (1)$$

$$\text{Using } \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\Rightarrow I = \int_0^{\pi} \frac{(\pi - x)}{1 + \sin(\pi - x)} dx$$

$$\text{It is known that } \sin(\pi - x) = \sin x$$

$$\Rightarrow I = \int_0^{\pi} \frac{(\pi - x)}{1 + \sin x} dx \rightarrow (2)$$

$$\text{Add equations } (1) + (2)$$

$$I + I = \int_0^{\pi} \frac{x}{1 + \sin x} dx + \int_0^{\pi} \frac{(\pi - x)}{1 + \sin x} dx$$

$$\Rightarrow 2I = \int_0^{\pi} \frac{x + \pi - x}{1 + \sin x} dx$$

$$= \int_0^{\pi} \frac{\pi}{1 + \sin x} dx$$

$$= \pi \int_0^{\pi} \frac{1}{1 + \sin x} dx$$

Multiply and divide by $(1 - \sin x)$

$$= \pi \int_0^{\pi} \frac{(1 - \sin x)}{(1 + \sin x)(1 - \sin x)} dx$$

$$= \pi \int_0^{\pi} \frac{(1 - \sin x)}{(1 - \sin^2 x)} dx$$

It is known that $-\sin^2 x + \cos^2 x = 1$

$$= \pi \int_0^{\pi} \frac{(1 - \sin x)}{(\cos^2 x)} dx$$

Simplifying

$$= \pi \int_0^{\pi} \left\{ \left(\frac{1}{\cos^2 x} \right) - \left(\frac{\sin x}{\cos^2 x} \right) \right\} dx$$

$$= \pi \int_0^{\pi} \{ \sec^2 x - \sec x \tan x \} dx$$

$$= \pi \left[\int_0^{\pi} \sec^2 x dx - \int_0^{\pi} \sec x \tan x dx \right]$$

$$= \pi [\tan x]_0^{\pi} - \pi [\sec x]_0^{\pi}$$

$$= \pi [\tan \pi - \tan 0] - \pi [\sec \pi - \sec 0]$$

$$= \pi [0 - 0] - \pi [-1 - 1]$$

$$\Rightarrow 2I = 2\pi$$

$$\Rightarrow I = \pi$$

Thus, $\int_0^{\pi} \frac{x}{1 + \sin x} dx = \pi$

(iii) $\int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$

Ans: Given integral – $\int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$

Say, $I = \int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx \rightarrow (1)$

Using – $\int_0^a f(x) dx = \int_0^a f(a - x) dx$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} \log \left[1 + \tan \left(\frac{\pi}{4} - x \right) \right] dx$$

$$= \int_0^{\frac{\pi}{4}} \log \left[1 + \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x} \right] dx$$

$$= \int_0^{\frac{\pi}{4}} \log \left\{ 1 + \frac{1 - \tan x}{1 + \tan x} \right\} dx$$

$$= \int_0^{\frac{\pi}{4}} \log \frac{2}{1 + \tan x} dx$$

$$= \int_0^{\frac{\pi}{4}} \log 2 dx - \int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$$

Comparing with (1)

$$= \int_0^{\frac{\pi}{4}} \log 2 dx - I$$

$$\Rightarrow 2I = \left[x \log 2 \right]_0^{\frac{\pi}{4}}$$

$$\Rightarrow 2I = \frac{\pi}{4} \log 2$$

$$\Rightarrow I = \frac{\pi}{8} \log 2$$

$$\text{Thus, } \int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx = \frac{\pi}{8} \log 2$$

$$\text{(iv) } \int_0^{\frac{\pi}{2}} \log \sin x dx$$

$$\text{Ans: Given integral} - \int_0^{\frac{\pi}{2}} \log \sin x dx$$

$$\text{Say, } I = \int_0^{\frac{\pi}{2}} \log \sin x dx \rightarrow (1)$$

$$\text{Using } - \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \log \sin \left(\frac{\pi}{2} - x \right) dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \log \cos x dx \rightarrow (2)$$

Adding equations – (1) + (2)

$$I + I = \int_0^{\frac{\pi}{2}} \log \sin x dx + \int_0^{\frac{\pi}{2}} \log \cos x dx$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} (\log \sin x + \log \cos x) dx$$

Add and subtract $\log 2$

$$= \int_0^{\frac{\pi}{2}} (\log \sin x + \log \cos x + \log 2 - \log 2) dx$$

It is known that $-\log m + \log n = \log \frac{n}{m}$

$$= \int_0^{\frac{\pi}{2}} (\log 2 \sin x \cos x - \log 2) dx$$

$$= \int_0^{\frac{\pi}{2}} (\log \sin 2x - \log 2) dx$$

$$\therefore 2I = \int_0^{\frac{\pi}{2}} \log \sin 2x dx - \int_0^{\frac{\pi}{2}} \log 2 dx$$

Say, $2I = I_1 - I_2$

With –

$$I_1 = \int_0^{\frac{\pi}{2}} \log \sin 2x dx$$

$$I_2 = \int_0^{\frac{\pi}{2}} \log 2 dx$$

Now,

$$I_1 = \int_0^{\frac{\pi}{2}} \log \sin 2x dx$$

Put $2x = t \Rightarrow 2dx = dt$

The limits –

- When $x = 0 \Rightarrow t = 0$
- When $x = \frac{\pi}{2} \Rightarrow t = \pi$

$$\Rightarrow I_1 = \frac{1}{2} \int_0^{\pi} \log \sin t dt$$

Using $\int_0^{2a} f(x)dx = 2\int_0^a f(x)dx$, when $f(2a - x) = f(x)$

Here, $\sin(\pi - t) = \sin t$

$$\Rightarrow I_1 = \frac{1}{2} \times 2 \int_0^{\frac{\pi}{2}} \log \sin t dt$$

$$\Rightarrow I_1 = \int_0^{\frac{\pi}{2}} \log \sin t dt$$

Comparing with equation (1)

$$\Rightarrow I_1 = I$$

$$\text{Now, } I_2 = \int_0^{\frac{\pi}{2}} \log 2 dx$$

$$= \log 2 \int_0^{\frac{\pi}{2}} dx$$

$$= \log 2 [x]_0^{\frac{\pi}{2}}$$

$$= \log 2 \left[\frac{\pi}{2} - 0 \right]$$

$$\Rightarrow I_2 = \frac{\pi}{2} \log 2$$

Substituting them –

$$2I = I_1 - I_2$$

$$2I = I - \frac{\pi}{2} \log 2$$

$$\Rightarrow I = -\frac{\pi}{2} \log 2$$

$$\text{Thus, } \int_0^{\frac{\pi}{2}} \log \sin x dx = -\frac{\pi}{2} \log 2$$

$$(v) \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

Ans: Given integral $-\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$

Say, $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \rightarrow (1)$

Using $-\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$$\therefore I = \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

Here, $\sin(\pi - x) = \sin x$ and $\cos(\pi - x) = -\cos x \Rightarrow \cos^2(\pi - x) = \cos^2 x$

$$= \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

Comparing with equation (1)

$$\Rightarrow I = \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - I$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

Say, $\cos x = t \Rightarrow -\sin x dx = dt$

The limits –

- When $x = 0 \Rightarrow t = 1$
- When $x = \pi \Rightarrow t = -1$

The integral becomes –

$$\Rightarrow I = -\frac{\pi}{2} \int_1^{\pi^{-1}} \frac{dt}{1+t^2}$$

Interchanging the limits,

$$\Rightarrow I = \frac{\pi}{2} \int_{-1}^1 \frac{dt}{1+t^2}$$

Using $-\int_{-a}^a f(x)dx = 2\int_0^a f(x)dx$, when $f(x)$ is even

Here, $\frac{1}{1+t^2}$ is an even function.

$$\Rightarrow I = \frac{\pi}{2} \times 2 \int_0^1 \frac{dt}{1+t^2}$$

$$= \pi \int_0^1 \frac{dt}{1+t^2}$$

$$= \pi [\tan^{-1} t]_0^1$$

$$= \pi [\tan^{-1} 1 - \tan^{-1} 0]$$

$$= \pi \left[\frac{\pi}{4} - 0 \right]$$

$$\Rightarrow I = \frac{\pi^2}{4}$$

$$\text{Thus, } \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$$

$$\text{(vi) } \int_{-2}^2 f(x)dx \text{ where } f(x) = \begin{cases} 2x - x^3 & \text{when } -2 \leq x < -1 \\ x^3 - 3x + 2 & \text{when } -1 \leq x < 1 \\ 3x - 2 & \text{when } 1 \leq x < 2 \end{cases}$$

$$\text{Ans: Given integral } - \int_{-2}^2 f(x)dx \text{ where } f(x) = \begin{cases} 2x - x^3 & \text{when } -2 \leq x < -1 \\ x^3 - 3x + 2 & \text{when } -1 \leq x < 1 \\ 3x - 2 & \text{when } 1 \leq x < 2 \end{cases}$$

Splitting based on its intervals –

$$\begin{aligned}
 \int_{-2}^2 f(x) dx &= \int_{-2}^{-1} (2x - x^3) dx + \int_{-1}^1 (x^3 - 3x + 2) dx + \int_1^2 (3x - 2) dx \\
 &= \left[2 \frac{x^2}{2} - \frac{x^4}{4} \right]_{-2}^{-1} + \left[\frac{x^4}{4} - 3 \frac{x^2}{2} + 2x \right]_{-1}^1 + \left[3 \frac{x^2}{2} - 2x \right]_1^2 \\
 &= \left[2 \times \frac{1}{2} - \frac{1}{4} - 2 \times \frac{4}{2} + \frac{16}{4} \right] + \left[\frac{1}{4} - 3 \times \frac{1}{2} + 2(1) - \frac{1}{4} + 3 \times \frac{1}{2} - 2(-1) \right] \\
 &\quad + \left[3 \times \frac{4}{2} - 2(2) - 3 \times \frac{1}{2} + 2(1) \right] \\
 &= \left[1 - \frac{1}{4} - 4 + 4 \right] + \left[\frac{1}{4} - \frac{3}{2} + 2 - \frac{1}{4} + \frac{3}{2} + 2 \right] + \left[6 - 4 - 3 \times \frac{1}{2} + 2 \right] \\
 &= \left[\frac{3}{4} \right] + [4] + \left[4 - \frac{3}{2} \right] \\
 &= \frac{3 + 16 + 16 - 6}{4} \\
 &= \frac{29}{4}
 \end{aligned}$$

Thus, $\int_{-2}^2 f(x) dx = \frac{29}{4}$

(vii) $\int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$

Ans: Given integral – $\int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$

Say, $I = \int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx \rightarrow (1)$

Using – $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$$I = \int_0^{\frac{\pi}{2}} \frac{\left(\frac{\pi}{2} - x\right) \sin\left(\frac{\pi}{2} - x\right) \cos\left(\frac{\pi}{2} - x\right)}{\sin^4\left(\frac{\pi}{2} - x\right) + \cos^4\left(\frac{\pi}{2} - x\right)} dx$$

Here, $\sin\left(\frac{\pi}{2} - x\right) = \cos x$ and $\cos\left(\frac{\pi}{2} - x\right) = \sin x$

$$\begin{aligned} \Rightarrow I &= \int_0^{\frac{\pi}{2}} \frac{\left(\frac{\pi}{2} - x\right) \cos x \sin x}{\cos^4 x + \sin^4 x} dx \\ &= \left[\int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2} \cos x \sin x}{\cos^4 x + \sin^4 x} dx \right] - \left[\int_0^{\frac{\pi}{2}} \frac{x \cos x \sin x}{\cos^4 x + \sin^4 x} dx \right] \end{aligned}$$

Comparing with equation (1)

$$\Rightarrow I = \left[\int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2} \cos x \sin x}{\cos^4 x + \sin^4 x} dx \right] - I$$

$$\Rightarrow 2I = \left[\frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{\cos x \sin x}{\cos^4 x + \sin^4 x} dx \right]$$

$$\Rightarrow I = \frac{\pi}{4} \int_0^{\frac{\pi}{2}} \frac{\cos x \sin x}{\cos^4 x + \sin^4 x} dx$$

Split the interval –

$$= \frac{\pi}{4} \left[\int_0^{\frac{\pi}{4}} \frac{\cos x \sin x}{\cos^4 x + \sin^4 x} dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos x \sin x}{\cos^4 x + \sin^4 x} dx \right]$$

Divide the numerator and denominator of the first term by $\cos^4 x$ and those of the second term by $\sin^4 x$

$$= \frac{\pi}{4} \left[\int_0^{\frac{\pi}{4}} \frac{\tan x \sec^2 x}{1 + \tan^4 x} dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cot x \operatorname{cosec}^2 x}{\cot^4 x + 1} dx \right]$$

Putting – $\tan^2 x = t \Rightarrow 2 \tan x \sec^2 x dx = dt$ in the first term, and

$$\cot^2 x = u \Rightarrow -2 \cot x \operatorname{cosec}^2 x dx = du$$

The limits –

- When $x = 0 \Rightarrow t = 0$
- When $x = \frac{\pi}{4} \Rightarrow t = 1$
- When $x = \frac{\pi}{4} \Rightarrow u = 1$
- When $x = \frac{\pi}{2} \Rightarrow u = 0$

The integral becomes –

$$= \frac{\pi}{4} \left[\frac{1}{2} \int_0^1 \frac{dt}{1+t^2} - \frac{1}{2} \int_1^0 \frac{du}{u^2+1} \right]$$

Interchanging the limits in the second term –

$$\begin{aligned} &= \frac{\pi}{4} \left[\frac{1}{2} \int_0^1 \frac{dt}{1+t^2} + \frac{1}{2} \int_0^1 \frac{du}{u^2+1} \right] \\ &= \frac{\pi}{4} \left[\frac{1}{2} (\tan^{-1} t)_0^1 + \frac{1}{2} (\tan^{-1} u)_0^1 \right] \\ &= \frac{\pi}{4} \left[\frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0) + \frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0) \right] \\ &= \frac{\pi}{4} \left[\frac{1}{2} \left(\frac{\pi}{4} - 0 \right) + \frac{1}{2} \left(\frac{\pi}{4} - 0 \right) \right] \\ &= \frac{\pi}{4} \times \frac{\pi}{4} \\ \therefore I &= \frac{\pi^2}{8} \end{aligned}$$

Thus, $\int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx = \frac{\pi^2}{8}$

(viii) $\int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$

Ans: Given integral $-\int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$

Say, $I = \int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx \rightarrow (1)$

Using $-\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$I = \int_0^{\pi} \frac{(\pi-x)}{a^2 \cos^2(\pi-x) + b^2 \sin^2(\pi-x)} dx$

Here, $\sin(\pi-x) = \sin x \Rightarrow \sin^2(\pi-x) = \sin^2 x$ and

$\cos(\pi-x) = -\cos x \Rightarrow \cos^2(\pi-x) = \cos^2 x$

$= \int_0^{\pi} \frac{(\pi-x)}{a^2 \cos^2 x + b^2 \sin^2 x} dx$

$= \int_0^{\pi} \frac{\pi}{a^2 \cos^2 x + b^2 \sin^2 x} dx - \int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$

Comparing with equation (1)

$\Rightarrow I = \int_0^{\pi} \frac{\pi}{a^2 \cos^2 x + b^2 \sin^2 x} dx - I$

$\Rightarrow 2I = \pi \int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$

$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$

Using $-\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$, when $f(2a-x) = f(x)$

$$\Rightarrow I = \frac{\pi}{2} \times 2 \int_0^{\frac{\pi}{2}} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$$

$$= \pi \int_0^{\frac{\pi}{2}} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$$

Split the intervals –

$$= \pi \left[\int_0^{\frac{\pi}{4}} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} \right]$$

Divide the numerator and denominator of the first term by $\cos^2 x$ and those of the second term by $\sin^2 x$

$$= \pi \left[\int_0^{\frac{\pi}{4}} \frac{\sec^2 x dx}{a^2 + b^2 \tan^2 x} + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\operatorname{cosec}^2 x dx}{a^2 \cot^2 x + b^2} \right]$$

Putting – $\tan x = t \Rightarrow \sec^2 x dx = dt$ in the first term, and

$\cot x = u \Rightarrow -\operatorname{cosec}^2 x dx = du$

The limits –

- When $x = 0 \Rightarrow t = 0$
- When $x = \frac{\pi}{4} \Rightarrow t = 1$
- When $x = \frac{\pi}{4} \Rightarrow u = 1$
- When $x = \frac{\pi}{2} \Rightarrow u = 0$

The integral becomes –

$$= \pi \left[\int_0^1 \frac{dt}{a^2 + b^2 t^2} - \int_1^0 \frac{du}{a^2 u^2 + b^2} \right]$$

Interchanging the limits in the second term –

$$= \pi \left[\int_0^1 \frac{dt}{a^2 + b^2 t^2} + \int_0^1 \frac{du}{a^2 u^2 + b^2} \right]$$

$$\begin{aligned}
 &= \pi \left[\frac{1}{ab} \left(\tan^{-1} \frac{bt}{a} \right)_0^1 + \frac{1}{ab} \left(\tan^{-1} \frac{au}{b} \right)_0^1 \right] \\
 &= \pi \left[\frac{1}{ab} \left(\tan^{-1} \frac{b}{a} - \tan^{-1} 0 \right) + \frac{1}{ab} \left(\tan^{-1} \frac{a}{b} - \tan^{-1} 0 \right) \right] \\
 &= \frac{\pi}{ab} \left[\left(\tan^{-1} \frac{b}{a} \right) + \left(\tan^{-1} \frac{a}{b} \right) \right] \\
 &= \frac{\pi}{ab} \left[\frac{\pi}{2} \right] \\
 \Rightarrow I &= \frac{\pi^2}{2ab}
 \end{aligned}$$

Thus, $\int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx = \frac{\pi^2}{2ab}$

47. Evaluate the following integrals

(i) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \sqrt{\tan x}}$

Ans: The given integral can be written as

$$\begin{aligned}
 I &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \sqrt{\tan x}} \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \sqrt{\frac{\sin x}{\cos x}}} \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x} \, dx}{\sqrt{\cos x} + \sqrt{\sin x}}
 \end{aligned}$$

That is, $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x} \, dx}{\sqrt{\cos x} + \sqrt{\sin x}} \dots\dots (i)$

Now, recall that, since, $\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx$.

So,

$$I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos\left(\frac{\pi}{6} + \frac{\pi}{3} - x\right)}}{\sqrt{\cos\left(\frac{\pi}{6} + \frac{\pi}{3} - x\right)} + \sqrt{\sin\left(\frac{\pi}{6} + \frac{\pi}{3} - x\right)}} dx$$

Again, it is known that, $\sin\left(\frac{\pi}{2} - x\right) = \cos x$ and $\cos\left(\frac{\pi}{2} - x\right) = \sin x$.

Therefore,

$$I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \dots\dots (ii)$$

Now, adding the equations (i) and (ii), we get

$$2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x} \, dx}{\sqrt{\cos x} + \sqrt{\sin x}} + \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sin x} \, dx}{\sqrt{\sin x} + \sqrt{\cos x}}$$

$$\Rightarrow 2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x} + \sqrt{\sin x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$$

$$\Rightarrow 2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1 \cdot dx$$

$$\Rightarrow 2I = \left[x \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

$$\Rightarrow I = \frac{\pi}{12}$$

Therefore,

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \sqrt{\tan x}} = \frac{\pi}{12}.$$

(ii) $\int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$

Ans: Let $x = \tan \theta$.

Then, $dx = \sec^2 \theta d\theta$

Also, $x \rightarrow 0 \Rightarrow \theta \rightarrow 0$ and

$$x \rightarrow 1 \Rightarrow \theta \rightarrow \frac{\pi}{4}.$$

Therefore, substituting the values, we get

$$\begin{aligned} \int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx &= \int_0^{\frac{\pi}{4}} \sin^{-1}\left(\frac{2 \tan \theta}{1 + \tan^2 \theta}\right) \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} \sin^{-1}(\sin 2\theta) \sec^2 \theta d\theta \quad \left[\text{since, } \sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}\right] \\ &= \int_0^{\frac{\pi}{4}} 2\theta \sec^2 \theta d\theta \\ &= 2 \int_0^{\frac{\pi}{4}} \theta \sec^2 \theta d\theta \\ &= 2 \left[\left[\theta \cdot \tan \theta \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} 1 \cdot \tan \theta d\theta \right] \quad [\text{applying integration by parts}] \\ &= 2 \left[\frac{\pi}{4} - \left[\ln |\sec \theta| \right]_0^{\frac{\pi}{4}} \right] \\ &= \frac{\pi}{2} - 2 \left[\ln(\sqrt{2}) - 0 \right] \end{aligned}$$

$$= \frac{\pi}{2} - \ln 2.$$

$$\text{Thus, } \int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx = \frac{\pi}{2} - \ln 2.$$

$$\text{(iii) } \int_{-1}^1 \log\left(\frac{1+\sin x}{1-\sin x}\right) dx$$

Ans: Recall that, if $f(x)$ is an odd function and continuous at the closed interval $[-a, a]$, then $\int_{-a}^a f(x) dx = 0$.

Let $f(x) = \log\left(\frac{1+\sin x}{1-\sin x}\right)$, which is continuous on $[-a, a]$.

$$\text{Then, } f(-x) = \log\left(\frac{1-\sin x}{1+\sin x}\right).$$

$$\text{Now, } f(x) + f(-x) = \log\left(\frac{1+\sin x}{1-\sin x}\right) + \log\left(\frac{1-\sin x}{1+\sin x}\right)$$

$$= \log\left(\frac{1+\sin x}{1-\sin x} \cdot \frac{1-\sin x}{1+\sin x}\right)$$

$$= \log(1)$$

$$= 0.$$

Therefore, the function $f(x) = \log\left(\frac{1+\sin x}{1-\sin x}\right)$ is an odd function.

$$\text{Hence, } \int_{-1}^1 \log\left(\frac{1+\sin x}{1-\sin x}\right) dx = 0.$$

$$\text{(iv) } \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}} dx$$

Ans: The given integral can be written as

$$I = \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}} dx \quad \dots\dots (i)$$

Now, recall that, $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$$\text{Then, } I = \int_0^{\pi} \frac{e^{\cos(\pi-x)}}{e^{\cos(\pi-x)} + e^{-\cos(\pi-x)}} dx$$

$$\Rightarrow I = \int_0^{\pi} \frac{e^{-\cos x}}{e^{-\cos x} + e^{\cos x}} dx \quad [\text{since } \cos(\pi-x) = -\cos x] \quad \dots\dots (ii)$$

Add the equations (i) and (ii). Then, we have

$$2I = \int_0^{\pi} \frac{e^{\cos x} + e^{-\cos x}}{e^{\cos x} + e^{-\cos x}} dx$$

$$\Rightarrow 2I = \int_0^{\pi} 1 \cdot dx = [x]_0^{\pi} = \pi$$

$$\Rightarrow I = \frac{\pi}{2}$$

$$\text{Thus, } \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}} dx = \frac{\pi}{2}.$$

$$(v) \int_0^{\pi} \frac{x \tan x}{\sec x \cdot \operatorname{cosec} x} dx$$

Ans: The given integral can be written as

$$I = \int_0^{\pi} \frac{x \tan x}{\sec x \cdot \operatorname{cosec} x} dx \quad \dots\dots (1)$$

Now, recall that, $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.

$$\text{Then, } I = \int_0^{\pi} \frac{(\pi-x) \tan(\pi-x)}{\sec(\pi-x) \cdot \operatorname{cosec}(\pi-x)} dx$$

It is known that, $\tan(\pi-x) = -\tan x$, $\operatorname{cosec}(\pi-x) = \operatorname{cosec} x$, and

$$\sec(\pi - x) = \sec x.$$

$$\text{Therefore, } I = \int_0^{\pi} \frac{(\pi - x) \tan x}{\sec x \cdot \operatorname{cosec} x} dx \dots\dots (2)$$

Adding the equations (1) and (2), gives

$$2I = \pi \int_0^{\pi} \frac{\tan x}{\sec x \cdot \operatorname{cosec} x} dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \sin^2 x dx$$

$$\Rightarrow 2I = \frac{2\pi}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2x) dx$$

$$\Rightarrow 2I = \pi \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$\Rightarrow 2I = \pi \left[\frac{\pi}{2} - \frac{1}{2} \sin \frac{2\pi}{2} - 0 \right]$$

$$\Rightarrow 2I = \frac{\pi^2}{2}$$

$$\Rightarrow I = \frac{\pi^2}{4}.$$

$$\text{Thus, } \int_0^{\pi} \frac{x \tan x}{\sec x \cdot \operatorname{cosec} x} dx = \frac{\pi^2}{4}.$$

$$\text{(vi) } \int_{-a}^a \sqrt{\frac{a-x}{a+x}} dx$$

Ans: Let $x = a \cos \theta$

Then, $dx = -a \sin \theta d\theta$

Also, $x \rightarrow -a \Rightarrow \theta \rightarrow \pi$

and $x \rightarrow a \Rightarrow \theta \rightarrow 0$.

Therefore, substituting the values into the given integral, we have

$$\begin{aligned}
 \int_{-a}^a \sqrt{\frac{a-x}{a+x}} dx &= -a \int_{\pi}^0 \sqrt{\frac{a-a\cos\theta}{a+a\cos\theta}} \cdot \sin\theta d\theta \\
 &= a \int_0^{\pi} \sqrt{\frac{1-\cos\theta}{1+\cos\theta}} \cdot \sin\theta d\theta \\
 &= a \int_0^{\pi} \sqrt{\frac{2\sin^2\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}} \cdot \sin\theta d\theta \\
 &= a \int_0^{\pi} \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}} \cdot 2\sin\frac{\theta}{2} \cos\frac{\theta}{2} d\theta \\
 &= a \int_0^{\pi} (1-\cos\theta) d\theta \quad [\text{since, } 1-\cos\theta = 2\sin^2\frac{\theta}{2}] \\
 &= a[\theta - \sin\theta]_0^{\pi} \\
 &= a[\pi - \sin\pi - 0] \\
 &= a\pi.
 \end{aligned}$$

Thus, $\int_{-a}^a \sqrt{\frac{a-x}{a+x}} dx = a\pi.$

48. $\int_0^1 [2x] dx$ where $[]$ is the greatest integer function.

Ans: First, let $2x = z.$

Then, $2dx = dz$

$$\Rightarrow dx = \frac{dz}{2}.$$

Also, $x \rightarrow 0 \Rightarrow z \rightarrow 0$ and

$x \rightarrow 1 \Rightarrow z \rightarrow 2.$

Therefore, substituting the values we get,

$$\int_0^1 [2x] dx = \frac{1}{2} \int_0^2 [z] dz \quad \dots\dots (i)$$

Now, note that, by the definition of greatest integer function,

$$[z] = \begin{cases} 0, & \text{when } 0 \leq z < 1 \\ 1, & \text{when } 1 \leq z < 2 \end{cases}$$

So, from the equation (i), we have

$$\begin{aligned} \frac{1}{2} \int_0^2 [z] dz &= \frac{1}{2} \left[\int_0^1 0 dz + \int_1^2 1 \cdot dz \right] \\ &= \frac{1}{2} [z]_1^2 \\ &= \frac{1}{2} (2 - 1) \\ &= \frac{1}{2} \end{aligned}$$

$$\text{Hence, } \int_0^1 [2x] dx = \frac{1}{2}.$$

49. $\int e^{\log x + \log \sin x} dx$

Ans: The given integral can be written as

$$\int e^{\log x + \log \sin x} dx = \int e^{\log(x \cdot \sin x)} dx \quad [\text{since, } \log(a) + \log(b) = \log(ab)]$$

$$= \int e^{\log_e(x \sin x)} dx$$

$$= \int x \sin x dx \quad [\text{since, } e^{\log_e x} = x]$$

$$= x(-\cos x) - \int 1 \cdot (-\cos x) dx + C \quad [\text{applying integration by parts}]$$

$$= -x \cos x + \sin x + C, \text{ where } C \text{ is an integration constant.}$$

$$\text{Thus, } \int e^{\log x + \log \sin x} dx = -x \cos x + \sin x + C, \text{ where } C \text{ is an integration constant.}$$

50. $\int e^{\log(x+1)-\log x} dx$

Ans: The given function can be written as

$$\int e^{\log(x+1)-\log x} dx = \int e^{\log\left(\frac{x+1}{x}\right)} dx \quad \left[\text{since, } \log(a) - \log(b) = \log\left(\frac{a}{b}\right)\right]$$

$$= \int e^{\log_e\left(\frac{x+1}{x}\right)} dx$$

$$= \int \frac{x+1}{x} dx \quad [\text{since, } e^{\log_e x} = x]$$

$$= \int \left(1 + \frac{1}{x}\right) dx$$

$= x + \log x + C$, where C is an integration constant.

Thus, $\int e^{\log(x+1)-\log x} dx = x + \log x + C$, where C is an integration constant.

51. $\int \frac{\sin x}{\sin 2x} dx$

Ans: Recall that, $\sin 2x = 2 \sin x \cos x$.

Then, we have

$$\int \frac{\sin x}{\sin 2x} dx = \int \frac{\cancel{\sin x}}{2 \cancel{\sin x} \cos x} dx$$

$$= \frac{1}{2} \int \sec x dx$$

$$= \frac{1}{2} \log|\sec x + \tan x| + C, \text{ where } C \text{ is an integration constant.}$$

$$\int \frac{\sin x}{\sin 2x} dx = \frac{1}{2} \log|\sec x + \tan x| + C, \text{ where } C \text{ is an integration constant.}$$

52. $\int \sin x \sin 2x dx$

Ans: Recall that, $\sin 2x = 2 \sin x \cos x$.

$$\text{Then, } \int \sin x \sin 2x \, dx = \int \sin x \cdot (2 \sin x \cos x) \, dx$$

$$= 2 \int \sin^2 x \, d(\sin x)$$

$$= \frac{2}{3} \sin^3 x + C, \text{ where } C \text{ is an integration constant.}$$

$$\text{Hence, } \int \sin x \sin 2x \, dx = \frac{2}{3} \sin^3 x + C, \text{ where } C \text{ is an integration constant.}$$

$$53. \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} |\sin x| \, dx$$

Ans: Observe that, the given function is an even function, since $|\sin(-x)| = |\sin x| = \sin x$.

Therefore, the given integral can be written as

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} |\sin x| \, dx = 2 \int_0^{\frac{\pi}{4}} |\sin x| \, dx \quad \left[\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx, \text{ when } f(x) \text{ is even} \right]$$

Now, since, $0 \leq x \leq \frac{\pi}{4}$, so $|\sin x| = \sin x$.

$$\text{Therefore, } \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} |\sin x| \, dx = 2 \int_0^{\frac{\pi}{4}} \sin x \, dx$$

$$= -2 [\cos x]_0^{\frac{\pi}{4}}$$

$$= -2 \left(\cos \frac{\pi}{4} - \cos 0 \right)$$

$$= -2 \left(\frac{1}{\sqrt{2}} - 1 \right)$$

$$= 2 - \sqrt{2}$$

Thus, $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} |\sin x| dx = 2 - \sqrt{2}.$

54. $\int_a^b f(x) dx + \int_b^a f(a+b-x) dx$

Ans: The given integral is $\int_a^b f(x) dx + \int_b^a f(a+b-x) dx.$

Now, recall that, $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx.$

Therefore, we have

$$\begin{aligned} \int_a^b f(x) dx + \int_a^b f(a+b-x) dx &= \int_a^b f(a+b-x) dx + \int_a^b f(a+b-x) dx \\ &= \int_a^b f(a+b-x) dx - \int_a^b f(a+b-x) dx \quad \left[\text{since, } \int_a^b f(x) dx = -\int_b^a f(x) dx \right] \\ &= 0 \end{aligned}$$

Hence, $\int_a^b f(x) + \int_a^b f(a+b-x) dx = 0.$

55. $\int \frac{1}{\sec x + \tan x} dx$

Ans: The given integral can be written as

$$\begin{aligned} \int \frac{1}{\sec x + \tan x} dx &= \int \frac{dx}{\frac{1}{\cos x} + \frac{\sin x}{\cos x}} \\ &= \int \frac{\cos x dx}{1 + \sin x} \end{aligned}$$

Let $1 + \sin x = z.$

Then, $\cos x dx = dz$

Therefore, we have

$$\int \frac{\cos x \, dx}{1 + \sin x} = \int \frac{dz}{z}$$

= $\log z + C$, where C is an integration constant.

$$= \log(1 + \sin x) + C.$$

Thus, $\int \frac{1}{\sec x + \tan x} \, dx = \log(1 + \sin x) + C$, where C is an integration constant.

56. $\int \frac{\sin^2 x}{1 + \cos x} \, dx$

Ans: The given equation can be written as

$$\int \frac{\sin^2 x}{1 + \cos x} \, dx = \int \frac{1 - \cos^2 x}{1 + \cos x} \, dx \quad [\text{since, } \sin^2 x + \cos^2 x = 1]$$

$$= \int \frac{(1 + \cos x)(1 - \cos x)}{1 + \cos x} \, dx \quad [\text{since, } a^2 - b^2 = (a + b)(a - b)]$$

$$= \int (1 - \cos x) \, dx$$

= $x - \sin x + C$, where C is an integration constant.

Thus, $\int \frac{\sin^2 x}{1 + \cos x} \, dx = x - \sin x + C$, where C is an integration constant.

57. $\int \frac{1 - \tan x}{1 + \tan x} \, dx$

Ans: The given integral can be written as

$$\int \frac{1 - \tan x}{1 + \tan x} \, dx = \int \frac{1 - \frac{\sin x}{\cos x}}{1 + \frac{\sin x}{\cos x}} \, dx$$

$$= \int \frac{\cos x - \sin x}{\cos x + \sin x} \, dx$$

Now, suppose that, $\cos x + \sin x = z$.

Then, $(-\sin x + \cos x)dx = dz$

Therefore, substituting the values, we get

$$\int \frac{\cos x - \sin x}{\cos x + \sin x} dx = \int \frac{dz}{z}$$

$= \log z + C$, where C is an integration constant.

$= \log(\cos x + \sin x) + C$, substituting the value of z .

Hence, $\int \frac{1 - \tan x}{1 + \tan x} dx = \log(\cos x + \sin x) + C$, where C is an integration constant.

58. $\int \frac{a^x + b^x}{c^x} dx$

Ans: The given integral can be written as

$$\begin{aligned} \int \frac{a^x + b^x}{c^x} dx &= \int \left(\frac{a^x}{c^x} + \frac{b^x}{c^x} \right) dx \\ &= \int \left(\frac{a}{c} \right)^x dx + \int \left(\frac{b}{c} \right)^x dx \\ &= \frac{\left(\frac{a}{c} \right)^x}{\log_e \left(\frac{a}{c} \right)} + \frac{\left(\frac{b}{c} \right)^x}{\log_e \left(\frac{b}{c} \right)} + C, \text{ where } C \text{ is an integration constant.} \end{aligned}$$

$$\text{Thus, } \int \frac{a^x + b^x}{c^x} dx = \frac{\left(\frac{a}{c} \right)^x}{\log_e \left(\frac{a}{c} \right)} + \frac{\left(\frac{b}{c} \right)^x}{\log_e \left(\frac{b}{c} \right)} + C, \text{ where } C \text{ is an integration constant.}$$

59. Evaluate

(i) $\int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx, x \in [0, 1]$

Ans: Recall that, $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

Then, the given integral can be written as

$$\begin{aligned} \int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx &= \int \frac{\sin^{-1} \sqrt{x} - \left(\frac{\pi}{2} - \sin^{-1} \sqrt{x} \right)}{\frac{\pi}{2}} \\ &= \frac{2}{\pi} \int \left(2 \sin^{-1} \sqrt{x} - \frac{\pi}{2} \right) dx \\ &= \int \left(\frac{4}{\pi} \sin^{-1} \sqrt{x} - 1 \right) dx \\ &= \frac{4}{\pi} I - x + C_1, \text{ where } C_1 \text{ is an integration constant and} \end{aligned}$$

$$I = \int \sin^{-1} \sqrt{x} dx.$$

Let $x = z^2$. Then, we get

$$dx = 2z dz$$

Therefore, we have

$$\begin{aligned} I &= 2 \int \sin^{-1} z \cdot z dz \\ &= 2 \left[\sin^{-1}(z) \cdot \frac{z^2}{2} - \int \frac{1}{\sqrt{1-z^2}} \cdot \frac{z^2}{2} dz \right] + C_2 \quad [\text{applying integration by parts}] \\ &= z^2 \sin^{-1} z - \int \frac{z^2}{\sqrt{1-z^2}} dz + C_2 \\ &= z^2 \sin^{-1} z + \int \frac{1-z^2-1}{\sqrt{1-z^2}} dz + C_2 \\ &= z^2 \sin^{-1} z + \int \left(\sqrt{1-z^2} - \frac{1}{\sqrt{1-z^2}} \right) dz + C_2 \end{aligned}$$

Now, recall that, $\int \sqrt{a^2 - x^2} = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$ and

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C.$$

Therefore,

$$I = z^2 \sin^{-1} z + \frac{z}{2} \sqrt{1 - z^2} + \frac{1}{2} \sin^{-1} z - \sin^{-1} z + C_2$$

$$\Rightarrow I = z^2 \sin^{-1} z + \frac{z}{2} \sqrt{1 - z^2} - \frac{1}{2} \sin^{-1} z + C_2$$

Thus,

$$\begin{aligned} \int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx &= \frac{4}{\pi} I - x + C_1 \\ &= \frac{4}{\pi} \left(z^2 \sin^{-1} z + \frac{z}{2} \sqrt{1 - z^2} - \frac{1}{2} \sin^{-1} z + C_2 \right) - x + C_2 \\ &= \frac{4}{\pi} \left(x \sin^{-1} \sqrt{x} + \frac{\sqrt{x - x^2}}{2} - \frac{1}{2} \sin^{-1} \sqrt{x} \right) - x + C, \text{ substituting the value of } z. \end{aligned}$$

$$\text{Hence, } \int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx = \sin^{-1} \sqrt{x} \left[\frac{2(2x - 1)}{\pi} \right] + \frac{2\sqrt{x - x^2}}{\pi} - x + C, \text{ where}$$

C is an integration constant.

$$(ii) \int \sqrt{\frac{1 - \sqrt{x}}{1 + \sqrt{x}}} dx$$

Ans: Let $x = \cos^2 \theta$

Then, $dx = -2 \cos \theta \sin \theta$.

Therefore, substituting the values we get,

$$\int \sqrt{\frac{1 - \sqrt{x}}{1 + \sqrt{x}}} dx = -2 \int \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \cdot \cos \theta \sin \theta d\theta$$

Now, recall that, $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$ and $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$.

So, we have

$$-2 \int \sqrt{\frac{1-\cos\theta}{1+\cos\theta}} \cdot \cos\theta \sin\theta d\theta = -2 \int \frac{\sqrt{2} \sin \frac{\theta}{2}}{\sqrt{2} \cos \frac{\theta}{2}} \cdot \cos\theta \left(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right) d\theta$$

$$= -2 \int 2 \sin^2 \frac{\theta}{2} \cdot \cos\theta d\theta$$

$$= -2 \int (1 - \cos\theta) \cdot \cos\theta d\theta$$

$$= -2 \int (\cos\theta - \cos^2\theta) d\theta$$

$$= -2 \sin\theta + \int (1 + \cos 2\theta) d\theta + C$$

$$= -2 \sin\theta + \theta + \frac{\sin 2\theta}{2} + C, \text{ where } C \text{ is an integration constant.}$$

$$= -2\sqrt{1-x} + \cos^{-1}\sqrt{x} + \frac{2\sqrt{1-x} \cdot \sqrt{x}}{2} + C$$

$$= \cos^{-1}\sqrt{x} - 2\sqrt{1-x} + \sqrt{x-x^2} + C.$$

Thus, $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx = \cos^{-1}\sqrt{x} - 2\sqrt{1-x} + \sqrt{x-x^2} + C$, where C is an integration constant.

(iii) $\int \frac{\sqrt{x^2+1} [\log(x^2+1) - 2\log x]}{x^4} dx$

Ans: The given integral can be written as

$$\int \frac{\sqrt{x^2+1} [\log(x^2+1) - 2\log x]}{x^4} dx = \int \sqrt{1 + \frac{1}{x^2}} \left[\log\left(1 + \frac{1}{x^2}\right) \right] \cdot \frac{1}{x^3} dx$$

Now, suppose that, $1 + \frac{1}{x^2} = z^2$

Then, $-\frac{2}{x^3} dx = 2z dz$

$$\Rightarrow \frac{1}{x^3} dx = -z dz$$

Therefore, substituting the values we get,

$$\begin{aligned} \int \sqrt{1 + \frac{1}{x^2}} \left[\log \left(1 + \frac{1}{x^2} \right) \right] \cdot \frac{1}{x^3} dx &= - \int z (2 \log z) z dz \\ &= -2 \int \log z \cdot z^2 dz \\ &= -2 \log z \cdot \frac{z^3}{3} + 2 \int \frac{1}{z} \cdot \frac{z^3}{3} dz \quad [\text{applying integration by parts}] \\ &= -\frac{2}{3} z^3 \log z + \frac{2}{9} z^3 + C \\ &= \frac{2}{3} \left(1 + \frac{1}{x^2} \right)^3 \left[\frac{1}{3} - \log \left(1 + \frac{1}{x^2} \right) \right] + C, \text{ substituting the value of } z. \end{aligned}$$

$$\text{Hence, } \int \frac{\sqrt{x^2+1} [\log(x^2+1) - 2\log x]}{x^4} dx = \frac{2}{3} \left(1 + \frac{1}{x^2} \right)^3 \left[\frac{1}{3} - \log \left(1 + \frac{1}{x^2} \right) \right] + C,$$

where C is an integration constant.

$$\text{(iv) } \int \frac{x^2}{(x \sin x + \cos x)^2} dx$$

Ans: The given integral can be written as

$$\int \frac{x^2}{(x \sin x + \cos x)^2} dx = \int \frac{x}{\cos x} \cdot \frac{x \cos x}{(x \sin x + \cos x)^2} dx \quad \dots\dots (i)$$

Now, note that, $\frac{d}{dx}(x \sin x + \cos x) = x \cos x + \sin x - \sin x = x \cos x$

Now, applying integration by parts on the equation (i), we have

$$\begin{aligned} &\int \frac{x}{\cos x} \cdot \frac{x \cos x}{(x \sin x + \cos x)^2} dx \\ &= \frac{x}{\cos x} \cdot \left(-\frac{1}{x \sin x + \cos x} \right) - \int \frac{\cos x + x \sin x}{\cos^2 x} \cdot \frac{-1}{x \sin x + \cos x} dx \end{aligned}$$

$$= -\frac{x}{\cos x(x \sin x + \cos x)} + \int \sec^2 x dx$$

$$= \frac{-x}{\cos x(x \sin x + \cos x)} + \tan x + C, \text{ where } C \text{ is an integration constant.}$$

$$= \frac{-x}{\cos x(x \sin x + \cos x)} + \frac{\sin x}{\cos x} + C$$

$$= \frac{-x + \sin x(x \sin x + \cos x)}{\cos x(x \sin x + \cos x)} + C$$

$$= \frac{\sin x - x \cos x}{\cos x + x \sin x} + C \quad [\text{simplify}]$$

Thus,

$$\int \frac{x^2}{(x \sin x + \cos x)^2} dx = \frac{\sin x - x \cos x}{\cos x + x \sin x} + C, \text{ where } C \text{ is an integration constant.}$$

$$(v) \int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$$

Ans: Let $x = a \tan^2 \theta$

Then, $dx = 2a \tan \theta \sec^2 \theta d\theta$

Therefore, substituting the values we get,

$$\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx = \int \sin^{-1} \sqrt{\frac{a \tan^2 \theta}{a(1+\tan^2 \theta)}} (2a \tan \theta \sec^2 \theta) d\theta$$

$$= 2a \int \sin^{-1} \left(\frac{\tan \theta}{\sec \theta} \right) \tan \theta \sec^2 \theta d\theta$$

$$= 2a \int \sin^{-1} (\sin \theta) \tan \theta \sec^2 \theta d\theta$$

$$= 2a \int \theta \tan \theta \sec^2 \theta d\theta$$

$$= 2a \int \theta \tan \theta d(\tan \theta)$$

$$= 2a \left[\theta \cdot \frac{\tan^2 \theta}{2} - \int 1 \cdot \frac{\tan^2 \theta}{2} d\theta \right] \quad [\text{applying integration by parts}]$$

$$= a\theta \tan^2 \theta - a \int (\sec^2 \theta - 1) d\theta$$

$$= a\theta \tan^2 \theta - a \tan \theta + a\theta + C, \text{ where } C \text{ is an integration constant.}$$

$$= a \tan^{-1} \sqrt{\frac{x}{a}} \cdot \frac{x}{a} - a \cdot \sqrt{\frac{x}{a}} + a \tan^{-1} \sqrt{\frac{x}{a}} + C, \text{ substituting the value of } \theta.$$

$$= (x + a) \tan^{-1} \sqrt{\frac{x}{a}} - \sqrt{ax} + C$$

Hence, $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx = (x + a) \tan^{-1} \sqrt{\frac{x}{a}} - \sqrt{ax} + C$, where C is an integration constant.

$$(vi) \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx$$

Ans: The given integral can be written as

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{1 - (1 - \sin 2x)}} dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{1 - (\sin^2 x + \cos^2 x - 2 \sin x \cos x)}} dx \quad [\text{since, } \sin 2x = 2 \sin x \cos x \text{ and } \sin^2 x + \cos^2 x = 1]$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} dx$$

Now, note that, $\frac{d}{dx}(\sin x - \cos x) = \cos x + \sin x$

So, we have

$$\begin{aligned} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} dx &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{d(\sin x - \cos x)}{\sqrt{1 - (\sin x - \cos x)^2}} \\ &= \left[\sin^{-1}(\sin x - \cos x) \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \quad \left[\text{since, } \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C \right] \\ &= \sin^{-1} \left(\sin \frac{\pi}{3} - \cos \frac{\pi}{3} \right) - \sin^{-1} \left(\sin \frac{\pi}{6} - \cos \frac{\pi}{6} \right) \\ &= \sin^{-1} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right) + \sin^{-1} \left(\frac{\sqrt{3}}{3} - \frac{1}{2} \right) \\ &= 2 \sin^{-1} \frac{1}{2} (\sqrt{3} - 1). \end{aligned}$$

Hence, $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = 2 \sin^{-1} \frac{1}{2} (\sqrt{3} - 1).$

(vii) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin |x| - \cos |x|) dx$

Ans: Note that, $|x| = \begin{cases} -x, & \text{when } -\frac{\pi}{2} \leq x \leq 0 \\ x, & \text{when } 0 \leq x \leq \frac{\pi}{2} \end{cases}$

Therefore, the given integral can be written as

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin |x| - \cos |x|) dx &= \int_{-\frac{\pi}{2}}^0 [\sin(-x) - \cos(-x)] dx + \int_0^{\frac{\pi}{2}} [\sin x - \cos x] dx \\ &= \int_{-\frac{\pi}{2}}^0 (-\sin x - \cos x) dx + [-\cos x - \sin x]_0^{\frac{\pi}{2}} \end{aligned}$$

$$= -[-\cos x + \sin x]_{-\frac{\pi}{2}}^0 - [0 + 1 - 1]$$

$$= -[-1 + 0 + 0 + 1] - 0$$

$$= 0$$

$$\text{Hence, } \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin|x| - \cos|x|) dx = 0.$$

(viii) $\int_1^2 [x^2] dx$, where $[x]$ is greatest integer function.

Ans: Suppose that, $x^2 = z$

Then, $2x dx = dz$

$$\Rightarrow dx = \frac{1}{2} \frac{dz}{\sqrt{z}}$$

Also, $x \rightarrow 1 \Rightarrow z \rightarrow 1$ and

$$x \rightarrow 2 \Rightarrow z \rightarrow 4.$$

Therefore, the given integral can be written as

$$\begin{aligned} \int_1^2 [x^2] dx &= \frac{1}{2} \int_1^4 \frac{[z] dz}{\sqrt{z}} \\ \int_1^2 [x^2] dx &= \frac{1}{2} \left[\int_1^2 \frac{[z] dz}{\sqrt{z}} + \int_2^3 \frac{[z] dz}{\sqrt{z}} + \int_3^4 \frac{[z] dz}{\sqrt{z}} \right] \quad \dots\dots (i) \end{aligned}$$

Now, recall the definition of greatest integer function.

$$[z] = \begin{cases} 1, & \text{when } 1 \leq z < 2 \\ 2, & \text{when } 2 \leq z < 3 \\ 3, & \text{when } 3 \leq z < 4 \end{cases}$$

Therefore, from the equation (i) we get,

$$\int_1^2 [x^2] dx = \frac{1}{2} \left[\int_1^2 \frac{1 \cdot dz}{\sqrt{z}} + \int_2^3 \frac{2}{\sqrt{z}} dz + \int_3^4 \frac{3}{\sqrt{z}} dz \right]$$

$$\begin{aligned}
 &= \frac{1}{2} \left[\left[2\sqrt{z} \right]_1^2 + 4 \left[\sqrt{z} \right]_2^3 + 6 \left[\sqrt{z} \right]_3^4 \right] \\
 &= \frac{1}{2} \left[2\sqrt{2} - 2 + 4\sqrt{3} - 4\sqrt{2} + 12 - 6\sqrt{3} \right] \\
 &= \frac{1}{2} \left[10 - 2\sqrt{2} - 2\sqrt{3} \right] \\
 &= 5 - \sqrt{2} - \sqrt{3}
 \end{aligned}$$

Hence, $\int_1^2 [x^2] dx = 5 - \sqrt{2} - \sqrt{3}.$

(ix) $\int_{-1}^{\frac{3}{2}} |x \sin \pi x| dx$

Ans: The given integral can be written as

$$\int_{-1}^{\frac{3}{2}} |x \sin \pi x| dx = \int_{-1}^1 |x \sin \pi x| dx + \int_1^{\frac{3}{2}} |x \sin \pi x| dx \quad \dots\dots (i)$$

Now, note that $|x| = -x$ and $|\sin \pi x| = -\sin \pi x$ for $-1 \leq x \leq 1$.

That is, $|x \sin \pi x| = (-x)(-\sin \pi x) = x \sin \pi x$ when $-1 \leq x \leq 1$.

Also, $|x| = x$ and $|\sin \pi x| = -\sin \pi x$ for $1 \leq x \leq \frac{3}{2}$.

That is, $|x \sin \pi x| = -x \sin \pi x$ when $1 \leq x \leq \frac{3}{2}$.

Therefore, from equation (i), we have

$$\int_{-1}^{\frac{3}{2}} |x \sin \pi x| dx = \int_{-1}^1 x \sin \pi x dx - \int_1^{\frac{3}{2}} x \sin \pi x dx \quad \dots\dots (ii)$$

Now, $\int x \sin \pi x dx = -\frac{x \cos \pi x}{\pi} + \frac{1}{\pi} \int 1 \cdot \cos \pi x dx$

$$\int x \sin \pi x dx = -\frac{x \cos \pi x}{\pi} + \frac{\sin \pi x}{\pi^2} + C.$$

Thus, the equation (ii) becomes

$$\begin{aligned}\int_{-1}^{\frac{3}{2}} |x \sin \pi x| dx &= \left[-\frac{x \cos \pi x}{\pi} + \frac{\sin \pi x}{\pi^2} \right]_{-1}^1 + \left[-\frac{x \cos \pi x}{\pi} + \frac{\sin \pi x}{\pi^2} \right]_{\frac{3}{2}}^1 \\&= \frac{1}{\pi} - \left(-\frac{1}{\pi} \right) - \left[0 - \frac{1}{\pi^2} - \frac{1}{\pi} - 0 \right] \\&= \frac{3}{\pi} + \frac{1}{\pi^2}\end{aligned}$$

Hence, $\int_{-1}^{\frac{3}{2}} |x \sin \pi x| dx = \frac{3}{\pi} + \frac{1}{\pi^2}.$

Long Answer Questions

5 Marks

60. Evaluate the following integrals:

(i) $\int \frac{x^5 + 4}{x^5 - x} dx$

Ans: The given integral can be written as

$$\begin{aligned}\int \frac{x^5 + 4}{x^5 - x} dx &= \int \left(\frac{x^5}{x^5 - x} + \frac{4}{x^5 - x} \right) dx \\&= \int \frac{x^5}{x^5 - x} dx + \int \frac{4}{x^5 - x} dx \\&= I_1 + I_2 \text{ (say)}\end{aligned}$$

Now, $I_1 = \int \frac{x^5}{x^5 - x} dx$

$$= \int \frac{x^5 dx}{x(x^4 - 1)}$$

$$= \int \frac{x^4}{x^4 - 1} dx$$

$$= \int \left[1 - \frac{1}{2(x^2 + 1)} - \frac{1}{4(x + 1)} + \frac{1}{4(x - 1)} \right] dx \quad [\text{Taking partial fraction}]$$

$$= \int dx - \int \frac{1}{2(x^2 + 1)} dx - \int \frac{1}{4(x+1)} dx - \int \frac{1}{4(x-1)} dx \quad [\text{sum rule of integration}]$$

$$= x - \frac{1}{2} \tan^{-1} x - \frac{1}{4} \log|x+1| + \frac{1}{4} \log|x-1| + C_1$$

$$\text{Also, } I_2 = \int \frac{4}{x^5 - x} dx$$

$$= 4 \int \left(-\frac{1}{x} + \frac{x}{2(x^2 + 1)} + \frac{1}{4(x+1)} + \frac{1}{4(x-1)} \right) dx$$

$$= 4 \left[-\int \frac{dx}{x} + \int \frac{d(x^2 + 1)}{4(x^2 + 1)} + \frac{1}{4} \int \frac{dx}{x+1} + \frac{1}{4} \int \frac{dx}{x-1} \right] \quad [\text{sum rule of integration}]$$

$$= -4 \log|x| + \log|x^2 + 1| + \log|x+1| + \log|x-1| + C_2$$

Therefore, substituting the values of I_1 and I_2

$$\begin{aligned} \int \frac{x^5 + 4}{x^5 - x} dx &= x - \frac{1}{2} \tan^{-1} x - \frac{1}{4} \log|x+1| + \frac{1}{4} \log|x-1| \\ &\quad - 4 \log|x| + \log|x^2 + 1| + \log|x+1| + \log|x-1| + C \end{aligned}$$

$$\Rightarrow \int \frac{x^5 + 4}{x^5 - x} dx = x - \frac{1}{2} \tan^{-1} x + \log|x^2 + 1| + \frac{3}{4} \log|x+1| + \frac{5}{4} \log|x-1| - 4 \log|x| + C,$$

where C is an integration constant.

$$(ii) \int \frac{dx}{(x-1)(x^2 + 4)}$$

Ans: The given integral can be written as

$$\int \frac{dx}{(x-1)(x^2 + 4)} = \int \left[\frac{1}{5(x-1)} - \frac{x}{5(x^2 + 4)} - \frac{1}{5(x^2 + 4)} \right] dx \quad [\text{taking partial fraction}]$$

$$\Rightarrow \int \frac{dx}{(x-1)(x^2+4)} = \frac{1}{5} \int \frac{dx}{x-1} - \frac{1}{5} \cdot \frac{1}{2} \int \frac{d(x^2+4)}{x^2+4} - \frac{1}{5} \int \frac{dx}{x^2+2^2}$$

$$\Rightarrow \int \frac{dx}{(x-1)(x^2+4)} = \frac{1}{5} \log|x-1| - \frac{1}{10} \log|x^2+4| - \frac{1}{5} \cdot \frac{1}{2} \tan^{-1} \frac{x}{2} + C$$

$$\Rightarrow \int \frac{dx}{(x-1)(x^2+4)} = \frac{1}{5} \log|x-1| - \frac{1}{10} \log|x^2+4| - \frac{1}{10} \tan^{-1} \frac{x}{2} + C, \text{ where } C \text{ is an integration constant.}$$

$$\text{(iii)} \int \frac{2x^3}{(x+1)(x-3)^2} dx$$

Ans: The given integral can be written as

$$\int \frac{2x^3}{(x+1)(x-3)^2} dx = \int \left[1 - \frac{1}{16(x+1)} + \frac{81}{16(x-3)} + \frac{27}{4(x-3)^2} \right] dx \quad [\text{taking partial fraction}]$$

$$\Rightarrow \int \frac{2x^3}{(x+1)(x-3)^2} dx = 2 \left[\int dx - \frac{1}{16} \int \frac{dx}{x+1} + \frac{81}{16} \int \frac{dx}{x-3} + \frac{27}{4} \int \frac{dx}{(x-3)^2} \right] \quad [\text{by sum rule of integration}]$$

$$\Rightarrow \int \frac{2x^3}{(x+1)(x-3)^2} dx = 2 \left[x - \frac{1}{16} \log|x+1| + \frac{81}{16} \log|x-3| - \frac{27}{4} \cdot \frac{1}{(x-3)} \right] + C,$$

where C is an integration constant.

$$\text{(iv)} \int \frac{x^4}{x^4-16} dx$$

Ans: The given integral can be written as

$$\int \frac{x^4}{x^4-16} dx = \int \frac{x^4-16+16}{x^4-16} dx$$

$$= \int \left(1 + \frac{16}{x^4-16} \right) dx$$

$$= \int dx + 16 \int \left[-\frac{1}{8(x^2 + 4)} - \frac{1}{32(x + 2)} + \frac{1}{32(x - 2)} \right] dx \quad [\text{taking partial fraction}]$$

$$= x - 2 \int \frac{dx}{x^2 + 2^2} - \frac{1}{2} \int \frac{dx}{x + 2} + \frac{1}{2} \int \frac{dx}{x - 2} + C, \text{ where } C \text{ is an integration constant.}$$

$$= x - \frac{2}{2} \tan^{-1} \frac{x}{2} - \frac{1}{2} \log|x + 2| + \frac{1}{2} |x - 2| + C$$

$$= x - \tan^{-1} \frac{x}{2} + \log \sqrt{\frac{x - 2}{x + 2}} + C$$

Hence, $\int \frac{x^4}{x^4 - 16} dx = x - \tan^{-1} \frac{x}{2} + \log \sqrt{\frac{x - 2}{x + 2}} + C$, where C is an integration constant.

$$(v) \int_0^{\frac{\pi}{2}} (\sqrt{\tan x} + \sqrt{\cot x}) dx$$

Ans: The given integral can be written as

$$\int_0^{\frac{\pi}{2}} (\sqrt{\tan x} + \sqrt{\cot x}) dx = \int_0^{\frac{\pi}{2}} \left(\sqrt{\frac{\sin x}{\cos x}} + \sqrt{\frac{\cos x}{\sin x}} \right) dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sqrt{\cos x \sin x}} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sqrt{-\frac{1}{2}(-2 \sin x \cos x)}} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sqrt{-\frac{1}{2}(1 - 2 \sin x \cos x - 1)}} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sqrt{-\frac{1}{2}(\sin^2 x + \cos^2 x - 2\sin x \cos x - 1)}} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sqrt{-\frac{1}{2}[(\sin x - \cos x)^2 - 1]}} dx$$

$$= \sqrt{2} \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} dx$$

Now, note that, $\frac{d}{dx}(\sin x - \cos x) = \cos x + \sin x$

Therefore, we have

$$\int_0^{\frac{\pi}{2}} (\sqrt{\tan x} + \sqrt{\cot x}) dx = \sqrt{2} \int_0^{\frac{\pi}{2}} \frac{d(\sin x - \cos x)}{\sqrt{1 - (\sin x - \cos x)^2}}$$

$$= \sqrt{2} \left[\sin^{-1}(\sin x - \cos x) \right]_0^{\frac{\pi}{2}}$$

$$= \sqrt{2} \left[\sin^{-1} \left(\sin \frac{\pi}{2} - \cos \frac{\pi}{2} \right) - \sin^{-1}(\sin 0 - \cos 0) \right]$$

$$= \sqrt{2} \left[\sin^{-1}(1 - 0) - \sin^{-1}(0 - 1) \right]$$

$$= \sqrt{2} \left(\frac{\pi}{2} + \frac{\pi}{2} \right)$$

$$= \sqrt{2}\pi$$

$$\text{Hence, } \int_0^{\frac{\pi}{2}} (\sqrt{\tan x} + \sqrt{\cot x}) dx = \sqrt{2}\pi.$$

$$\text{(vi) } \int \frac{dx}{x^4 + 1}$$

Ans: Note that, $(x^4 + 1) = (x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1)$.

Therefore, by using partial fraction, it can be obtained that,

$$\frac{1}{x^4 + 1} = \frac{-\frac{1}{2\sqrt{2}}x + \frac{1}{2}}{x^2 - \sqrt{2}x + 1} + \frac{\frac{1}{2\sqrt{2}}x + \frac{1}{2}}{x^2 + \sqrt{2}x + 1}$$

$$\Rightarrow \frac{1}{x^4 + 1} = \frac{1}{2\sqrt{2}} \frac{-x + \sqrt{2}}{x^2 - \sqrt{2}x + 1} + \frac{1}{2\sqrt{2}} \cdot \frac{x + \sqrt{2}}{x^2 + \sqrt{2}x + 1}$$

Then, the given integral can be written as

$$\int \frac{dx}{x^4 + 1} = \int \left[\frac{1}{2\sqrt{2}} \frac{-x + \sqrt{2}}{x^2 - \sqrt{2}x + 1} + \frac{1}{2\sqrt{2}} \cdot \frac{x + \sqrt{2}}{x^2 + \sqrt{2}x + 1} \right] dx$$

$$\Rightarrow \int \frac{dx}{x^4 + 1} = \frac{1}{2\sqrt{2}} \int \frac{-x + \sqrt{2}}{x^2 - \sqrt{2}x + 1} dx + \frac{1}{2\sqrt{2}} \int \frac{x + \sqrt{2}}{x^2 + \sqrt{2}x + 1} dx$$

$$= \frac{1}{4\sqrt{2}} \left(-\int \frac{2x - \sqrt{2}}{x^2 - \sqrt{2}x + 1} dx + \int \frac{\sqrt{2}}{x^2 - \sqrt{2}x + 1} dx + \int \frac{2x + \sqrt{2}}{x^2 + \sqrt{2}x + 1} dx + \int \frac{\sqrt{2} dx}{x^2 + \sqrt{2}x + 1} \right)$$

$$= \frac{1}{4\sqrt{2}} \left(\log|x^2 + \sqrt{2}x + 1| - \log|x^2 - \sqrt{2}x + 1| \right) + \frac{1}{2} \int \frac{dx}{x^2 + \sqrt{2}x + 1}$$

$$+ \frac{1}{2} \int \frac{dx}{x^2 - \sqrt{2}x + 1}$$

$$= \frac{1}{4\sqrt{2}} \log \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| + \frac{1}{4} \int \frac{dx}{\left(x + \frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} + \frac{1}{2} \int \frac{dx}{\left(x - \frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$$

$$= \frac{1}{4\sqrt{2}} \log \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| + \frac{1}{2\sqrt{2}} \int \frac{\sqrt{2} dx}{(\sqrt{2}x + 1)^2 + 1^2} + \frac{1}{2\sqrt{2}} \int \frac{\sqrt{2} dx}{(\sqrt{2}x - 1)^2 + 1^2}$$

Hence,

$$\int \frac{dx}{x^4+1} = \frac{1}{4\sqrt{2}} \log \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| + \frac{1}{2\sqrt{2}} \left[\tan^{-1}(\sqrt{2}x + 1) + \tan^{-1}(\sqrt{2}x - 1) \right] + C$$

By simplifying, we get

$$\int \frac{dx}{x^4+1} = \frac{1}{4\sqrt{2}} \log \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| + \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{2x}{1-x^2} \right) + C, \text{ where } C \text{ is an}$$

integration constant.

$$\text{(vii)} \int_0^{\infty} \frac{x \tan^{-1} x}{(1+x^2)^2} dx$$

Ans: Suppose that, $x = \tan z$

Then, $dx = \sec^2 z dz$

Also, $x \rightarrow 0 \Rightarrow z \rightarrow 0$ and

$$x \rightarrow \infty \Rightarrow z \rightarrow \frac{\pi}{2}.$$

Therefore, substituting the values we get,

$$\begin{aligned} \int_0^{\infty} \frac{x \tan^{-1} x}{(1+x^2)^2} dx &= \int_0^{\frac{\pi}{2}} \frac{z \tan z \cdot \sec^2 z}{(1+\tan^2 z)^2} dz \\ &= \int_0^{\frac{\pi}{2}} \frac{z \tan z}{\sec^2 z} dz \quad [\sec^2 z - \tan^2 z = 1] \\ &= \int_0^{\frac{\pi}{2}} \frac{z \sin z \cdot \cos^2 z}{\cos z} dz \\ &= \int_0^{\frac{\pi}{2}} z \sin z \cos z dz \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} z \sin 2z dz \end{aligned}$$

$$= \frac{1}{2} \left[\left[z \cdot \left(-\frac{\cos 2z}{2} \right) \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} 1 \cdot \frac{\cos 2z}{2} dz \right] \quad [\text{applying integration by parts}]$$

$$= \frac{1}{2} \left[-\frac{\pi}{4} \cos \pi + \frac{1}{2} \left[\frac{\sin 2\pi}{2} \right]_0^{\frac{\pi}{2}} \right]$$

$$= \frac{1}{2} \left(\frac{\pi}{2} + 0 \right)$$

$$= \frac{\pi}{8}$$

$$\text{Hence, } \int_0^{\infty} \frac{x \tan^{-1} x}{(1+x^2)^2} dx = \frac{\pi}{8}$$

61. Evaluate the following integrals as limit of sums:

$$(i) \int_2^4 (2x+1) dx$$

Ans: Recall the definition of definite integral as limit of sums:

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} h \left[f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h) \right]$$

$$\text{and } h = \frac{b-a}{n}.$$

Now, in the given integral, $a=2$, $b=4$, $f(x)=2x+1$.

$$\text{Therefore, } h = \frac{4-2}{n} = \frac{2}{n}$$

So, the given integral can be expressed as

$$\int_2^4 (2x+1) dx = \lim_{h \rightarrow 0} h \left[f(2) + f(2+h) + f(2+2h) + \dots + f(2+(n-1)h) \right]$$

$$= \lim_{h \rightarrow 0} h \left[\{2(2)+1\} + \{2(2+h)+1\} + \{2(2+2h)+1\} + \dots + \{2(2+(n-1)h)+1\} \right]$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} h \left[5 + (5+h) + (5+2h) + \dots + (5+(n-1)h) \right] \\
 &= \lim_{h \rightarrow 0} h \left[5n + h(1+2+3+\dots+(n-1)) \right] \\
 &= \lim_{h \rightarrow 0} h \left[5n + h \cdot \frac{n(n-1)}{2} \right] \quad \left[\text{since, } \sum_{i=1}^{n-1} i = 1+2+\dots+(n-1) = \frac{n(n-1)}{2} \right]
 \end{aligned}$$

Now, since, $h \rightarrow 0$ and $h = \frac{2}{n} \Rightarrow n \rightarrow \infty$.

Therefore,

$$\begin{aligned}
 \int_2^4 (2x+1) dx &= \lim_{n \rightarrow \infty} \left[5n \cdot \frac{2}{n} + \frac{2^2}{n^2} \cdot \frac{n(n-1)}{2} \right] \\
 \Rightarrow \int_2^4 (2x+1) dx &= \lim_{n \rightarrow \infty} \left[10 + \frac{4(n-1)}{n} \right] = \lim_{n \rightarrow \infty} \left[10 + 4 \left(1 - \frac{1}{n} \right) \right] \\
 \Rightarrow \int_2^4 (2x+1) dx &= 14
 \end{aligned}$$

(ii) $\int_0^2 (x^2 + 3) dx$

Ans: Recall the definition of definite integral as limit of sums:

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} h \left[f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h) \right]$$

and $h = \frac{b-a}{n}$.

Now, in the given integral, $a=0$, $b=2$, $f(x) = x^2 + 3$.

Therefore, $h = \frac{2-0}{n} = \frac{2}{n}$

So, the given integral can be expressed as

$$\int_0^2 (x^2 + 3) dx = \lim_{h \rightarrow 0} h \left[f(0) + f(0+h) + f(0+2h) + \dots + f(0+(n-1)h) \right]$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} h \left[3 + \{h^2 + 3\} + \{(2h)^2 + 3\} + \dots + \{(n-1)h\}^2 + 3 \right] \\
 &= \lim_{h \rightarrow 0} h \left[3n + h^2 \{1^2 + 2^2 + 3^2 + \dots + (n-1)^2\} \right] \\
 &= \lim_{h \rightarrow 0} h \left[3n + h^2 \cdot \frac{n(n-1)(2n-1)}{6} \right] \left[\because 1^2 + 2^2 + \dots + n^2 = \frac{n(n-1)(2n-1)}{6} \right]
 \end{aligned}$$

Now, since, $h \rightarrow 0$ and $h = \frac{2}{n}$, so $n \rightarrow \infty$.

Therefore, we have

$$\begin{aligned}
 \int_0^2 (x^2 + 3) dx &= \lim_{n \rightarrow \infty} \left[3n \cdot \frac{2}{n} + \left(\frac{2}{n} \right)^3 \cdot \frac{n(n-1)(2n-1)}{6} \right] \\
 &= \lim_{n \rightarrow \infty} \left[6 + \frac{8}{6} \left(1 - \frac{1}{n} \right) \left(2 - \frac{1}{n} \right) \right] \\
 &= 6 + \frac{4}{3}(2)
 \end{aligned}$$

Thus,

$$\int_0^2 (x^2 + 3) dx = \frac{26}{3}.$$

(iii) $\int_1^3 (3x^2 - 2x + 4) dx$

Ans: Recall the definition of definite integral as limit of sums:

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} h \left[f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h) \right]$$

and $h = \frac{b-a}{n}$.

Now, in the given integral, $a=1$, $b=3$, $f(x) = 3x^2 - 2x + 4$.

Therefore, $h = \frac{3-1}{n} = \frac{2}{n}$

So, the given integral can be expressed as

$$\begin{aligned}
 \int_1^3 (3x^2 - 2x + 4) dx &= \lim_{h \rightarrow 0} h \left[f(1) + f(1+h) + f(1+2h) + \dots + f(1+(n-1)h) \right] \\
 &= \lim_{h \rightarrow 0} h \left[\{3(1)^2 - 2(1) + 4\} + \{3(1+h)^2 - 2(1+h) + 4\} \right. \\
 &\quad \left. + \{3(1+2h)^2 - 2(1+2h) + 4\} + \dots + \{3(1+(n-1)h)^2 - 2(1+(n-1)h) + 4\} \right] \\
 &= \lim_{h \rightarrow 0} h \left[5 + (5 + 4h + 3h^2) + (5 + 8h + 12h^2) + \dots + (5 + 4(n-1)h + 3(n-1)^2 h^2) \right] \\
 &= \lim_{h \rightarrow 0} h \left[5n + \{4h + 8h + \dots + 4(n-1)h\} + \{3h^2 + 12h^2 + \dots + 3(n-1)^2 h^2\} \right] \\
 &= \lim_{h \rightarrow 0} h \left[5n + 4h \{1 + 2 + \dots + (n-1)\} + 3h^2 \{1^2 + 2^2 + \dots + (n-1)^2\} \right] \\
 &= \lim_{h \rightarrow 0} h \left[5n + 4h \cdot \frac{n(n-1)}{2} + 3h^2 \cdot \frac{n(n-1)(2n-1)}{6} \right]
 \end{aligned}$$

Now, since, $h \rightarrow 0$ and $h = \frac{2}{n}$, so $n \rightarrow \infty$.

Therefore, we have

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} \left[5n \cdot \frac{2}{n} + 4 \cdot \frac{2^2}{n^2} \cdot \frac{n(n-1)}{2} + 3 \cdot \frac{2^3}{n^3} \cdot \frac{n(n-1)(2n-1)}{6} \right] \\
 &= \lim_{n \rightarrow \infty} \left[10 + 8 \cdot \left(1 - \frac{1}{n}\right) + 4 \left(1 - \frac{1}{n}\right) \left(2 - \frac{1}{n}\right) \right] \\
 &= 10 + 8 + 8 \\
 &= 26
 \end{aligned}$$

Hence, $\int_1^3 (3x^2 - 2x + 4) dx = 26$

(iv) $\int_0^4 (3x^2 + e^{2x}) dx$

Ans: Recall the definition of definite integral as limit of sums:

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} h [f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h)]$$

and $h = \frac{b-a}{n}$.

Now, in the given integral, $a=0$, $b=4$, $f(x) = 3x^2 + e^{2x}$.

Therefore, $h = \frac{4-0}{n} = \frac{4}{n}$

So, the given integral can be expressed as

$$\begin{aligned} \int_0^4 (3x^2 + e^{2x}) dx &= \lim_{h \rightarrow 0} h [f(0) + f(0+h) + f(0+2h) + \dots + f(0+(n-1)h)] \\ &= \lim_{h \rightarrow 0} h [3 \cdot 0^2 + e^{2 \cdot 0} + (3h^2 + e^{2h}) + \{3(2h)^2 + e^{2(2h)}\} + \dots + \{3(n-1)^2 h^2 + e^{2(n-1)h}\}] \\ &= \lim_{h \rightarrow 0} h [1 + 3h^2 + e^{2h} + 12h^2 + e^{4h} + \dots + 3(n-1)^2 h^2 + e^{2(n-1)h}] \\ &= \lim_{h \rightarrow 0} h [3h^2 \{1^2 + 2^2 + \dots + (n-1)^2\} + \{1 + e^{2h} + e^{4h} + \dots + e^{2(n-1)h}\}] \end{aligned}$$

Now, observe that, the numbers $1, e^{2h}, e^{4h}, \dots, e^{2(n-1)h}$ forms a G.P series with the common ratio $r = \frac{e^{2h}}{1} = e^{2h}$.

Therefore, $S = 1 + e^{2h} + e^{4h} + \dots + e^{2(n-1)h} = 1 \left\{ \frac{(e^{2h})^n - 1}{e^{2h} - 1} \right\} = \frac{e^{2nh} - 1}{e^{2h} - 1}$

Thus, $\int_0^4 (3x^2 + e^{2x}) dx = \lim_{h \rightarrow 0} h \left[3h^2 \cdot \frac{n(n-1)(2n-1)}{6} + \frac{e^{2nh} - 1}{e^{2h} - 1} \right]$

Now, since, $h \rightarrow 0$ and $h = \frac{4}{n}$, so $n \rightarrow \infty$.

Therefore, we have

$$\int_0^4 (3x^2 + e^{2x}) dx = \lim_{n \rightarrow \infty} \left[3 \cdot \frac{4^3}{n^3} \cdot \frac{n(n-1)(2n-1)}{6} \right] + \lim_{n \rightarrow \infty} \frac{4}{n} \left[\frac{e^{2nh} - 1}{2h \cdot \frac{e^{2h} - 1}{2h}} \right]$$

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} \left[32 \left(1 - \frac{1}{n} \right) \left(2 - \frac{1}{n} \right) \right] + 4 \lim_{n \rightarrow \infty} \left(\frac{2nh - 1}{2nh} \cdot \frac{1}{\frac{e^{2h} - 1}{2h}} \right) \\
 &= 64 + 4 \lim_{n \rightarrow \infty} \frac{2nh - 1}{2nh} \cdot \lim_{n \rightarrow \infty} \frac{1}{\frac{e^{2h} - 1}{2h}} \\
 &= 64 + 4 \lim_{n \rightarrow \infty} \frac{e^{2nh} - 1}{2nh} \cdot 1 \quad [\text{since, } \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1] \\
 &= 64 + 4 \lim_{n \rightarrow \infty} \frac{e^{2n \cdot \frac{4}{n}} - 1}{2n \left(\frac{4}{n} \right)} \\
 &= 64 + 4 \lim_{n \rightarrow \infty} \frac{e^8 - 1}{8} \\
 &= 64 + \frac{e^8 - 1}{2}.
 \end{aligned}$$

Hence, $\int_0^4 (3x^2 + e^{2x}) dx = 64 + \frac{e^8 - 1}{2}.$

(v) $\int_2^5 (x^2 + 3x) dx$

Ans: Recall the definition of definite integral as limit of sums:

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} h \left[f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h) \right]$$

and $h = \frac{b-a}{n}.$

Now, in the given integral, $a = 2$, $b = 5$, $f(x) = x^2 + 3x$.

Therefore, $h = \frac{5-2}{n} = \frac{3}{n}$

So, the given integral can be expressed as

$$\begin{aligned}
 \int_2^5 (x^2 + 3x) dx &= \lim_{h \rightarrow 0} h \left[f(2) + f(2+h) + f(2+2h) + \dots + f(2+(n-1)h) \right] \\
 &= \lim_{h \rightarrow 0} h \left[\{2^2 + 3(2)\} + \{(2+h)^2 + 3(2+h)\} + \{(2+2h)^2 + 3(2+2h)\} \right. \\
 &\quad \left. + \dots + \{(2+(n-1)h)^2 + 3(2+(n-1)h)\} \right] \\
 &= \lim_{h \rightarrow 0} h \left[10 + (10 + 7h + h^2) + (10 + 14h + 4h^2) + \dots + (10 + 7(n-1)h + 4(n-1)h^2) \right] \\
 &= \lim_{h \rightarrow 0} h \left[10n + 7h \{1 + 2 + 3 + \dots + (n-1)\} + h^2 \{1^2 + 2^2 + \dots + (n-1)^2\} \right]
 \end{aligned}$$

It is known that, $1 + 2 + 3 + \dots + (n-1) = \frac{n(n-1)}{2}$ and

$$1^2 + 2^2 + \dots + (n-1)^2 = \frac{n(n-1)(2n-1)}{6}.$$

Therefore, we have

$$\int_2^5 (x^2 + 3x) dx = \lim_{h \rightarrow 0} h \left[10n + 7h \cdot \frac{n(n-1)}{2} + h^2 \cdot \frac{n(n-1)(2n-1)}{6} \right]$$

Now, since, $h \rightarrow 0$ and $h = \frac{3}{n}$, so $n \rightarrow \infty$.

Therefore, we have

$$\begin{aligned}
 \int_2^5 (x^2 + 3x) dx &= \lim_{n \rightarrow \infty} \left[10n \cdot \frac{3}{n} + 7 \cdot \frac{3^2}{n^2} \cdot \frac{n(n-1)}{2} + \frac{3^3}{n^3} \cdot \frac{n(n-1)(2n-1)}{6} \right] \\
 \Rightarrow \int_2^5 (x^2 + 3x) dx &= \lim_{n \rightarrow \infty} \left[30 + \frac{63}{2} \cdot \left(1 - \frac{1}{n}\right) + \frac{9}{2} \left(1 - \frac{1}{n}\right) \left(2 - \frac{1}{n}\right) \right] \\
 \Rightarrow \int_2^5 (x^2 + 3x) dx &= 30 + \frac{63}{2} + 9 = \frac{141}{2}.
 \end{aligned}$$

62. Evaluate

(i) $\int_0^1 \cot^{-1}(1-x+x^2) dx$

Ans: The given integral can be written as

$$\begin{aligned}
 \int_0^1 \cot^{-1}(1-x+x^2) dx &= \int_0^1 \tan^{-1}\left(\frac{1}{1-x+x^2}\right) dx \\
 &= \int_0^1 \tan^{-1}\left[\frac{x+1-x}{1-x(1-x)}\right] dx \\
 &= \int_0^1 [\tan^{-1} x + \tan^{-1}(1-x)] dx \quad [\text{since, } \tan^{-1}\left(\frac{a+b}{1-ab}\right) = \tan^{-1} a + \tan^{-1} b] \\
 &= \int_0^1 \tan^{-1}(x) dx + \int_0^1 \tan^{-1}(1-x) dx \\
 &= \int_0^1 \tan^{-1}(x) dx + \int_0^1 \tan^{-1}\{1-(1-x)\} dx \quad [\text{since } \int_0^a f(x) dx = \int_0^a f(a-x) dx] \\
 &= 2 \int_0^1 \tan^{-1}(x) dx \\
 &= 2 \int_0^1 \tan^{-1}(x) \cdot 1 dx \\
 &= 2 \left[[\tan^{-1}(x) \cdot x]_0^1 - \int_0^1 \frac{x}{1+x^2} dx \right] \\
 &= 2 \left[[\tan^{-1}(x) \cdot x]_0^1 - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx \right] \\
 &= 2 \left[[\tan^{-1}(x) \cdot x]_0^1 - \frac{1}{2} \int_0^1 \frac{d(1+x^2)}{1+x^2} dx \right] \\
 &= 2 \left[\frac{\pi}{4} - \left[\frac{1}{2} \log|1+x^2| \right]_0^1 \right] \\
 &= \frac{\pi}{2} - (\log 2 - \log 1) \\
 &= \frac{\pi}{2} - \log 2.
 \end{aligned}$$

Hence, $\int_0^1 \cot^{-1}(1-x+x^2) dx = \frac{\pi}{2} - \log 2$.

(ii) $\int \frac{dx}{(\sin x - 2\cos x)(2\sin x + \cos x)}$

Ans: The given integral can be written as

$$\begin{aligned} \int \frac{dx}{(\sin x - 2\cos x)(2\sin x + \cos x)} &= \int \frac{dx}{2\sin^2 x + \sin x \cos x - 4\sin x \cos x - 2\cos^2 x} \\ &= \int \frac{dx}{2\sin^2 x - 3\sin x \cos x - 2\cos^2 x} \\ &= \int \frac{\sec^2 x \, dx}{2\tan^2 x - 3\tan x - 2} \quad [\text{divide the numerator and denominator by } \cos^2 x] \\ &= \frac{1}{2} \int \frac{d(\tan x)}{\tan^2 x - \frac{3}{2}\tan x - 1} \quad [\text{since, } \frac{d}{dx}(\tan x) = \sec^2 x] \\ &= \frac{1}{2} \int \frac{d(\tan x)}{\left(\tan x - \frac{3}{4}\right)^2 - \left(\frac{5}{4}\right)^2} \\ &= \frac{1}{2} \times \frac{1}{2\left(\frac{5}{4}\right)} \log \left| \frac{\tan x - \frac{3}{4} - \frac{5}{4}}{\tan x - \frac{3}{4} + \frac{5}{4}} \right| + C \quad [\text{since, } \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C] \\ &= \frac{1}{5} \log \left| \frac{2\tan x - 4}{2\tan x + 1} \right| + C, \text{ where } C \text{ is an integration constant.} \end{aligned}$$

Hence, $\int \frac{dx}{(\sin x - 2\cos x)(2\sin x + \cos x)} = \frac{1}{5} \log \left| \frac{2\tan x - 4}{2\tan x + 1} \right| + C$, where C is an integration constant.

(iii) $\int_0^1 \frac{\log(1+x)}{1+x^2} dx$

Ans: Suppose that, $x = \tan \theta$

Then, $dx = \sec^2 \theta d\theta$

Also, $x \rightarrow 0 \Rightarrow \theta \rightarrow 0$ and

$$x \rightarrow 1 \Rightarrow \theta \rightarrow \frac{\pi}{4}$$

Therefore, substituting the values in the given integral we get,

$$\begin{aligned} \int_0^1 \frac{\log(1+x)}{1+x^2} dx &= \int_0^{\frac{\pi}{4}} \frac{\log(1+\tan \theta) \sec^2 \theta d\theta}{1+\tan^2 \theta} \\ &= \int_0^{\frac{\pi}{4}} \frac{\log(1+\tan \theta) \cancel{\sec^2 \theta} d\theta}{\cancel{\sec^2 \theta}} \quad [\text{since, } \sec^2 \theta - \tan^2 \theta = 1] \\ &= \int_0^{\frac{\pi}{4}} \log(1+\tan \theta) d\theta = I \text{ (say)} \end{aligned}$$

Now, since, $\int_0^a f(x) dx = \int_0^a f(a-x) dx$, so we have

$$\begin{aligned} I &= \int_0^{\frac{\pi}{4}} \log \left[1 + \tan \left(\frac{\pi}{4} - \theta \right) \right] d\theta \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log \left[1 + \frac{\tan \left(\frac{\pi}{4} \right) - \tan \theta}{1 + \tan \left(\frac{\pi}{4} \right) \tan \theta} \right] d\theta \quad [\text{since, } \tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}] \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log \left[1 + \frac{1 - \tan \theta}{1 + \tan \theta} \right] d\theta \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log \left(\frac{2}{1 + \tan \theta} \right) d\theta \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} [\log 2 - \log(1 + \tan \theta)] d\theta \end{aligned}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} \log 2 d\theta - \int_0^{\frac{\pi}{4}} \log(1 + \tan \theta) d\theta$$

$$\Rightarrow I = \log 2 \left[\theta \right]_0^{\frac{\pi}{4}} - I$$

$$\Rightarrow 2I = \frac{\pi}{4} \log 2$$

$$\Rightarrow I = \frac{\pi}{8} \log 2$$

$$\text{Hence, } \int_0^1 \frac{\log(1+x)}{1+x^2} dx = \frac{\pi}{8} \log 2.$$

$$\text{(iv) } \int_0^{\frac{\pi}{2}} (2 \log \sin x - \log \sin 2x) dx$$

Ans: Recall that, $\sin 2x = 2 \sin x \cos x$ and $\log(ab) = \log a + \log b$.

Then, the given integral can be written as

$$I = \int_0^{\frac{\pi}{2}} (2 \log \sin x - \log \sin 2x) dx$$

$$= \int_0^{\frac{\pi}{2}} [2 \log \sin x - \log(2 \sin x \cos x)] dx$$

$$= \int_0^{\frac{\pi}{2}} [2 \log \sin x - \log 2 - \log \sin x - \log \cos x] dx$$

$$= \int_0^{\frac{\pi}{2}} (\log \sin x) dx - \int_0^{\frac{\pi}{2}} \log 2 dx - \int_0^{\frac{\pi}{2}} \log(\cos x) dx$$

$$= \int_0^{\frac{\pi}{2}} \log \left[\sin \left(\frac{\pi}{2} - x \right) \right] dx - \log 2 \left[x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \log(\cos x) dx \quad \left[\int_0^a f(x) dx = \int_0^a f(a-x) dx \right]$$

$$= \int_0^{\frac{\pi}{2}} \log(\cos x) - \frac{\pi}{2} \log 2 - \int_0^{\frac{\pi}{2}} \log(\cos x)$$

$$\Rightarrow I = -\frac{\pi}{2} \log 2 = \frac{\pi}{2} \log \frac{1}{2}$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} (2 \log \sin x - \log \sin 2x) dx = \frac{\pi}{2} \log \frac{1}{2}.$$

63. $\int \frac{1}{\sin x + \sin 2x} dx$

Ans: The given integral can be written as

$$\begin{aligned} \int \frac{1}{\sin x + \sin 2x} dx &= \int \frac{dx}{\sin x + 2 \sin x \cos x} \\ &= \int \frac{dx}{\sin x (1 + 2 \cos x)} \\ &= \int \frac{\sin x dx}{\sin^2 x (1 + 2 \cos x)} \quad [\text{multiply numerator and denominator by } \sin x] \\ &= \int \frac{\sin x dx}{(1 - \cos^2 x)(1 + 2 \cos x)} \end{aligned}$$

Substitute $\cos x = z$

Then, $-\sin x dx = dz$

$$\begin{aligned} &= -\int \frac{dz}{(1 - z^2)(1 + 2z)} \\ &= -\int \left(-\frac{1}{6} \cdot \frac{1}{z-1} - \frac{1}{2} \cdot \frac{1}{z+1} + \frac{4}{3} \cdot \frac{1}{2z+1} \right) dz \\ &= \frac{1}{6} \int \frac{dz}{z-1} - \frac{1}{2} \int \frac{dz}{z+1} + \frac{4}{3} \int \frac{dz}{2z+1} \end{aligned}$$

$= \frac{1}{6} \log|z-1| - \frac{1}{2} \log|z+1| + \frac{4}{3} \cdot \frac{1}{2} \log|2z+1| + C$, where C is an integration constant.

Hence, by the substituting the value of z we get,

$\int \frac{1}{\sin x + \sin 2x} dx = \frac{1}{6} \log|\cos x - 1| - \frac{1}{2} \log|\cos x + 1| + \frac{2}{3} \log|2\cos x + 1| + C$, where C is an integration constant.

64. $\int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta} d\theta$

Ans: The given integral can be written as

$$\begin{aligned} \int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta} d\theta &= \int \frac{(3\sin\theta - 2)\cos\theta}{5 - (1 - \sin^2\theta) - 4\sin\theta} d\theta \\ &= \int \frac{(3\sin\theta - 2)\cos\theta}{4 + \sin^2\theta - 4\sin\theta} d\theta \\ &= \int \frac{(3\sin\theta - 2)\cos\theta}{(2 - \sin\theta)^2} d\theta \end{aligned}$$

Suppose that, $2 - \sin\theta = z$

Then, $-\cos\theta d\theta = dz$

Therefore, substituting the values the integral becomes

$$\begin{aligned} \int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta} d\theta &= - \int \frac{[3(2 - z) - 2] dz}{z^2} \\ &= -4 \int \frac{(4 - 3z) dz}{z^2} \\ &= -4 \int \frac{dz}{z^2} + 3 \int \frac{z dz}{z^2} \\ &= \frac{4}{z} + 3 \log|z| + C \end{aligned}$$

Thus, substituting the value of z , we get

$\int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta} d\theta = \frac{4}{2 - \sin\theta} + 3\log|2 - \sin\theta| + C$, where C is an integration constant.

65. $\int \sec^3 x dx$

Ans: The given integral can be written as

$$\begin{aligned} I &= \int \sec^3 x dx \\ &= \int \sec x \cdot \sec^2 x dx \\ &= \sec x \int \sec^2 x dx - \int \left\{ \frac{d}{dx}(\sec x) \cdot \int \sec^2 x dx \right\} dx \quad [\text{Apply integration by parts}] \\ &= \sec x \tan x - \int \sec x \tan^2 x dx \\ &= \sec x \tan x - \int \sec x (\sec^2 x - 1) dx \quad [\text{since, } \sec^2 x - \tan^2 x = 1] \\ &= \sec x \tan x - \int (\sec^3 x - \sec x) dx \\ &= \sec x \tan x + \int \sec x dx - \int \sec^3 x dx \\ &= \sec x \tan x + \log|\sec x + \tan x| - I \quad [\text{since, } I = \int \sec^3 x dx] \\ \Rightarrow 2I &= \sec x \tan x + \log|\sec x + \tan x| + C \\ \Rightarrow I &= \frac{1}{2} [\sec x \tan x + \log|\sec x + \tan x|] + C \end{aligned}$$

Hence, $\int \sec^3 x dx = \frac{1}{2} [\sec x \tan x + \log|\sec x + \tan x|] + C$, where C is an integration constant.

66. $\int e^{2x} \cos 3x dx$

Ans: The given integral can be written as

$$I = \int e^{2x} \cos 3x dx$$

$$= \frac{1}{3} e^{2x} \sin 3x - \frac{2}{3} \int e^{2x} \sin 3x dx \quad [\text{Applying integration by parts}]$$

$$= \frac{1}{3} e^{2x} \sin 3x - \frac{2}{3} \left(-\frac{1}{3} e^{2x} \cos 3x + \frac{2}{3} \int e^{2x} \cos 3x dx \right) \quad [\text{again, apply integration by parts}]$$

$$\Rightarrow I = \frac{1}{3} e^{2x} \sin 3x + \frac{2}{9} e^{2x} \cos 3x - \frac{4}{9} \int e^{2x} \cos 3x dx$$

$$\Rightarrow I = \frac{1}{3} e^{2x} \sin 3x + \frac{2}{9} e^{2x} \cos 3x - \frac{4}{9} I$$

$$\Rightarrow \frac{13}{9} I = \frac{1}{3} e^{2x} \sin 3x + \frac{2}{9} e^{2x} \cos 3x$$

$$\Rightarrow I = \frac{9}{13} \left(\frac{1}{3} e^{2x} \sin 3x + \frac{2}{9} e^{2x} \cos 3x \right) + C$$

$$\Rightarrow \int e^{2x} \cos 3x dx = \frac{e^{2x}}{13} (3 \sin 3x + 2 \cos 3x) + C, \quad \text{where } C \text{ is an integration constant.}$$